

VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY  
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY  
DEPARTMENT OF ELECTRONICS



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BIGPROJECT REPORT

**Thiết kế mạch đo nhiệt độ sử dụng  
Thermistor**

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SUBJECT: Applied Electronics (EE3129)

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Thiết kế mạch đo nhiệt độ sử dụng Thermistor. Mạch có tầm đo  $20 - 80^{\circ}\text{C}$ , độ phân giải  $0.1^{\circ}\text{C}$ , ngõ ra là điện áp tương ứng từ  $0 - 5\text{V}$ :

1. Lựa chọn và mô phỏng Thermistor 2 dây (có điện trở dây dẫn là  $10\Omega$ ).
2. Lựa chọn điện trở song song để tuyến tính hóa.
3. Lựa chọn OPAMP sử dụng, mô phỏng OPAMP theo datasheet.
4. Đảm bảo sai số (giữa giá trị lý thuyết và thực tế đo qua mạch)  $\pm 5\%$ .
5. Thiết kế mạch đọc điện áp cho ngõ ra ADC. Tiến hành lựa chọn ADC (số bit, nguồn).  
Mô phỏng mạch đọc và kiểm chứng.

## Chapter 1. Lựa chọn và Mô phỏng Thermistor 2 dây

### 1.1 Lựa chọn Thermistor

Với Thermistor 2 dây, với mỗi dây có điện trở dây dẫn là  $10\Omega$ , với tầm đo là  $20^{\circ}C \rightarrow 80^{\circ}C$  và span là  $60^{\circ}C$ . Nhóm em chọn con Thermistor NTC là **NTCLE100E3** với các thông số tổng quan như sau:



### LINKS TO ADDITIONAL RESOURCES



| QUICK REFERENCE DATA                                                       |                                                      |                    |
|----------------------------------------------------------------------------|------------------------------------------------------|--------------------|
| PARAMETER                                                                  | VALUE                                                | UNIT               |
| Resistance value at 25 °C                                                  | 3.3 to 470K                                          | $\Omega$           |
| Tolerance on $R_{25}$ -value                                               | $\pm 2$ ; $\pm 3$ ; $\pm 5$                          | %                  |
| $B_{25/85}$ -value                                                         | 2880 to 4570                                         | K                  |
| Tolerance on $B_{25/85}$ -value                                            | $\pm 0.5$ to $\pm 3$                                 | %                  |
| Operating temperature range:<br>At zero power dissipation;<br>continuously | -40 to +125                                          | $^{\circ}\text{C}$ |
| At zero power dissipation;<br>for short periods                            | $\leq 150$                                           |                    |
| Response time (in oil)                                                     | $\approx 1.2$                                        | s                  |
| Thermal time constant $\tau$<br>(for information only)                     | 15                                                   | s                  |
| Dissipation factor $\delta$<br>(for information only)                      | 7<br>8.5<br>(for $R_{25}$ -value $\leq 680 \Omega$ ) | mW/K               |
| Maximum power dissipation<br>at 55 °C                                      | 500                                                  | mW                 |
| Climatic category<br>(LCT / UCT / days)                                    | 40 / 125 / 56                                        | -                  |
| Weight                                                                     | $\approx 0.3$                                        | g                  |

Hình 1.1: Bảng Datasheet tổng quát của NTCLE100E3.

## 1.2 Mô phỏng Thermistor

Đầu tiên, ta có mô hình SPICE mô phỏng NTCLE100E3103 như sau:

```

1 .subckt NTC 1 2 Params:B0=-14.6337, B1=4791.842, B2=-115334, B3=-3.730535E+06, R25=10k,
2 + CTH=0.105, GTH0=0.007, GTH1=0, T=1,
3 + TK = 273.15
4 *****
5 *
```

```

6  Gthem 1 4 Value={V(1,4)/( R25*T*exp( (((B3/(V(3)+TK+TEMP)+B2)/(V(3)+TK+TEMP)+B1)/(V(3)+TK+TEMP) + B0) )
7  ) }
8  *
9  RP 1 4 1T
10 RS 4 2 1n
11 *
12 Gtmp1 0 3 Value={V(1,4)*V(1,4)/( R25*T*exp( (((B3/(V(3)+TK+TEMP)+B2)/(V(3)+TK+TEMP)+B1)/(V(3)+TK+TEMP) +
13 B0) ) ) }
14 C_par 3 0 {CTH}
15 R_par 3 0 1T
16 Gpar 3 0 Value={V(3)*(GTH0+(GTH1*(V(3)+TEMP)))}
17 *
18 .ends

```

Listing 1: Mô hình SPICE NTCLE100E3103.

Trong đó,

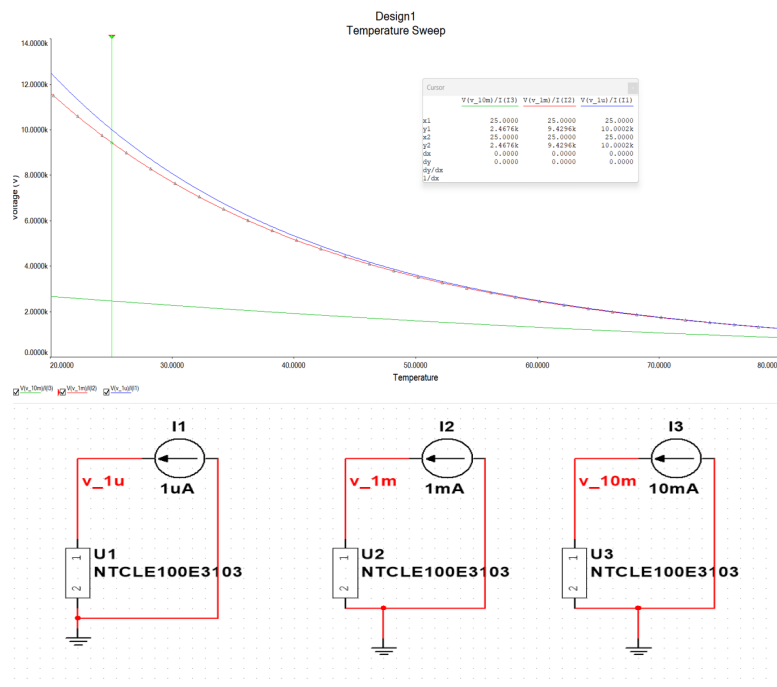
-  $R_{Thermal} = \frac{1}{CTH} = \frac{1}{0.007} = 148.57 \Omega$  trong đó GTH chính là hệ số tiêu tán của Thermistor.

-  $\tau = CTH \times R_{Thermal}$

| QUICK REFERENCE DATA                                                       |                                                             |          |
|----------------------------------------------------------------------------|-------------------------------------------------------------|----------|
| PARAMETER                                                                  | VALUE                                                       | UNIT     |
| Resistance value at 25 °C                                                  | 3.3 to 470K                                                 | $\Omega$ |
| Tolerance on $R_{25}$ -value                                               | $\pm 2; \pm 3; \pm 5$                                       | %        |
| $B_{25/85}$ -value                                                         | 2880 to 4570                                                | K        |
| Tolerance on $B_{25/85}$ -value                                            | $\pm 0.5$ to $\pm 3$                                        | %        |
| Operating temperature range:<br>At zero power dissipation;<br>continuously | -40 to +125                                                 | °C       |
| At zero power dissipation;<br>for short periods                            | $\leq 150$                                                  |          |
| Response time (in oil)                                                     | $\approx 1.2$                                               | s        |
| Thermal time constant $\tau$<br>(for information only)                     | 15                                                          | s        |
| Dissipation factor $\delta$<br>(for information only)                      | $\frac{1}{8.5}$<br>(for $R_{25}$ -value $\leq 680 \Omega$ ) | mW/K     |
| Maximum power dissipation<br>at 55 °C                                      | 500                                                         | mW       |
| Climatic category<br>(LCT / UCT / days)                                    | 40 / 125 / 56                                               | -        |
| Weight                                                                     | $\approx 0.3$                                               | g        |

$$CTH = \frac{\tau}{R_{Thermal}} = \frac{15}{148.57} = 0.105 \text{ F.}$$

GTH và CTH là hai đại lượng đặc trưng cho sự thay đổi điện trở do chính dòng điện chạy qua NTC đốt nóng nó.



Hình 1.2: Mô phỏng đồ thị có hiệu ứng nhiệt.

**Nhận xét:** Ta thấy với dòng càng lớn chảy qua NTC điện trở sau khi ổn định sẽ khá thấp do sự tỏa nhiệt lớn chính vì thế cần chọn dòng vừa đủ để hiệu ứng nhiệt được giảm bớt:

## Chapter 2. Lựa chọn điện trở song song để tuyến tính hóa

Theo datasheet của NTCLE100E3,

$$R_{(T)} = R_{ref} \times e^{A+B/T^1+C/T^2+D/T^3} \quad (2.1)$$

$$T_{(R)} = \left( A_1 + B_1 \ln^2 \frac{R}{R_{ref}} - C_1 \ln^2 \frac{R}{R_{ref}} - D_1 \ln^3 \frac{R}{R_{ref}} \right)^{-1}$$

Trong đó:

- $A, B, C, D, A_1, B_1, C_1$  và  $D_1$  là các hằng số dựa vào thông số của vật liệu.
- $R_{ref}$  là giá trị điện trở tại nhiệt độ tham chiếu (thường chọn ở  $T = 25^\circ C$ ,  $R_{ref} = R_{25}$ ).
- $T$  là nhiệt độ Kelvin (K).  $T(^{\circ}C) = T(K) - 273.15$ .

Từ phương trình 2.1,

$$R_{TH} = R_{25} \times e^{A+B/T^1+C/T^2+D/T^3}$$

$$V_e = I \times \frac{R_{TH} \times R_p}{R_{TH} + R_p}$$

Ta có:

$$\begin{aligned} \frac{dV_e}{dT} &= \frac{I \times R_p \left( R_{TH} \left( -\frac{B}{T^2} - \frac{2C}{T^3} - \frac{3D}{T^4} \right) (R_{TH} + R_p) - R_{TH} \left( -\frac{B}{T^2} - \frac{2C}{T^3} - \frac{3D}{T^4} \right) R_{TH} \right)}{(R_{TH} + R_p)^2} \\ &= \frac{I R_p^2 R_{TH} \left( -\frac{B}{T^2} - \frac{2C}{T^3} - \frac{3D}{T^4} \right)}{(R_{TH} + R_p)^2} \\ \frac{d^2 V_e}{dT^2} &= I R_p^2 \left( \frac{R_{TH} (A_1^2 + A_2) (R_{TH} + R_p)^2 - 2(R_{TH} + R_p) R_{TH}^2 A_1^2}{(R_{TH} + R_p)^4} \right) \end{aligned}$$

Trong đó,

$$- A_1 = -\frac{B}{T^2} - \frac{2C}{T^3} - \frac{3D}{T^4}$$

$$- A_2 = \frac{2B}{T^3} + \frac{6C}{T^4} + \frac{12D}{T^5}$$

$$\Rightarrow \frac{d^2 V_e}{d^2 T} = I R_p^2 \left( \frac{R_{TH}(A_1^2 + A_2)(R_{TH} + R_p) - 2R_{TH}^2 A_1^2}{(R_{TH} + R_p)^3} \right)$$

Từ Datasheet của NTCLE100E3 với  $R_{25} = 10 \text{ k}\Omega$ :

| ELECTRICAL DATA AND ORDERING INFORMATION |                              |                    |                                 |                                                                                                      | SAP MATERIAL AND ORDERING NUMBER <sup>(1)</sup>                          |                                            |
|------------------------------------------|------------------------------|--------------------|---------------------------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|--------------------------------------------|
| $R_{25}$<br>( $\Omega$ )                 | $R_{25}$ -TOL.<br>( $\pm$ %) | $B_{25/85}$<br>(K) | $B_{25/85}$ -TOL.<br>( $\pm$ %) | UL<br>RECOG.<br> US | NTCLE100E3...B0/T1/T2<br>RoHS-COMPLIANT<br>WITH EXEMPTION <sup>(2)</sup> | NTCLE100E3...B0A/T1A/T2A<br>RoHS-COMPLIANT |
| 330                                      | 5                            | 3560               | 1.5                             | ✓                                                                                                    | 331*B0                                                                   | 331*B0A                                    |
| 470                                      | 5                            | 3560               | 1.5                             | ✓                                                                                                    | 471*B0                                                                   | 471*B0A                                    |
| 680                                      | 5                            | 3560               | 1.5                             | ✓                                                                                                    | 681*B0                                                                   | 681*B0A                                    |
| 1000                                     | 2, 3, 5                      | 3528               | 0.5                             | ✓                                                                                                    | 102*B0                                                                   | 102*B0A                                    |
| 1500                                     | 2, 3, 5                      | 3528               | 0.5                             | ✓                                                                                                    | 152*B0                                                                   | 152*B0A                                    |
| 2000                                     | 2, 3, 5                      | 3528               | 0.5                             | ✓                                                                                                    | 202*B0                                                                   | 202*B0A                                    |
| 2200                                     | 2, 3, 5                      | 3977               | 0.75                            | ✓                                                                                                    | 222*B0                                                                   | 222*B0A                                    |
| 2700                                     | 2, 3, 5                      | 3977               | 0.75                            | ✓                                                                                                    | 272*B0                                                                   | 272*B0A                                    |
| 3300                                     | 2, 3, 5                      | 3977               | 0.75                            | ✓                                                                                                    | 332*B0                                                                   | 332*B0A                                    |
| 4700                                     | 2, 3, 5                      | 3977               | 0.75                            | ✓                                                                                                    | 472*B0                                                                   | 472*B0A                                    |
| 5000                                     | 2, 3, 5                      | 3977               | 0.75                            | ✓                                                                                                    | 502*B0                                                                   | 502*B0A                                    |
| 6800                                     | 2, 3, 5                      | 3977               | 0.75                            | ✓                                                                                                    | 682*B0                                                                   | 682*B0A                                    |
| 10 000                                   | 2, 3, 5                      | 3977               | 0.75                            | ✓                                                                                                    | 103*B0                                                                   | 103*B0A                                    |
| 12 000                                   | 2, 3, 5                      | 3740               | 2                               | ✓                                                                                                    | 123*B0                                                                   | 123*B0A                                    |
| 15 000                                   | 2, 3, 5                      | 3740               | 2                               | ✓                                                                                                    | 153*B0                                                                   | 153*B0A                                    |
| 22 000                                   | 2, 3, 5                      | 3740               | 2                               | ✓                                                                                                    | 223*B0                                                                   | 223*B0A                                    |
| 33 000                                   | 2, 3, 5                      | 4090               | 1.5                             | ✓                                                                                                    | 333*B0                                                                   | 333*B0A                                    |
| 47 000                                   | 2, 3, 5                      | 4090               | 1.5                             | ✓                                                                                                    | 473*B0                                                                   | 473*B0A                                    |
| 50 000                                   | 2, 3, 5                      | 4190               | 1.5                             | ✓                                                                                                    | 503*B0                                                                   | 503*B0A                                    |
| 68 000                                   | 2, 3, 5                      | 4190               | 1.5                             | ✓                                                                                                    | 683*B0                                                                   | 683*B0A                                    |
| 100 000                                  | 2, 3, 5                      | 4190               | 1.5                             | ✓                                                                                                    | 104*B0                                                                   | 104*B0A                                    |
| 150 000                                  | 2, 3, 5                      | 4370               | 2.5                             | ✓                                                                                                    | 154*B0                                                                   | 154*B0A                                    |
| 220 000                                  | 2, 3, 5                      | 4370               | 2.5                             | ✓                                                                                                    | 224*B0                                                                   | 224*B0A                                    |
| 330 000                                  | 2, 3, 5                      | 4570               | 1.5                             |                                                                                                      | 334*B0                                                                   | 334*B0A                                    |
| 470 000                                  | 2, 3, 5                      | 4570               | 1.5                             |                                                                                                      | 474*B0                                                                   | 474*B0A                                    |

Ta chọn được  $B_{25/85} = 3977$ . Từ đó ta chọn được các giá trị của  $A$ ,  $B$ ,  $C$  và  $D$  như sau:

$$- A = -14.6337$$

$$- B = 4791,842$$

$$- C = -115334$$

$$- D = -3730535$$

| PARAMETER FOR DETERMINING NOMINAL RESISTANCE VALUES |                           |                           |               |           |          |                        |                        |                |                                      |                                      |                                      |
|-----------------------------------------------------|---------------------------|---------------------------|---------------|-----------|----------|------------------------|------------------------|----------------|--------------------------------------|--------------------------------------|--------------------------------------|
| NUMBER                                              | B <sub>25/85</sub><br>(K) | NAME                      | TOL. B<br>(%) | A         | B<br>(K) | C<br>(K <sup>2</sup> ) | D<br>(K <sup>3</sup> ) | A <sub>1</sub> | B <sub>1</sub><br>(K <sup>-1</sup> ) | C <sub>1</sub><br>(K <sup>-2</sup> ) | D <sub>1</sub><br>(K <sup>-3</sup> ) |
| 1                                                   | 2880                      | Mat O. with<br>Bn = 2880K | 3             | - 9.094   | 2251.74  | 229098                 | - 2.744820E+07         | 3.354016E-03   | 3.495020E-04                         | 2.095959E-06                         | 4.260615E-07                         |
| 2                                                   | 2990                      | Mat P. with<br>Bn = 3990K | 3             | - 10.2296 | 2887.62  | 132336                 | - 2.502510E+07         | 3.354016E-03   | 3.415560E-04                         | 4.955455E-06                         | 4.364236E-07                         |
| 3                                                   | 3041                      | Mat Q. with<br>Bn = 3041K | 3             | - 11.1334 | 3658.73  | - 102895               | 5.166520E+05           | 3.354016E-03   | 3.349290E-04                         | 3.683843E-06                         | 7.050455E-07                         |
| 4                                                   | 3136                      | Mat R. with<br>Bn = 3136K | 3             | - 12.4493 | 4702.74  | - 402687               | 3.196830E+07           | 3.354016E-03   | 3.243880E-04                         | 2.658012E-06                         | - 2.701560E-07                       |
| 5                                                   | 3390                      | Mat S. with<br>Bn = 3390K | 3             | - 12.6814 | 4391.97  | - 232807               | 1.509643E+07           | 3.354016E-03   | 2.993410E-04                         | 2.135133E-06                         | - 5.672000E-09                       |
| 6                                                   | 3528 <sup>(1)</sup>       | Mat I. with<br>Bn = 3528K | 0.5           | - 12.0596 | 3687.667 | - 7617.13              | - 5.914730E+06         | 3.354016E-03   | 2.909670E-04                         | 1.632136E-06                         | 7.192200E-08                         |
|                                                     | 3528 <sup>(2)</sup>       |                           |               | - 21.0704 | 11903.95 | - 2504699              | 2.470338E+08           | 3.354016E-03   | 2.933908E-04                         | 3.494314E-06                         | - 7.712690E-07                       |
| 7                                                   | 3560                      | Mat H. with<br>Bn = 3560K | 1.5           | - 13.0723 | 4190.574 | - 47158.4              | - 1.199256E+07         | 3.354016E-03   | 2.884193E-04                         | 4.118032E-06                         | 1.786790E-07                         |
| 8                                                   | 3740                      | Mat B. with<br>Bn = 3740K | 2             | - 13.8973 | 4557.725 | - 98275                | - 7.522357E+06         | 3.354016E-03   | 2.744032E-04                         | 3.666944E-06                         | 1.375492E-07                         |
| 9                                                   | 3977                      | Mat A. with<br>Bn = 3977K | 0.75          | - 14.6337 | 4791.842 | - 115334               | - 3.730535E+06         | 3.354016E-03   | 2.569850E-04                         | 2.620131E-06                         | 6.383091E-08                         |
| 10                                                  | 4090                      | Mat C. with<br>Bn = 4090K | 1.5           | - 15.5322 | 5229.973 | - 160451               | - 5.414091E+06         | 3.354016E-03   | 2.519107E-04                         | 3.510939E-06                         | 1.105179E-07                         |
| 11                                                  | 4190                      | Mat D. with<br>Bn = 4190K | 1.5           | - 16.0349 | 5459.339 | - 191141               | - 3.328322E+06         | 3.354016E-03   | 2.460382E-04                         | 3.405377E-06                         | 1.034240E-07                         |
| 12                                                  | 4370                      | Mat E. with<br>Bn = 4370K | 2.5           | - 16.8717 | 5759.15  | - 194267               | - 6.869149E+06         | 3.354016E-03   | 2.367720E-04                         | 3.585140E-06                         | 1.255349E-07                         |
| 13                                                  | 4570                      | Mat F. with<br>Bn = 4570K | 1.5           | - 17.6439 | 6022.726 | - 203157               | - 7.183526E+06         | 3.354016E-03   | 2.264097E-04                         | 3.278184E-06                         | 1.097628E-07                         |

Notes

<sup>(1)</sup> Temperature < 25 °C

<sup>(2)</sup> Temperature ≥ 25 °C

$$\Rightarrow \begin{cases} A_1 = -0.03802560045 \\ A_2 = 2.078383337 \end{cases} \quad \text{Tại } T = 50^\circ\text{C} = 323.15\text{K}$$

$$\text{Xét } \frac{V^2 V_e}{d^2 T} = 0 \quad \text{Tại } T = 50^\circ\text{C} = 323.15\text{K}.$$

Từ Datasheets,

| RESISTANCE VALUES AT INTERMEDIATE TEMPERATURES WITH $R_{25}$ AT (2.2, 2.7, 3.3, 4.7, 5.0, 6.8, 10) k $\Omega$ |                                                         |                                                          |                                                          |                                                          |                                                          |                                                          |                                                          |                                                               |
|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------|
| T <sub>OPER</sub><br>(°C)                                                                                     | PART NUMBER<br>NTCLE100E322***<br>$R_T$<br>( $\Omega$ ) | PART NUMBER<br>NTCLE100E3272***<br>$R_T$<br>( $\Omega$ ) | PART NUMBER<br>NTCLE100E3332***<br>$R_T$<br>( $\Omega$ ) | PART NUMBER<br>NTCLE100E3472***<br>$R_T$<br>( $\Omega$ ) | PART NUMBER<br>NTCLE100E3502***<br>$R_T$<br>( $\Omega$ ) | PART NUMBER<br>NTCLE100E3682***<br>$R_T$<br>( $\Omega$ ) | PART NUMBER<br>NTCLE100E3103***<br>$R_T$<br>( $\Omega$ ) | TCR<br>(%/K)<br>$\Delta R/R$<br>DUE<br>TO<br>$B_{tot}$<br>(%) |
| -40                                                                                                           | 73 061                                                  | 89 665                                                   | 109 591                                                  | 156 084                                                  | 166 047                                                  | 225 824                                                  | 332 094                                                  | -6.62 2.79                                                    |
| -35                                                                                                           | 52 778                                                  | 64 773                                                   | 79 167                                                   | 112 753                                                  | 119 950                                                  | 163 132                                                  | 239 900                                                  | -6.39 2.52                                                    |
| -30                                                                                                           | 38 544                                                  | 47 304                                                   | 57 816                                                   | 82 344                                                   | 87 600                                                   | 119 136                                                  | 175 200                                                  | -6.18 2.26                                                    |
| -25                                                                                                           | 28 443                                                  | 34 907                                                   | 42 665                                                   | 60 765                                                   | 64 643                                                   | 87 915                                                   | 129 287                                                  | -5.98 2.02                                                    |
| -20                                                                                                           | 21 199                                                  | 26 017                                                   | 31 798                                                   | 45 288                                                   | 48 179                                                   | 65 524                                                   | 96 358                                                   | -5.78 1.78                                                    |
| -15                                                                                                           | 15 950                                                  | 19 575                                                   | 23 925                                                   | 34 075                                                   | 36 250                                                   | 49 300                                                   | 72 500                                                   | -5.60 1.55                                                    |
| -10                                                                                                           | 12 110                                                  | 14 862                                                   | 18 165                                                   | 25 872                                                   | 27 523                                                   | 37 431                                                   | 55 046                                                   | -5.42 1.33                                                    |
| -5                                                                                                            | 9275                                                    | 11 382                                                   | 13 912                                                   | 19 814                                                   | 21 078                                                   | 28 667                                                   | 42 157                                                   | -5.25 1.12                                                    |
| 0                                                                                                             | 7162                                                    | 8790                                                     | 10 743                                                   | 15 300                                                   | 16 277                                                   | 22 137                                                   | 32 554                                                   | -5.09 0.92                                                    |
| 5                                                                                                             | 5574                                                    | 6841                                                     | 8362                                                     | 11 909                                                   | 12 669                                                   | 17 230                                                   | 25 339                                                   | -4.93 0.72                                                    |
| 10                                                                                                            | 4372                                                    | 5365                                                     | 6558                                                     | 9340                                                     | 9936                                                     | 13 513                                                   | 19 872                                                   | -4.79 0.53                                                    |
| 15                                                                                                            | 3454                                                    | 4239                                                     | 5180                                                     | 7378                                                     | 7849                                                     | 10 675                                                   | 15 698                                                   | -4.64 0.35                                                    |
| 20                                                                                                            | 2747                                                    | 3372                                                     | 4121                                                     | 5869                                                     | 6244                                                     | 8492                                                     | 12 488                                                   | -4.51 0.17                                                    |
| 25                                                                                                            | 2200                                                    | 2700                                                     | 3300                                                     | 4700                                                     | 5000                                                     | 6800                                                     | 10 000                                                   | -4.38 0.00                                                    |
| 30                                                                                                            | 1773                                                    | 2176                                                     | 2659                                                     | 3788                                                     | 4030                                                     | 5480                                                     | 8059                                                     | -4.25 0.17                                                    |
| 35                                                                                                            | 1438                                                    | 1764                                                     | 2156                                                     | 3071                                                     | 3267                                                     | 4444                                                     | 6535                                                     | -4.13 0.32                                                    |
| 40                                                                                                            | 1173                                                    | 1439                                                     | 1759                                                     | 2505                                                     | 2665                                                     | 3624                                                     | 5330                                                     | -4.02 0.48                                                    |
| 45                                                                                                            | 961.8                                                   | 1180                                                     | 1443                                                     | 2055                                                     | 2186                                                     | 2973                                                     | 4372                                                     | -3.94 0.63                                                    |
| 50                                                                                                            | 793.2                                                   | 973.4                                                    | 1190                                                     | 1694                                                     | 1803                                                     | 2452                                                     | 3605                                                     | -3.80 0.77                                                    |
| 55                                                                                                            | 657.5                                                   | 806.9                                                    | 986.3                                                    | 1405                                                     | 1494                                                     | 2032                                                     | 2989                                                     | -3.70 0.91                                                    |
| 60                                                                                                            | 547.8                                                   | 672.3                                                    | 821.7                                                    | 1170                                                     | 1245                                                     | 1693                                                     | 2490                                                     | -3.60 1.05                                                    |

Ta có  $R_{50} = 3605 \Omega$ .

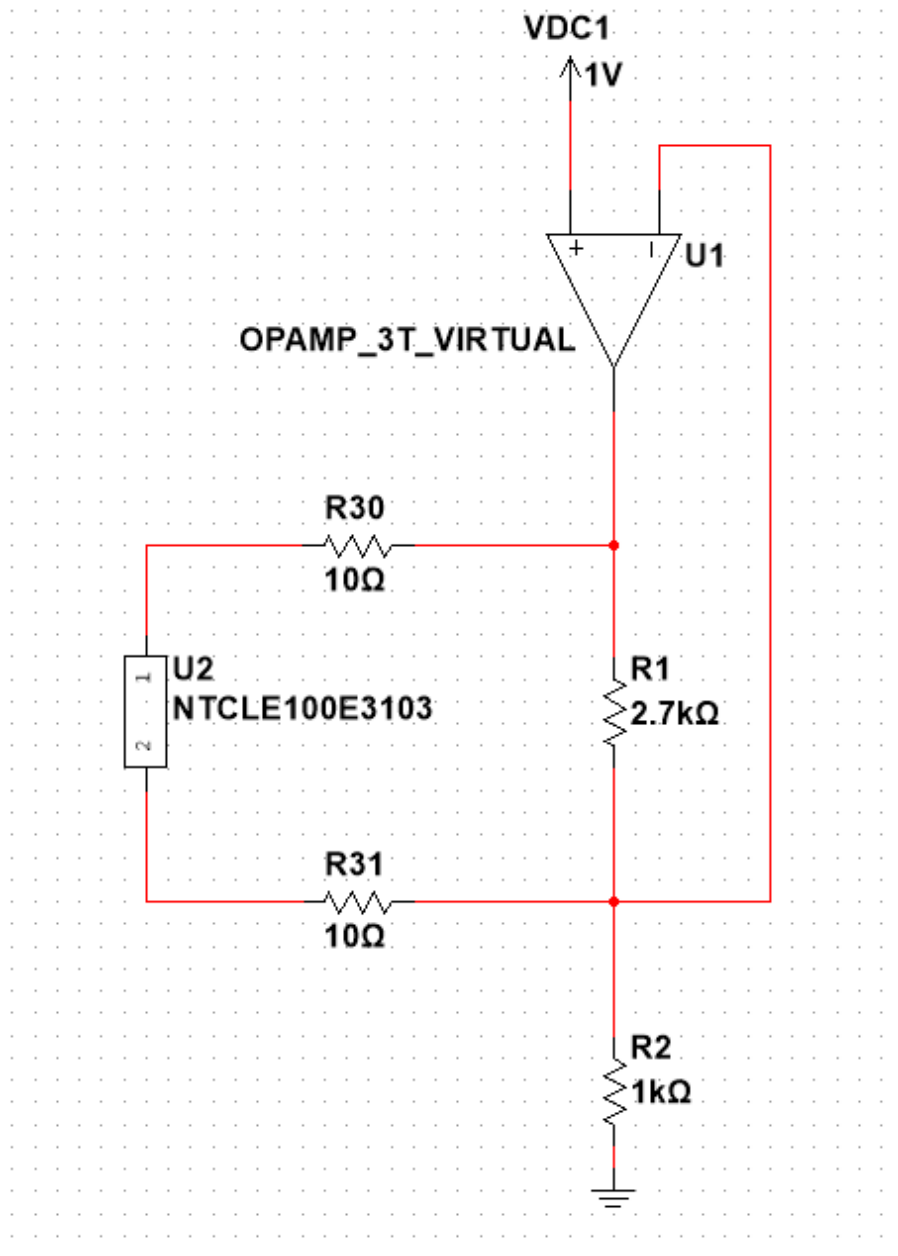
$$\Rightarrow R_p^2 R_{TH} (A_1^2 + A_2) (R_{TH} + R_p) - 2R_p^2 R_{TH}^2 A_1^2 = 0$$

$$\Leftrightarrow R_p^2 R_{50} (A_1^2 + A_2) (R_{50} + R_p) - 2R_p^2 R_{50}^2 A_1^2 = 0$$

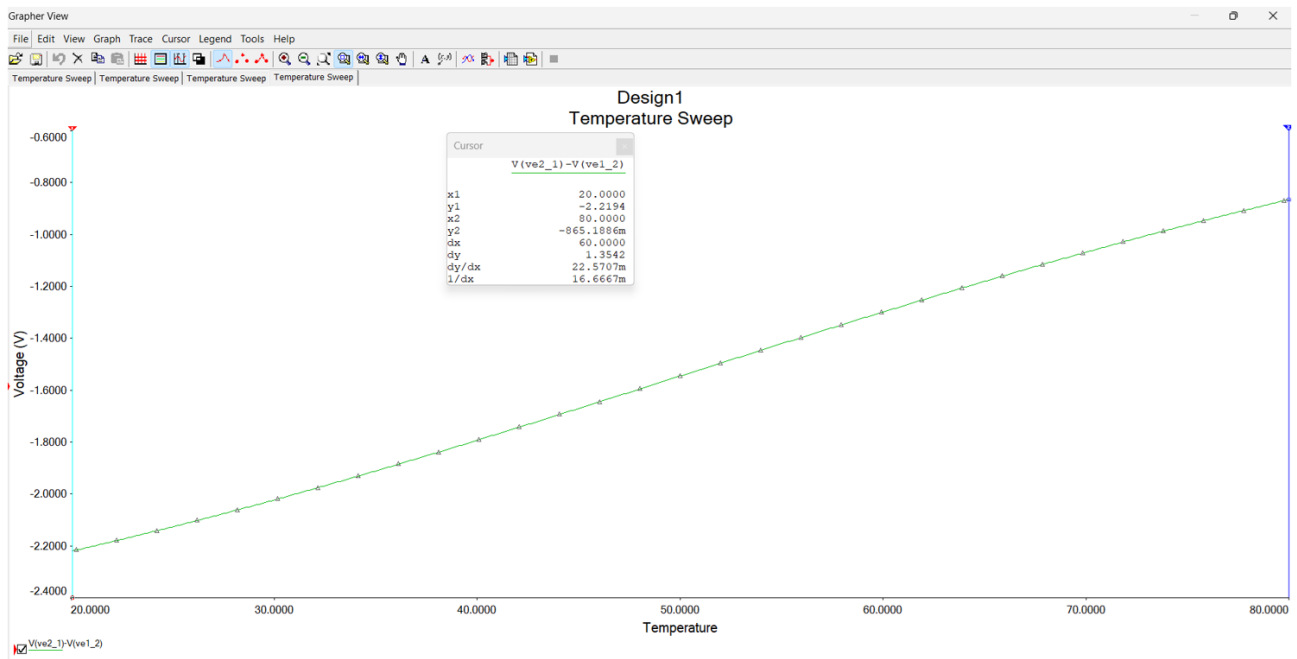
$$\Leftrightarrow R_p^3 5.962 - 16090 R_p^2 = 0$$

$$\Rightarrow R_p = 2.698 \text{ k}\Omega \approx 2.7 \text{ k}\Omega$$

## 2.1 Thiết kế mạch cho việc đo Sensor NTCLE100E3 ở lý tưởng



Hình 2.1: Mạch sau khi tuyến tính hóa bằng điện trở song song.



Hình 2.2: Điện áp sau khi tuyến tính hóa.

Giá trị tính được thể hiện qua bảng excel sau:

| t    | T      | Variable, Temperature setting   | Operating point value | Req(sim) | Req(cal)    | %error      | RTH          | Req(sim)=-23.64T+9180.2 | %error1 |                                       |          |          |          |          |
|------|--------|---------------------------------|-----------------------|----------|-------------|-------------|--------------|-------------------------|---------|---------------------------------------|----------|----------|----------|----------|
| 20   | 293.15 | V(ve1)-V(ve2), Temperature=20   | 2.21865               | 2218.65  | 2220.015444 | 0.061506046 | 12487.98866  | 2250.134                | 1.3567% | MAX ERROR                             | A        | B        | C        | D        |
| 20.1 | 293.25 | V(ve1)-V(ve2), Temperature=20.1 | 2.21711               | 2217.11  | 2218.233943 | 0.050668379 | 12431.825700 | 2247.77                 | 1.3315% | 2.9959%                               | -14.6337 | 4791.842 | -115334  | -3730535 |
| 20.2 | 293.35 | V(ve1)-V(ve2), Temperature=20.2 | 2.21556               | 2215.56  | 2216.448326 | 0.040078797 | 12375.948217 | 2245.406                | 1.3065% | MIN ERROR                             |          |          |          |          |
| 20.3 | 293.45 | V(ve1)-V(ve2), Temperature=20.3 | 2.214                 | 2214     | 2214.658596 | 0.029738019 | 12320.354608 | 2243.042                | 1.2816% | 0.0027%                               |          |          |          |          |
| 20.4 | 293.55 | V(ve1)-V(ve2), Temperature=20.4 | 2.21191               | 2211.91  | 2212.864757 | 0.043145724 | 12265.043276 | 2240.678                | 1.2569% |                                       |          |          |          |          |
| 20.5 | 293.65 | V(ve1)-V(ve2), Temperature=20.5 | 2.21034               | 2210.34  | 2211.066813 | 0.032871584 | 12210.012634 | 2238.314                | 1.2323% |                                       |          |          |          |          |
| 20.6 | 293.75 | V(ve1)-V(ve2), Temperature=20.6 | 2.2085                | 2208.5   | 2209.264768 | 0.034616415 | 12155.261103 | 2235.95                 | 1.2079% | Req(sim)=-23.64T+9180.2               | t        | T        | Req(sim) | Ve       |
| 20.7 | 293.85 | V(ve1)-V(ve2), Temperature=20.7 | 2.20665               | 2206.65  | 2207.458627 | 0.036631603 | 12100.787115 | 2233.586                | 1.1836% | Ve(T) = -0.02364T+9.1802              | 20       | 293.15   | 2250.134 | 2.250134 |
| 20.8 | 293.95 | V(ve1)-V(ve2), Temperature=20.8 | 2.20453               | 2204.53  | 2205.648395 | 0.050705938 | 12046.589112 | 2231.222                | 1.1595% |                                       | 80       | 353.15   | 831.73   | 0.83173  |
| 20.9 | 294.05 | V(ve1)-V(ve2), Temperature=20.9 | 2.20293               | 2202.93  | 2203.834074 | 0.041022797 | 11992.665545 | 2228.858                | 1.1355% | Veconvert(T) = T/12 -24.429           |          |          |          |          |
| 21   | 294.15 | V(ve1)-V(ve2), Temperature=21   | 2.20106               | 2201.06  | 2202.015671 | 0.043399825 | 11939.014874 | 2226.494                | 1.1116% |                                       | t        | T        |          | Ve       |
| 21.1 | 294.25 | V(ve1)-V(ve2), Temperature=21.1 | 2.19944               | 2199.44  | 2200.193189 | 0.034232858 | 11885.635569 | 2224.13                 | 1.0879% | Veconvert(T) = Ve(T) * -3.525 + 7.931 | 20       | 293.15   |          | 0        |
| 21.2 | 294.35 | V(ve1)-V(ve2), Temperature=21.2 | 2.19756               | 2197.56  | 2198.366633 | 0.036692388 | 11832.526109 | 2221.766                | 1.0644% |                                       | 80       | 353.15   |          | 5        |
| 21.3 | 294.45 | V(ve1)-V(ve2), Temperature=21.3 | 2.19593               | 2195.93  | 2196.536008 | 0.027589271 | 11779.684982 | 2219.402                | 1.0410% |                                       |          |          |          |          |
| 21.4 | 294.55 | V(ve1)-V(ve2), Temperature=21.4 | 2.19376               | 2193.76  | 2194.701319 | 0.042890533 | 11727.110688 | 2217.038                | 1.0178% |                                       |          |          |          |          |
| 21.5 | 294.65 | V(ve1)-V(ve2), Temperature=21.5 | 2.19186               | 2191.86  | 2192.86257  | 0.045719714 | 11674.801732 | 2214.674                | 0.9947% |                                       |          |          |          |          |
| 21.6 | 294.75 | V(ve1)-V(ve2), Temperature=21.6 | 2.18994               | 2189.94  | 2191.019768 | 0.049281505 | 11622.756631 | 2212.31                 | 0.9717% |                                       |          |          |          |          |
| 21.7 | 294.85 | V(ve1)-V(ve2), Temperature=21.7 | 2.18828               | 2188.28  | 2189.172915 | 0.040787787 | 11570.973910 | 2209.946                | 0.9489% |                                       |          |          |          |          |
| 21.8 | 294.95 | V(ve1)-V(ve2), Temperature=21.8 | 2.18635               | 2186.35  | 2187.322019 | 0.044438754 | 11519.452103 | 2207.582                | 0.9262% |                                       |          |          |          |          |
| 21.9 | 295.05 | V(ve1)-V(ve2), Temperature=21.9 | 2.18467               | 2184.67  | 2185.467083 | 0.036471983 | 11468.189754 | 2205.218                | 0.9037% |                                       |          |          |          |          |
| 22   | 295.15 | V(ve1)-V(ve2), Temperature=22   | 2.18273               | 2182.73  | 2183.608114 | 0.040213904 | 11417.185414 | 2202.854                | 0.8814% |                                       |          |          |          |          |
| 22.1 | 295.25 | V(ve1)-V(ve2), Temperature=22.1 | 2.18104               | 2181.04  | 2181.745117 | 0.03231893  | 11366.437644 | 2200.49                 | 0.8592% |                                       |          |          |          |          |
| 22.2 | 295.35 | V(ve1)-V(ve2), Temperature=22.2 | 2.17856               | 2178.56  | 2179.878097 | 0.060466525 | 11315.945015 | 2198.126                | 0.8371% |                                       |          |          |          |          |
| 22.3 | 295.45 | V(ve1)-V(ve2), Temperature=22.3 | 2.17737               | 2177.37  | 2178.007059 | 0.029249636 | 11265.706103 | 2195.762                | 0.8152% |                                       |          |          |          |          |
| 22.4 | 295.55 | V(ve1)-V(ve2), Temperature=22.4 | 2.17488               | 2174.88  | 2176.13201  | 0.057533741 | 11215.719497 | 2193.398                | 0.7934% |                                       |          |          |          |          |
| 22.5 | 295.65 | V(ve1)-V(ve2), Temperature=22.5 | 2.17316               | 2173.16  | 2174.252955 | 0.050268086 | 11165.983792 | 2191.034                | 0.7718% |                                       |          |          |          |          |
| 22.6 | 295.75 | V(ve1)-V(ve2), Temperature=22.6 | 2.17117               | 2171.17  | 2172.3699   | 0.05523463  | 11116.497593 | 2188.67                 | 0.7503% |                                       |          |          |          |          |
| 22.7 | 295.85 | V(ve1)-V(ve2), Temperature=22.7 | 2.16943               | 2169.43  | 2170.482851 | 0.048507705 | 11067.259511 | 2186.306                | 0.7290% |                                       |          |          |          |          |
| 22.8 | 295.95 | V(ve1)-V(ve2), Temperature=22.8 | 2.16743               | 2167.43  | 2168.591814 | 0.053574588 | 11018.268167 | 2183.942                | 0.7078% |                                       |          |          |          |          |
| 22.9 | 296.05 | V(ve1)-V(ve2), Temperature=22.9 | 2.16568               | 2165.68  | 2166.696795 | 0.046928335 | 10969.522193 | 2181.578                | 0.6868% |                                       |          |          |          |          |
| 23   | 296.15 | V(ve1)-V(ve2), Temperature=23   | 2.16366               | 2163.66  | 2164.797799 | 0.052559148 | 10921.020224 | 2179.214                | 0.6659% |                                       |          |          |          |          |
| 23.1 | 296.25 | V(ve1)-V(ve2), Temperature=23.1 | 2.16189               | 2161.89  | 2162.894834 | 0.046457832 | 10872.760908 | 2176.85                 | 0.6452% |                                       |          |          |          |          |
| 23.2 | 296.35 | V(ve1)-V(ve2), Temperature=23.2 | 2.15986               | 2159.86  | 2160.987905 | 0.052193964 | 10824.742897 | 2174.486                | 0.6246% |                                       |          |          |          |          |
| 23.3 | 296.45 | V(ve1)-V(ve2), Temperature=23.3 | 2.15782               | 2157.82  | 2159.077019 | 0.058220218 | 10776.964856 | 2172.122                | 0.6042% |                                       |          |          |          |          |
| 23.4 | 296.55 | V(ve1)-V(ve2), Temperature=23.4 | 2.15603               | 2156.03  | 2157.162183 | 0.052484824 | 10729.425454 | 2169.758                | 0.5839% |                                       |          |          |          |          |
| 23.5 | 296.65 | V(ve1)-V(ve2), Temperature=23.5 | 2.15424               | 2154.24  | 2155.243402 | 0.046556323 | 10682.123370 | 2167.394                | 0.5638% |                                       |          |          |          |          |
| 23.6 | 296.75 | V(ve1)-V(ve2), Temperature=23.6 | 2.15217               | 2152.17  | 2153.320684 | 0.053437647 | 10635.057291 | 2165.03                 | 0.5438% |                                       |          |          |          |          |
| 23.7 | 296.85 | V(ve1)-V(ve2), Temperature=23.7 | 2.15011               | 2150.11  | 2151.394035 | 0.059683855 | 10588.225910 | 2162.666                | 0.5239% |                                       |          |          |          |          |
| 23.8 | 296.95 | V(ve1)-V(ve2), Temperature=23.8 | 2.14829               | 2148.29  | 2149.463462 | 0.054593247 | 10541.627931 | 2160.302                | 0.5042% |                                       |          |          |          |          |
| 23.9 | 297.05 | V(ve1)-V(ve2), Temperature=23.9 | 2.14646               | 2146.46  | 2147.528972 | 0.049776825 | 10495.262063 | 2157.938                | 0.4847% |                                       |          |          |          |          |
| 24   | 297.15 | V(ve1)-V(ve2), Temperature=24   | 2.14437               | 2144.37  | 2145.590571 | 0.056887437 | 10449.127026 | 2155.574                | 0.4653% |                                       |          |          |          |          |
| 24.1 | 297.25 | V(ve1)-V(ve2), Temperature=24.1 | 2.14278               | 2142.78  | 2143.648268 | 0.04050422  | 10403.221543 | 2153.21                 | 0.4460% |                                       |          |          |          |          |
| 24.2 | 297.35 | V(ve1)-V(ve2), Temperature=24.2 | 2.14093               | 2140.93  | 2141.702069 | 0.036049297 | 10357.544349 | 2150.846                | 0.4269% |                                       |          |          |          |          |
| 24.3 | 297.45 | V(ve1)-V(ve2), Temperature=24.3 | 2.13882               | 2138.82  | 2139.75198  | 0.043555529 | 10312.094186 | 2148.482                | 0.4080% |                                       |          |          |          |          |
| 24.4 | 297.55 | V(ve1)-V(ve2), Temperature=24.4 | 2.1367                | 2136.7   | 2137.79801  | 0.05136175  | 10266.869801 | 2146.118                | 0.3892% |                                       |          |          |          |          |
| 24.5 | 297.65 | V(ve1)-V(ve2), Temperature=24.5 | 2.13483               | 2134.83  | 2135.840167 | 0.047295978 | 10221.869951 | 2143.754                | 0.3705% |                                       |          |          |          |          |
| 24.6 | 297.75 | V(ve1)-V(ve2), Temperature=24.6 | 2.13295               | 2132.95  | 2133.878456 | 0.043510249 | 10177.093400 | 2141.39                 | 0.3520% |                                       |          |          |          |          |
| 24.7 | 297.85 | V(ve1)-V(ve2), Temperature=24.7 | 2.13055               | 2130.55  | 2131.912886 | 0.063927844 | 10132.538918 | 2139.026                | 0.3336% |                                       |          |          |          |          |
| 24.8 | 297.95 | V(ve1)-V(ve2), Temperature=24.8 | 2.12866               | 2128.66  | 2129.943465 | 0.060258149 | 10088.205285 | 2136.662                | 0.3154% |                                       |          |          |          |          |
| 24.9 | 298.05 | V(ve1)-V(ve2), Temperature=24.9 | 2.12726               | 2127.26  | 2127.970199 | 0.033374489 | 10044.091286 | 2134.298                | 0.2974% |                                       |          |          |          |          |
| 25   | 298.15 | V(ve1)-V(ve2), Temperature=25   | 2.12485               | 2124.85  | 2125.993098 | 0.053767709 | 10000.195715 | 2131.934                | 0.2794% |                                       |          |          |          |          |
| 25.1 | 298.25 | V(ve1)-V(ve2), Temperature=25.1 | 2.12293               | 2122.93  | 2124.012168 | 0.050949244 | 9956.517372  | 2129.57                 | 0.2617% |                                       |          |          |          |          |
| 25.2 | 298.35 | V(ve1)-V(ve2), Temperature=25.2 | 2.121                 | 2121     | 2122.027418 | 0.048416819 | 9913.050664  | 2127.206                | 0.2440% |                                       |          |          |          |          |
| 25.3 | 298.45 | V(ve1)-V(ve2), Temperature=25.3 | 2.11907               | 2119.07  | 2120.038856 | 0.045699916 | 9869.807607  | 2124.842                | 0.2266% |                                       |          |          |          |          |
| 25.4 | 298.55 | V(ve1)-V(ve2), Temperature=25.4 | 2.11688               | 2116.88  | 2118.04649  | 0.055073842 | 9826.773822  | 2122.478                | 0.2092% |                                       |          |          |          |          |
| 25.5 | 298.65 | V(ve1)-V(ve2), Temperature=25.5 | 2.11468               | 2114.68  | 2116.050327 | 0.064758724 | 9783.952538  | 2120.114                | 0.1920% |                                       |          |          |          |          |
| 25.6 | 298.75 | V(ve1)-V(ve2), Temperature=25.6 | 2.11323               | 2113.23  | 2114.050377 | 0.03880594  | 9741.342592  | 2117.75                 | 0.1750% |                                       |          |          |          |          |
| 25.7 | 298.85 | V(ve1)-V(ve2), Temperature=25.7 | 2.11076               | 2110.76  | 2112.046648 | 0.06091947  | 9698.942825  | 2115.386                | 0.1581% |                                       |          |          |          |          |
| 25.8 | 298.95 | V(ve1)-V(ve2), Temperature=25.8 | 2.1088                | 2108.8   | 2110.039147 | 0.058726269 | 9656.752090  | 2113.022                | 0.1414% |                                       |          |          |          |          |
| 25.9 | 299.05 | V(ve1)-V(ve2), Temperature=25.9 | 2.10682               | 2106.82  | 2108.027884 | 0.057299264 | 9614.769241  | 2110.658                | 0.1248% |                                       |          |          |          |          |
| 26   | 299.15 | V(ve1)-V(ve2), Temperature=26   | 2.10483               | 2104.83  | 2106.012868 | 0.056166219 | 9572.993142  | 2108.294                | 0.1083% |                                       |          |          |          |          |
| 26.1 | 299.25 | V(ve1)-V(ve2), Temperature=26.1 | 2.10284               | 2102.84  | 2103.994106 | 0.054853102 | 9531.422665  | 2105.93                 | 0.0920% |                                       |          |          |          |          |
| 26.2 | 299.35 | V(ve1)-V(ve2), Temperature=26.2 | 2.10034               | 2100.34  | 2101.971608 | 0.077622736 | 9490.056686  | 2103.566                | 0.0759% |                                       |          |          |          |          |
| 26.3 | 299.45 | V(ve1)-V(ve2), Temperature=26.3 | 2.09884               | 2098.84  | 2099.945382 | 0.052638614 | 9448.894089  | 2101.202                | 0.0598% |                                       |          |          |          |          |
| 26.4 | 299.55 | V(ve1)-V(ve2), Temperature=26.4 | 2.09657               | 2096.57  | 2097.915438 | 0.064132125 | 9407.933764  | 2098.838                | 0.0440% |                                       |          |          |          |          |
| 26.5 | 299.65 | V(ve1)-V(ve2), Temperature=26.5 | 2.0943                | 2094.3   | 2095.881784 | 0.075471041 | 9367.174610  | 2096.474                | 0.0283% |                                       |          |          |          |          |

|      |        |                                 |  |         |         |             |             |             |  |          |         |
|------|--------|---------------------------------|--|---------|---------|-------------|-------------|-------------|--|----------|---------|
| 26.6 | 299.75 | V(ve1)-V(ve2), Temperature=26.6 |  | 2.09277 | 2092.77 | 2093.844429 | 0.05131371  | 9326.615529 |  | 2094.11  | 0.0127% |
| 26.7 | 299.85 | V(ve1)-V(ve2), Temperature=26.7 |  | 2.09049 | 2090.49 | 2091.803383 | 0.06278713  | 9286.255431 |  | 2091.746 | 0.0027% |
| 26.8 | 299.95 | V(ve1)-V(ve2), Temperature=26.8 |  | 2.08844 | 2088.44 | 2089.758655 | 0.063100838 | 9246.093234 |  | 2089.382 | 0.0180% |
| 26.9 | 300.05 | V(ve1)-V(ve2), Temperature=26.9 |  | 2.08614 | 2086.14 | 2087.710254 | 0.075214188 | 9206.127861 |  | 2087.018 | 0.0332% |
| 27   | 300.15 | V(ve1)-V(ve2), Temperature=27   |  | 2.08433 | 2084.33 | 2085.65819  | 0.063682055 | 9166.358241 |  | 2084.654 | 0.0481% |
| 27.1 | 300.25 | V(ve1)-V(ve2), Temperature=27.1 |  | 2.08227 | 2082.27 | 2083.602472 | 0.06395038  | 9126.783310 |  | 2082.29  | 0.0630% |
| 27.2 | 300.35 | V(ve1)-V(ve2), Temperature=27.2 |  | 2.07995 | 2079.95 | 2081.543109 | 0.076535002 | 9087.402011 |  | 2079.926 | 0.0777% |
| 27.3 | 300.45 | V(ve1)-V(ve2), Temperature=27.3 |  | 2.07811 | 2078.11 | 2079.480112 | 0.065887222 | 9048.213292 |  | 2077.562 | 0.0922% |
| 27.4 | 300.55 | V(ve1)-V(ve2), Temperature=27.4 |  | 2.07652 | 2076.52 | 2077.413489 | 0.043009698 | 9009.216108 |  | 2075.198 | 0.1066% |
| 27.5 | 300.65 | V(ve1)-V(ve2), Temperature=27.5 |  | 2.07417 | 2074.17 | 2075.343252 | 0.056532897 | 8970.409420 |  | 2072.834 | 0.1209% |
| 27.6 | 300.75 | V(ve1)-V(ve2), Temperature=27.6 |  | 2.07207 | 2072.07 | 2073.269409 | 0.057851079 | 8931.792195 |  | 2070.47  | 0.1350% |
| 27.7 | 300.85 | V(ve1)-V(ve2), Temperature=27.7 |  | 2.07021 | 2070.21 | 2071.19197  | 0.047410887 | 8893.363408 |  | 2068.106 | 0.1490% |
| 27.8 | 300.95 | V(ve1)-V(ve2), Temperature=27.8 |  | 2.06784 | 2067.84 | 2069.110947 | 0.061424781 | 8855.122036 |  | 2065.742 | 0.1628% |
| 27.9 | 301.05 | V(ve1)-V(ve2), Temperature=27.9 |  | 2.06572 | 2065.72 | 2067.026348 | 0.06319939  | 8817.067065 |  | 2063.378 | 0.1765% |
| 28   | 301.15 | V(ve1)-V(ve2), Temperature=28   |  | 2.06334 | 2063.34 | 2064.938184 | 0.077396228 | 8779.197488 |  | 2061.014 | 0.1900% |
| 28.1 | 301.25 | V(ve1)-V(ve2), Temperature=28.1 |  | 2.06169 | 2061.69 | 2062.846466 | 0.056061648 | 8741.512301 |  | 2058.65  | 0.2034% |
| 28.2 | 301.35 | V(ve1)-V(ve2), Temperature=28.2 |  | 2.05955 | 2059.55 | 2060.751203 | 0.058289562 | 8704.010508 |  | 2056.286 | 0.2167% |
| 28.3 | 301.45 | V(ve1)-V(ve2), Temperature=28.3 |  | 2.05739 | 2057.39 | 2058.652406 | 0.061321961 | 8666.691118 |  | 2053.922 | 0.2298% |
| 28.4 | 301.55 | V(ve1)-V(ve2), Temperature=28.4 |  | 2.05523 | 2055.23 | 2056.550086 | 0.064189333 | 8629.553146 |  | 2051.558 | 0.2427% |
| 28.5 | 301.65 | V(ve1)-V(ve2), Temperature=28.5 |  | 2.05306 | 2053.06 | 2054.444253 | 0.06737845  | 8592.595613 |  | 2049.194 | 0.2556% |
| 28.6 | 301.75 | V(ve1)-V(ve2), Temperature=28.6 |  | 2.05088 | 2050.88 | 2052.334917 | 0.07089084  | 8555.817546 |  | 2046.83  | 0.2682% |
| 28.7 | 301.85 | V(ve1)-V(ve2), Temperature=28.7 |  | 2.04919 | 2049.19 | 2050.222091 | 0.050340442 | 8519.217978 |  | 2044.466 | 0.2808% |
| 28.8 | 301.95 | V(ve1)-V(ve2), Temperature=28.8 |  | 2.04675 | 2046.75 | 2048.105784 | 0.066196954 | 8482.795946 |  | 2042.102 | 0.2931% |
| 28.9 | 302.05 | V(ve1)-V(ve2), Temperature=28.9 |  | 2.04431 | 2044.31 | 2045.986007 | 0.081916822 | 8446.550494 |  | 2039.738 | 0.3054% |
| 29   | 302.15 | V(ve1)-V(ve2), Temperature=29   |  | 2.04259 | 2042.59 | 2043.862771 | 0.062272825 | 8410.480672 |  | 2037.374 | 0.3175% |
| 29.1 | 302.25 | V(ve1)-V(ve2), Temperature=29.1 |  | 2.04038 | 2040.38 | 2041.736088 | 0.066418373 | 8374.585536 |  | 2035.01  | 0.3294% |
| 29.2 | 302.35 | V(ve1)-V(ve2), Temperature=29.2 |  | 2.03816 | 2038.16 | 2039.605968 | 0.070894484 | 8338.864145 |  | 2032.646 | 0.3412% |
| 29.3 | 302.45 | V(ve1)-V(ve2), Temperature=29.3 |  | 2.03594 | 2035.94 | 2037.472423 | 0.075211964 | 8303.315565 |  | 2030.282 | 0.3529% |
| 29.4 | 302.55 | V(ve1)-V(ve2), Temperature=29.4 |  | 2.03394 | 2033.94 | 2035.335464 | 0.068561864 | 8267.938869 |  | 2027.918 | 0.3644% |
| 29.5 | 302.65 | V(ve1)-V(ve2), Temperature=29.5 |  | 2.03146 | 2031.46 | 2033.195102 | 0.085338697 | 8232.733134 |  | 2025.554 | 0.3758% |
| 29.6 | 302.75 | V(ve1)-V(ve2), Temperature=29.6 |  | 2.02969 | 2029.69 | 2031.051349 | 0.067026826 | 8197.697442 |  | 2023.19  | 0.3871% |
| 29.7 | 302.85 | V(ve1)-V(ve2), Temperature=29.7 |  | 2.02744 | 2027.44 | 2028.904217 | 0.072167853 | 8162.830881 |  | 2020.826 | 0.3982% |
| 29.8 | 302.95 | V(ve1)-V(ve2), Temperature=29.8 |  | 2.02518 | 2025.18 | 2026.753716 | 0.077647116 | 8128.132544 |  | 2018.462 | 0.4091% |
| 29.9 | 303.05 | V(ve1)-V(ve2), Temperature=29.9 |  | 2.02339 | 2023.39 | 2024.599859 | 0.05975791  | 8093.601530 |  | 2016.098 | 0.4199% |
| 30   | 303.15 | V(ve1)-V(ve2), Temperature=30   |  | 2.02087 | 2020.87 | 2022.442657 | 0.077760257 | 8059.236943 |  | 2013.734 | 0.4306% |
| 30.1 | 303.25 | V(ve1)-V(ve2), Temperature=30.1 |  | 2.01882 | 2018.82 | 2020.282122 | 0.072372158 | 8025.037892 |  | 2011.37  | 0.4411% |
| 30.2 | 303.35 | V(ve1)-V(ve2), Temperature=30.2 |  | 2.01701 | 2017.01 | 2018.118266 | 0.054915807 | 7991.003492 |  | 2009.006 | 0.4515% |
| 30.3 | 303.45 | V(ve1)-V(ve2), Temperature=30.3 |  | 2.01471 | 2014.71 | 2015.951101 | 0.061564043 | 7957.132861 |  | 2006.642 | 0.4618% |
| 30.4 | 303.55 | V(ve1)-V(ve2), Temperature=30.4 |  | 2.01241 | 2012.41 | 2013.780639 | 0.068062979 | 7923.425125 |  | 2004.278 | 0.4719% |
| 30.5 | 303.65 | V(ve1)-V(ve2), Temperature=30.5 |  | 2.01033 | 2010.33 | 2011.606892 | 0.063476234 | 7889.879414 |  | 2001.914 | 0.4818% |
| 30.6 | 303.75 | V(ve1)-V(ve2), Temperature=30.6 |  | 2.00801 | 2008.01 | 2009.429873 | 0.070660479 | 7856.494862 |  | 1999.55  | 0.4917% |
| 30.7 | 303.85 | V(ve1)-V(ve2), Temperature=30.7 |  | 2.00592 | 2005.92 | 2007.249593 | 0.066239533 | 7823.270609 |  | 1997.186 | 0.5014% |
| 30.8 | 303.95 | V(ve1)-V(ve2), Temperature=30.8 |  | 2.00359 | 2003.59 | 2005.066065 | 0.073616758 | 7790.205801 |  | 1994.822 | 0.5109% |
| 30.9 | 304.05 | V(ve1)-V(ve2), Temperature=30.9 |  | 2.00149 | 2001.49 | 2002.879301 | 0.069365178 | 7757.299587 |  | 1992.458 | 0.5203% |
| 31   | 304.15 | V(ve1)-V(ve2), Temperature=31   |  | 1.99937 | 1999.37 | 2000.689314 | 0.065942955 | 7724.551122 |  | 1990.094 | 0.5296% |
| 31.1 | 304.25 | V(ve1)-V(ve2), Temperature=31.1 |  | 1.99678 | 1996.78 | 1998.496116 | 0.085870363 | 7691.959567 |  | 1987.73  | 0.5387% |
| 31.2 | 304.35 | V(ve1)-V(ve2), Temperature=31.2 |  | 1.99466 | 1994.66 | 1996.29972  | 0.082137963 | 7659.524085 |  | 1985.366 | 0.5477% |
| 31.3 | 304.45 | V(ve1)-V(ve2), Temperature=31.3 |  | 1.99276 | 1992.76 | 1994.100139 | 0.067205182 | 7627.243847 |  | 1983.002 | 0.5565% |
| 31.4 | 304.55 | V(ve1)-V(ve2), Temperature=31.4 |  | 1.99015 | 1990.15 | 1991.897385 | 0.087724637 | 7595.118028 |  | 1980.638 | 0.5653% |
| 31.5 | 304.65 | V(ve1)-V(ve2), Temperature=31.5 |  | 1.988   | 1988    | 1989.691471 | 0.085011721 | 7563.145805 |  | 1978.274 | 0.5738% |
| 31.6 | 304.75 | V(ve1)-V(ve2), Temperature=31.6 |  | 1.98608 | 1986.08 | 1987.48241  | 0.070562154 | 7531.326365 |  | 1975.91  | 0.5823% |
| 31.7 | 304.85 | V(ve1)-V(ve2), Temperature=31.7 |  | 1.98392 | 1983.92 | 1985.270216 | 0.06801169  | 7499.658894 |  | 1973.546 | 0.5906% |
| 31.8 | 304.95 | V(ve1)-V(ve2), Temperature=31.8 |  | 1.98128 | 1981.28 | 1983.0549   | 0.08950334  | 7468.142587 |  | 1971.182 | 0.5987% |
| 31.9 | 305.05 | V(ve1)-V(ve2), Temperature=31.9 |  | 1.9791  | 1979.1  | 1980.836477 | 0.087663836 | 7436.776643 |  | 1968.818 | 0.6067% |
| 32   | 305.15 | V(ve1)-V(ve2), Temperature=32   |  | 1.97715 | 1977.15 | 1978.614959 | 0.074039643 | 7405.560263 |  | 1966.454 | 0.6146% |
| 32.1 | 305.25 | V(ve1)-V(ve2), Temperature=32.1 |  | 1.97496 | 1974.96 | 1976.39036  | 0.072372363 | 7374.492656 |  | 1964.09  | 0.6224% |
| 32.2 | 305.35 | V(ve1)-V(ve2), Temperature=32.2 |  | 1.97229 | 1972.29 | 1974.162693 | 0.09486013  | 7343.573033 |  | 1961.726 | 0.6300% |
| 32.3 | 305.45 | V(ve1)-V(ve2), Temperature=32.3 |  | 1.97032 | 1970.32 | 1971.931972 | 0.0817458   | 7312.800611 |  | 1959.362 | 0.6374% |
| 32.4 | 305.55 | V(ve1)-V(ve2), Temperature=32.4 |  | 1.96788 | 1967.88 | 1969.698209 | 0.092309    | 7282.174612 |  | 1956.998 | 0.6448% |
| 32.5 | 305.65 | V(ve1)-V(ve2), Temperature=32.5 |  | 1.96542 | 1965.42 | 1967.461418 | 0.103758994 | 7251.694260 |  | 1954.634 | 0.6520% |
| 32.6 | 305.75 | V(ve1)-V(ve2), Temperature=32.6 |  | 1.96343 | 1963.43 | 1965.221614 | 0.091165985 | 7221.358787 |  | 1952.27  | 0.6590% |
| 32.7 | 305.85 | V(ve1)-V(ve2), Temperature=32.7 |  | 1.96143 | 1961.43 | 1962.978809 | 0.07890094  | 7191.167427 |  | 1949.906 | 0.6660% |
| 32.8 | 305.95 | V(ve1)-V(ve2), Temperature=32.8 |  | 1.95942 | 1959.42 | 1960.733017 | 0.066965629 | 7161.119419 |  | 1947.542 | 0.6728% |
| 32.9 | 306.05 | V(ve1)-V(ve2), Temperature=32.9 |  | 1.95694 | 1956.94 | 1958.484253 | 0.078849383 | 7131.214006 |  | 1945.178 | 0.6794% |
| 33   | 306.15 | V(ve1)-V(ve2), Temperature=33   |  | 1.95445 | 1954.45 | 1956.232529 | 0.091120526 | 7101.450437 |  | 1942.814 | 0.6859% |
| 33.1 | 306.25 | V(ve1)-V(ve2), Temperature=33.1 |  | 1.95242 | 1952.42 | 1953.977861 | 0.079727663 | 7071.827963 |  | 1940.45  | 0.6923% |
| 33.2 | 306.35 | V(ve1)-V(ve2), Temperature=33.2 |  | 1.94992 | 1949.92 | 1951.720261 | 0.092239725 | 7042.345842 |  | 1938.086 | 0.6986% |

|      |        |                                 |         |         |             |             |             |          |         |
|------|--------|---------------------------------|---------|---------|-------------|-------------|-------------|----------|---------|
| 33.3 | 306.45 | V(ve1)-V(ve2), Temperature=33.3 | 1.94787 | 1947.87 | 1949.459745 | 0.081547969 | 7013.003333 | 1935.722 | 0.7047% |
| 33.4 | 306.55 | V(ve1)-V(ve2), Temperature=33.4 | 1.94582 | 1945.82 | 1947.196325 | 0.070682414 | 6983.799702 | 1933.358 | 0.7107% |
| 33.5 | 306.65 | V(ve1)-V(ve2), Temperature=33.5 | 1.9433  | 1943.3  | 1944.930017 | 0.083808531 | 6954.734219 | 1930.994 | 0.7165% |
| 33.6 | 306.75 | V(ve1)-V(ve2), Temperature=33.6 | 1.94123 | 1941.23 | 1942.660835 | 0.073653343 | 6925.806156 | 1928.63  | 0.7222% |
| 33.7 | 306.85 | V(ve1)-V(ve2), Temperature=33.7 | 1.93869 | 1938.69 | 1940.388792 | 0.087549049 | 6897.014792 | 1926.266 | 0.7278% |
| 33.8 | 306.95 | V(ve1)-V(ve2), Temperature=33.8 | 1.93661 | 1936.61 | 1938.113903 | 0.077596235 | 6868.359408 | 1923.902 | 0.7333% |
| 33.9 | 307.05 | V(ve1)-V(ve2), Temperature=33.9 | 1.93406 | 1934.06 | 1935.836184 | 0.091752787 | 6839.839291 | 1921.538 | 0.7386% |
| 34   | 307.15 | V(ve1)-V(ve2), Temperature=34   | 1.93196 | 1931.96 | 1933.555647 | 0.082523977 | 6811.453730 | 1919.174 | 0.7438% |
| 34.1 | 307.25 | V(ve1)-V(ve2), Temperature=34.1 | 1.92986 | 1929.86 | 1931.272308 | 0.073128377 | 6783.202019 | 1916.81  | 0.7488% |
| 34.2 | 307.35 | V(ve1)-V(ve2), Temperature=34.2 | 1.92729 | 1927.29 | 1928.986182 | 0.087931248 | 6755.083458 | 1914.446 | 0.7538% |
| 34.3 | 307.45 | V(ve1)-V(ve2), Temperature=34.3 | 1.92471 | 1924.71 | 1926.697282 | 0.103144497 | 6727.097348 | 1912.082 | 0.7586% |
| 34.4 | 307.55 | V(ve1)-V(ve2), Temperature=34.4 | 1.92259 | 1922.59 | 1924.405625 | 0.094347292 | 6699.242995 | 1909.718 | 0.7632% |
| 34.5 | 307.65 | V(ve1)-V(ve2), Temperature=34.5 | 1.92046 | 1920.46 | 1922.111224 | 0.085906768 | 6671.519710 | 1907.354 | 0.7678% |
| 34.6 | 307.75 | V(ve1)-V(ve2), Temperature=34.6 | 1.91831 | 1918.31 | 1919.814094 | 0.078345824 | 6643.926807 | 1904.99  | 0.7722% |
| 34.7 | 307.85 | V(ve1)-V(ve2), Temperature=34.7 | 1.91594 | 1915.94 | 1917.514251 | 0.082098532 | 6616.463603 | 1902.626 | 0.7764% |
| 34.8 | 307.95 | V(ve1)-V(ve2), Temperature=34.8 | 1.9131  | 1913.1  | 1915.211709 | 0.110259843 | 6589.129422 | 1900.262 | 0.7806% |
| 34.9 | 308.05 | V(ve1)-V(ve2), Temperature=34.9 | 1.91117 | 1911.17 | 1912.906484 | 0.090777274 | 6561.923590 | 1897.898 | 0.7846% |
| 35   | 308.15 | V(ve1)-V(ve2), Temperature=35   | 1.90877 | 1908.77 | 1910.598591 | 0.09570774  | 6534.845434 | 1895.534 | 0.7885% |
| 35.1 | 308.25 | V(ve1)-V(ve2), Temperature=35.1 | 1.90637 | 1906.37 | 1908.288044 | 0.10051124  | 6507.894291 | 1893.17  | 0.7922% |
| 35.2 | 308.35 | V(ve1)-V(ve2), Temperature=35.2 | 1.90464 | 1904.64 | 1905.974859 | 0.070035505 | 6481.069496 | 1890.806 | 0.7959% |
| 35.3 | 308.45 | V(ve1)-V(ve2), Temperature=35.3 | 1.90222 | 1902.22 | 1903.659052 | 0.075593978 | 6454.370392 | 1888.442 | 0.7994% |
| 35.4 | 308.55 | V(ve1)-V(ve2), Temperature=35.4 | 1.89935 | 1899.35 | 1901.340637 | 0.104696472 | 6427.796322 | 1886.078 | 0.8027% |
| 35.5 | 308.65 | V(ve1)-V(ve2), Temperature=35.5 | 1.89737 | 1897.37 | 1899.01963  | 0.086867435 | 6401.346637 | 1883.714 | 0.8060% |
| 35.6 | 308.75 | V(ve1)-V(ve2), Temperature=35.6 | 1.89493 | 1894.93 | 1896.696046 | 0.093111706 | 6375.020687 | 1881.35  | 0.8091% |
| 35.7 | 308.85 | V(ve1)-V(ve2), Temperature=35.7 | 1.89271 | 1892.71 | 1894.369901 | 0.08762288  | 6348.817830 | 1878.986 | 0.8121% |
| 35.8 | 308.95 | V(ve1)-V(ve2), Temperature=35.8 | 1.89004 | 1890.04 | 1892.041211 | 0.105769972 | 6322.737426 | 1876.622 | 0.8150% |
| 35.9 | 309.05 | V(ve1)-V(ve2), Temperature=35.9 | 1.88825 | 1888.25 | 1889.709992 | 0.077260084 | 6296.778837 | 1874.258 | 0.8177% |
| 36   | 309.15 | V(ve1)-V(ve2), Temperature=36   | 1.88579 | 1885.79 | 1887.376257 | 0.084045639 | 6270.941431 | 1871.894 | 0.8203% |
| 36.1 | 309.25 | V(ve1)-V(ve2), Temperature=36.1 | 1.88332 | 1883.32 | 1885.040025 | 0.091246064 | 6245.224578 | 1869.53  | 0.8228% |
| 36.2 | 309.35 | V(ve1)-V(ve2), Temperature=36.2 | 1.88106 | 1881.06 | 1882.70131  | 0.087178436 | 6219.627653 | 1867.166 | 0.8252% |
| 36.3 | 309.45 | V(ve1)-V(ve2), Temperature=36.3 | 1.87835 | 1878.35 | 1880.360127 | 0.106901197 | 6194.150034 | 1864.802 | 0.8274% |
| 36.4 | 309.55 | V(ve1)-V(ve2), Temperature=36.4 | 1.87609 | 1876.09 | 1878.016494 | 0.102581338 | 6168.791102 | 1862.438 | 0.8295% |
| 36.5 | 309.65 | V(ve1)-V(ve2), Temperature=36.5 | 1.87381 | 1873.81 | 1875.670426 | 0.099187271 | 6143.550241 | 1860.074 | 0.8315% |
| 36.6 | 309.75 | V(ve1)-V(ve2), Temperature=36.6 | 1.87153 | 1871.53 | 1873.321939 | 0.095655701 | 6118.426841 | 1857.71  | 0.8334% |
| 36.7 | 309.85 | V(ve1)-V(ve2), Temperature=36.7 | 1.86902 | 1869.02 | 1870.971049 | 0.104280028 | 6093.420293 | 1855.346 | 0.8351% |
| 36.8 | 309.95 | V(ve1)-V(ve2), Temperature=36.8 | 1.8665  | 1866.5  | 1868.617772 | 0.113333617 | 6068.529992 | 1852.982 | 0.8368% |
| 36.9 | 310.05 | V(ve1)-V(ve2), Temperature=36.9 | 1.86464 | 1864.64 | 1866.262124 | 0.086918355 | 6043.755337 | 1850.618 | 0.8383% |
| 37   | 310.15 | V(ve1)-V(ve2), Temperature=37   | 1.86189 | 1861.89 | 1863.904122 | 0.108059317 | 6019.095731 | 1848.254 | 0.8396% |
| 37.1 | 310.25 | V(ve1)-V(ve2), Temperature=37.1 | 1.86001 | 1860.01 | 1861.543781 | 0.082392981 | 5994.550578 | 1845.89  | 0.8409% |
| 37.2 | 310.35 | V(ve1)-V(ve2), Temperature=37.2 | 1.85725 | 1857.25 | 1859.181119 | 0.103869318 | 5970.119288 | 1843.526 | 0.8420% |
| 37.3 | 310.45 | V(ve1)-V(ve2), Temperature=37.3 | 1.85536 | 1855.36 | 1856.81615  | 0.078421896 | 5945.801273 | 1841.162 | 0.8431% |
| 37.4 | 310.55 | V(ve1)-V(ve2), Temperature=37.4 | 1.85258 | 1852.58 | 1854.448893 | 0.100778877 | 5921.595948 | 1838.798 | 0.8440% |
| 37.5 | 310.65 | V(ve1)-V(ve2), Temperature=37.5 | 1.85024 | 1850.24 | 1852.079362 | 0.099313363 | 5897.502733 | 1836.434 | 0.8447% |
| 37.6 | 310.75 | V(ve1)-V(ve2), Temperature=37.6 | 1.84811 | 1848.11 | 1849.707575 | 0.08636908  | 5873.521049 | 1834.07  | 0.8454% |
| 37.7 | 310.85 | V(ve1)-V(ve2), Temperature=37.7 | 1.84553 | 1845.53 | 1847.333549 | 0.097629835 | 5849.650322 | 1831.706 | 0.8460% |
| 37.8 | 310.95 | V(ve1)-V(ve2), Temperature=37.8 | 1.84295 | 1842.95 | 1844.957299 | 0.108799198 | 5825.889981 | 1829.342 | 0.8464% |
| 37.9 | 311.05 | V(ve1)-V(ve2), Temperature=37.9 | 1.8408  | 1840.8  | 1842.578842 | 0.096540894 | 5802.239457 | 1826.978 | 0.8467% |
| 38   | 311.15 | V(ve1)-V(ve2), Temperature=38   | 1.8382  | 1838.2  | 1840.198196 | 0.108585886 | 5778.698186 | 1824.614 | 0.8469% |
| 38.1 | 311.25 | V(ve1)-V(ve2), Temperature=38.1 | 1.83603 | 1836.03 | 1837.815376 | 0.097146623 | 5755.265605 | 1822.25  | 0.8469% |
| 38.2 | 311.35 | V(ve1)-V(ve2), Temperature=38.2 | 1.83342 | 1833.42 | 1835.430399 | 0.109532849 | 5731.941157 | 1819.886 | 0.8469% |
| 38.3 | 311.45 | V(ve1)-V(ve2), Temperature=38.3 | 1.83103 | 1831.03 | 1833.043283 | 0.10983282  | 5708.724286 | 1817.522 | 0.8467% |
| 38.4 | 311.55 | V(ve1)-V(ve2), Temperature=38.4 | 1.82884 | 1828.84 | 1830.654044 | 0.099092685 | 5685.614440 | 1815.158 | 0.8465% |
| 38.5 | 311.65 | V(ve1)-V(ve2), Temperature=38.5 | 1.82621 | 1826.21 | 1828.262699 | 0.112275944 | 5662.611068 | 1812.794 | 0.8461% |
| 38.6 | 311.75 | V(ve1)-V(ve2), Temperature=38.6 | 1.82401 | 1824.01 | 1825.869265 | 0.101829048 | 5639.713627 | 1810.43  | 0.8456% |
| 38.7 | 311.85 | V(ve1)-V(ve2), Temperature=38.7 | 1.8218  | 1821.8  | 1823.473759 | 0.091789594 | 5616.921571 | 1808.066 | 0.8450% |
| 38.8 | 311.95 | V(ve1)-V(ve2), Temperature=38.8 | 1.81916 | 1819.16 | 1821.076198 | 0.105223385 | 5594.234362 | 1805.702 | 0.8442% |
| 38.9 | 312.05 | V(ve1)-V(ve2), Temperature=38.9 | 1.81693 | 1816.93 | 1818.676599 | 0.096036806 | 5571.651463 | 1803.338 | 0.8434% |
| 39   | 312.15 | V(ve1)-V(ve2), Temperature=39   | 1.81427 | 1814.27 | 1816.274979 | 0.110389607 | 5549.172339 | 1800.974 | 0.8424% |
| 39.1 | 312.25 | V(ve1)-V(ve2), Temperature=39.1 | 1.81204 | 1812.04 | 1813.871355 | 0.100963884 | 5526.796460 | 1798.61  | 0.8414% |
| 39.2 | 312.35 | V(ve1)-V(ve2), Temperature=39.2 | 1.80937 | 1809.37 | 1811.465745 | 0.115693302 | 5504.523298 | 1796.246 | 0.8402% |
| 39.3 | 312.45 | V(ve1)-V(ve2), Temperature=39.3 | 1.80712 | 1807.12 | 1809.058165 | 0.107136667 | 5482.352327 | 1793.882 | 0.8389% |
| 39.4 | 312.55 | V(ve1)-V(ve2), Temperature=39.4 | 1.80486 | 1804.86 | 1806.648633 | 0.09900279  | 5460.283026 | 1791.518 | 0.8375% |
| 39.5 | 312.65 | V(ve1)-V(ve2), Temperature=39.5 | 1.80217 | 1802.17 | 1804.237166 | 0.114572835 | 5438.314876 | 1789.154 | 0.8360% |
| 39.6 | 312.75 | V(ve1)-V(ve2), Temperature=39.6 | 1.80011 | 1800.11 | 1801.823781 | 0.095113696 | 5416.447360 | 1786.79  | 0.8344% |
| 39.7 | 312.85 | V(ve1)-V(ve2), Temperature=39.7 | 1.7972  | 1797.2  | 1799.408497 | 0.122734589 | 5394.679966 | 1784.426 | 0.8326% |
| 39.8 | 312.95 | V(ve1)-V(ve2), Temperature=39.8 | 1.79513 | 1795.13 | 1796.991329 | 0.103580309 | 5373.012182 | 1782.062 | 0.8308% |
| 39.9 | 313.05 | V(ve1)-V(ve2), Temperature=39.9 | 1.79263 | 1792.63 | 1794.572296 | 0.108231715 | 5351.443501 | 1779.698 | 0.8288% |

|      |        |                                 |         |         |             |             |             |          |         |
|------|--------|---------------------------------|---------|---------|-------------|-------------|-------------|----------|---------|
| 40   | 313.15 | V(ve1)-V(ve2), Temperature=40   | 1.79012 | 1790.12 | 1792.151416 | 0.113350676 | 5329.973419 | 1777.334 | 0.8268% |
| 40.1 | 313.25 | V(ve1)-V(ve2), Temperature=40.1 | 1.7876  | 1787.6  | 1789.728705 | 0.118940077 | 5308.601433 | 1774.97  | 0.8246% |
| 40.2 | 313.35 | V(ve1)-V(ve2), Temperature=40.2 | 1.7855  | 1785.5  | 1787.304181 | 0.100944247 | 5287.327044 | 1772.606 | 0.8224% |
| 40.3 | 313.45 | V(ve1)-V(ve2), Temperature=40.3 | 1.78255 | 1782.55 | 1784.877862 | 0.130421342 | 5266.149757 | 1770.242 | 0.8200% |
| 40.4 | 313.55 | V(ve1)-V(ve2), Temperature=40.4 | 1.78044 | 1780.44 | 1782.449765 | 0.112752958 | 5245.069077 | 1767.878 | 0.8175% |
| 40.5 | 313.65 | V(ve1)-V(ve2), Temperature=40.5 | 1.7779  | 1777.9  | 1780.019908 | 0.119094618 | 5224.084513 | 1765.514 | 0.8149% |
| 40.6 | 313.75 | V(ve1)-V(ve2), Temperature=40.6 | 1.77578 | 1775.78 | 1777.588309 | 0.101728205 | 5203.195578 | 1763.15  | 0.8122% |
| 40.7 | 313.85 | V(ve1)-V(ve2), Temperature=40.7 | 1.7728  | 1772.8  | 1775.154985 | 0.132663613 | 5182.401786 | 1760.786 | 0.8094% |
| 40.8 | 313.95 | V(ve1)-V(ve2), Temperature=40.8 | 1.77066 | 1770.66 | 1772.719954 | 0.11620301  | 5161.702655 | 1758.422 | 0.8066% |
| 40.9 | 314.05 | V(ve1)-V(ve2), Temperature=40.9 | 1.7681  | 1768.1  | 1770.283234 | 0.123326814 | 5141.097704 | 1756.058 | 0.8036% |
| 41   | 314.15 | V(ve1)-V(ve2), Temperature=41   | 1.76574 | 1765.74 | 1767.844842 | 0.119062624 | 5120.586456 | 1753.694 | 0.8005% |
| 41.1 | 314.25 | V(ve1)-V(ve2), Temperature=41.1 | 1.76337 | 1763.37 | 1765.404797 | 0.115259542 | 5100.168437 | 1751.33  | 0.7973% |
| 41.2 | 314.35 | V(ve1)-V(ve2), Temperature=41.2 | 1.76099 | 1760.99 | 1762.963117 | 0.111920478 | 5079.843174 | 1748.966 | 0.7940% |
| 41.3 | 314.45 | V(ve1)-V(ve2), Temperature=41.3 | 1.75861 | 1758.61 | 1760.519818 | 0.108480361 | 5059.610199 | 1746.602 | 0.7906% |
| 41.4 | 314.55 | V(ve1)-V(ve2), Temperature=41.4 | 1.75581 | 1755.81 | 1758.07492  | 0.128829538 | 5039.469044 | 1744.238 | 0.7870% |
| 41.5 | 314.65 | V(ve1)-V(ve2), Temperature=41.5 | 1.75341 | 1753.41 | 1755.628439 | 0.12636155  | 5019.419245 | 1741.874 | 0.7834% |
| 41.6 | 314.75 | V(ve1)-V(ve2), Temperature=41.6 | 1.75101 | 1751.01 | 1753.180395 | 0.123797577 | 4999.460341 | 1739.51  | 0.7797% |
| 41.7 | 314.85 | V(ve1)-V(ve2), Temperature=41.7 | 1.7486  | 1748.6  | 1750.730804 | 0.121709417 | 4979.591872 | 1737.146 | 0.7760% |
| 41.8 | 314.95 | V(ve1)-V(ve2), Temperature=41.8 | 1.74619 | 1746.19 | 1748.279686 | 0.119528107 | 4959.813382 | 1734.782 | 0.7721% |
| 41.9 | 315.05 | V(ve1)-V(ve2), Temperature=41.9 | 1.74377 | 1743.77 | 1745.827057 | 0.117827076 | 4940.124417 | 1732.418 | 0.7681% |
| 42   | 315.15 | V(ve1)-V(ve2), Temperature=42   | 1.74134 | 1741.34 | 1743.372936 | 0.116609377 | 4920.524525 | 1730.054 | 0.7640% |
| 42.1 | 315.25 | V(ve1)-V(ve2), Temperature=42.1 | 1.73891 | 1738.91 | 1740.917342 | 0.115303686 | 4901.013258 | 1727.69  | 0.7598% |
| 42.2 | 315.35 | V(ve1)-V(ve2), Temperature=42.2 | 1.73606 | 1736.06 | 1738.460292 | 0.138069972 | 4881.590169 | 1725.326 | 0.7555% |
| 42.3 | 315.45 | V(ve1)-V(ve2), Temperature=42.3 | 1.73382 | 1733.82 | 1736.001804 | 0.125679811 | 4862.254814 | 1722.962 | 0.7511% |
| 42.4 | 315.55 | V(ve1)-V(ve2), Temperature=42.4 | 1.73158 | 1731.58 | 1733.541896 | 0.113172713 | 4843.006751 | 1720.598 | 0.7467% |
| 42.5 | 315.65 | V(ve1)-V(ve2), Temperature=42.5 | 1.72891 | 1728.91 | 1731.080588 | 0.125389183 | 4823.845542 | 1718.234 | 0.7421% |
| 42.6 | 315.75 | V(ve1)-V(ve2), Temperature=42.6 | 1.72665 | 1726.65 | 1728.617896 | 0.113842164 | 4804.770749 | 1715.87  | 0.7375% |
| 42.7 | 315.85 | V(ve1)-V(ve2), Temperature=42.7 | 1.72398 | 1723.98 | 1726.153839 | 0.125935434 | 4785.781938 | 1713.506 | 0.7327% |
| 42.8 | 315.95 | V(ve1)-V(ve2), Temperature=42.8 | 1.72171 | 1721.71 | 1723.688436 | 0.114779214 | 4766.878678 | 1711.142 | 0.7279% |
| 42.9 | 316.05 | V(ve1)-V(ve2), Temperature=42.9 | 1.71902 | 1719.02 | 1721.221704 | 0.127915212 | 4748.060539 | 1708.778 | 0.7230% |
| 43   | 316.15 | V(ve1)-V(ve2), Temperature=43   | 1.71674 | 1716.74 | 1718.753663 | 0.117158297 | 4729.327093 | 1706.414 | 0.7179% |
| 43.1 | 316.25 | V(ve1)-V(ve2), Temperature=43.1 | 1.71404 | 1714.04 | 1716.284329 | 0.130766726 | 4710.677916 | 1704.05  | 0.7128% |
| 43.2 | 316.35 | V(ve1)-V(ve2), Temperature=43.2 | 1.71174 | 1711.74 | 1713.813722 | 0.121000406 | 4692.112585 | 1701.686 | 0.7076% |
| 43.3 | 316.45 | V(ve1)-V(ve2), Temperature=43.3 | 1.70903 | 1709.03 | 1711.341859 | 0.135090437 | 4673.630681 | 1699.322 | 0.7024% |
| 43.4 | 316.55 | V(ve1)-V(ve2), Temperature=43.4 | 1.70652 | 1706.52 | 1708.86876  | 0.137445309 | 4655.231785 | 1696.958 | 0.6970% |
| 43.5 | 316.65 | V(ve1)-V(ve2), Temperature=43.5 | 1.7044  | 1704.4  | 1706.394442 | 0.116880491 | 4636.915482 | 1694.594 | 0.6915% |
| 43.6 | 316.75 | V(ve1)-V(ve2), Temperature=43.6 | 1.70188 | 1701.88 | 1703.918924 | 0.119660874 | 4618.681358 | 1692.23  | 0.6860% |
| 43.7 | 316.85 | V(ve1)-V(ve2), Temperature=43.7 | 1.69935 | 1699.35 | 1701.442225 | 0.122967717 | 4600.529003 | 1689.866 | 0.6804% |
| 43.8 | 316.95 | V(ve1)-V(ve2), Temperature=43.8 | 1.69681 | 1696.81 | 1698.964362 | 0.126804411 | 4582.458008 | 1687.502 | 0.6747% |
| 43.9 | 317.05 | V(ve1)-V(ve2), Temperature=43.9 | 1.69427 | 1694.27 | 1696.485354 | 0.130584912 | 4564.467966 | 1685.138 | 0.6689% |
| 44   | 317.15 | V(ve1)-V(ve2), Temperature=44   | 1.69192 | 1691.92 | 1694.00522  | 0.123094054 | 4546.558474 | 1682.774 | 0.6630% |
| 44.1 | 317.25 | V(ve1)-V(ve2), Temperature=44.1 | 1.68937 | 1689.37 | 1691.523977 | 0.12733945  | 4528.729128 | 1680.41  | 0.6570% |
| 44.2 | 317.35 | V(ve1)-V(ve2), Temperature=44.2 | 1.68681 | 1686.81 | 1689.041645 | 0.132124954 | 4510.979530 | 1678.046 | 0.6510% |
| 44.3 | 317.45 | V(ve1)-V(ve2), Temperature=44.3 | 1.68464 | 1684.64 | 1686.558243 | 0.113737112 | 4493.309282 | 1675.682 | 0.6449% |
| 44.4 | 317.55 | V(ve1)-V(ve2), Temperature=44.4 | 1.68207 | 1682.07 | 1684.073787 | 0.118984514 | 4475.717987 | 1673.318 | 0.6387% |
| 44.5 | 317.65 | V(ve1)-V(ve2), Temperature=44.5 | 1.67949 | 1679.49 | 1681.588297 | 0.124780688 | 4458.205254 | 1670.954 | 0.6324% |
| 44.6 | 317.75 | V(ve1)-V(ve2), Temperature=44.6 | 1.6771  | 1677.1  | 1679.101792 | 0.119218035 | 4440.770690 | 1668.59  | 0.6260% |
| 44.7 | 317.85 | V(ve1)-V(ve2), Temperature=44.7 | 1.67471 | 1674.71 | 1676.61429  | 0.113579474 | 4423.413906 | 1666.226 | 0.6196% |
| 44.8 | 317.95 | V(ve1)-V(ve2), Temperature=44.8 | 1.67192 | 1671.92 | 1674.125809 | 0.131758845 | 4406.134517 | 1663.862 | 0.6131% |
| 44.9 | 318.05 | V(ve1)-V(ve2), Temperature=44.9 | 1.66951 | 1669.51 | 1671.636368 | 0.12720277  | 4388.932136 | 1661.498 | 0.6065% |
| 45   | 318.15 | V(ve1)-V(ve2), Temperature=45   | 1.6671  | 1667.1  | 1669.145985 | 0.122576774 | 4371.806382 | 1659.134 | 0.5998% |
| 45.1 | 318.25 | V(ve1)-V(ve2), Temperature=45.1 | 1.66429 | 1664.29 | 1666.65468  | 0.141881807 | 4354.756873 | 1656.77  | 0.5931% |
| 45.2 | 318.35 | V(ve1)-V(ve2), Temperature=45.2 | 1.66227 | 1662.27 | 1664.16247  | 0.113719062 | 4337.783232 | 1654.406 | 0.5863% |
| 45.3 | 318.45 | V(ve1)-V(ve2), Temperature=45.3 | 1.65944 | 1659.44 | 1661.669374 | 0.134164727 | 4320.885081 | 1652.042 | 0.5794% |
| 45.4 | 318.55 | V(ve1)-V(ve2), Temperature=45.4 | 1.65701 | 1657.01 | 1659.175411 | 0.130511288 | 4304.062047 | 1649.678 | 0.5724% |
| 45.5 | 318.65 | V(ve1)-V(ve2), Temperature=45.5 | 1.65457 | 1654.57 | 1656.680599 | 0.127399292 | 4287.313756 | 1647.314 | 0.5654% |
| 45.6 | 318.75 | V(ve1)-V(ve2), Temperature=45.6 | 1.65193 | 1651.93 | 1654.184957 | 0.136318337 | 4270.639839 | 1644.95  | 0.5583% |
| 45.7 | 318.85 | V(ve1)-V(ve2), Temperature=45.7 | 1.64967 | 1649.67 | 1651.688504 | 0.122208517 | 4254.039928 | 1642.586 | 0.5511% |
| 45.8 | 318.95 | V(ve1)-V(ve2), Temperature=45.8 | 1.64683 | 1646.83 | 1649.191257 | 0.143176692 | 4237.513655 | 1640.222 | 0.5439% |
| 45.9 | 319.05 | V(ve1)-V(ve2), Temperature=45.9 | 1.64436 | 1644.36 | 1646.693237 | 0.141692243 | 4221.060657 | 1637.858 | 0.5365% |
| 46   | 319.15 | V(ve1)-V(ve2), Temperature=46   | 1.6419  | 1641.9  | 1644.19446  | 0.139549176 | 4204.680571 | 1635.494 | 0.5292% |
| 46.1 | 319.25 | V(ve1)-V(ve2), Temperature=46.1 | 1.63923 | 1639.23 | 1641.694946 | 0.150146399 | 4188.373037 | 1633.13  | 0.5217% |
| 46.2 | 319.35 | V(ve1)-V(ve2), Temperature=46.2 | 1.63694 | 1636.94 | 1639.194713 | 0.137550058 | 4172.137697 | 1630.766 | 0.5142% |
| 46.3 | 319.45 | V(ve1)-V(ve2), Temperature=46.3 | 1.63465 | 1634.65 | 1636.693781 | 0.124872514 | 4155.974194 | 1628.402 | 0.5066% |
| 46.4 | 319.55 | V(ve1)-V(ve2), Temperature=46.4 | 1.63216 | 1632.16 | 1634.192167 | 0.124352972 | 4139.882173 | 1626.038 | 0.4990% |
| 46.5 | 319.65 | V(ve1)-V(ve2), Temperature=46.5 | 1.62947 | 1629.47 | 1631.689889 | 0.136048485 | 4123.861281 | 1623.674 | 0.4913% |
| 46.6 | 319.75 | V(ve1)-V(ve2), Temperature=46.6 | 1.62715 | 1627.15 | 1629.186968 | 0.12502973  | 4107.911169 | 1621.31  | 0.4835% |

|      |        |                                 |         |         |             |             |             |          |         |
|------|--------|---------------------------------|---------|---------|-------------|-------------|-------------|----------|---------|
| 46.7 | 319.85 | V(ve1)-V(ve2), Temperature=46.7 | 1.62445 | 1624.45 | 1626.683421 | 0.137299068 | 4092.031488 | 1618.946 | 0.4757% |
| 46.8 | 319.95 | V(ve1)-V(ve2), Temperature=46.8 | 1.62213 | 1622.13 | 1624.179267 | 0.12617246  | 4076.221889 | 1616.582 | 0.4678% |
| 46.9 | 320.05 | V(ve1)-V(ve2), Temperature=46.9 | 1.61923 | 1619.23 | 1621.674524 | 0.150740746 | 4060.482029 | 1614.218 | 0.4598% |
| 47   | 320.15 | V(ve1)-V(ve2), Temperature=47   | 1.61689 | 1616.89 | 1619.169212 | 0.140764266 | 4044.811564 | 1611.854 | 0.4518% |
| 47.1 | 320.25 | V(ve1)-V(ve2), Temperature=47.1 | 1.61455 | 1614.55 | 1616.663348 | 0.130722794 | 4029.210152 | 1609.49  | 0.4437% |
| 47.2 | 320.35 | V(ve1)-V(ve2), Temperature=47.2 | 1.61201 | 1612.01 | 1614.156951 | 0.133007553 | 4013.677455 | 1607.126 | 0.4356% |
| 47.3 | 320.45 | V(ve1)-V(ve2), Temperature=47.3 | 1.60947 | 1609.47 | 1611.65004  | 0.135267557 | 3998.213135 | 1604.762 | 0.4274% |
| 47.4 | 320.55 | V(ve1)-V(ve2), Temperature=47.4 | 1.60692 | 1606.92 | 1609.142633 | 0.138125281 | 3982.816855 | 1602.398 | 0.4191% |
| 47.5 | 320.65 | V(ve1)-V(ve2), Temperature=47.5 | 1.60437 | 1604.37 | 1606.634749 | 0.140962272 | 3967.488282 | 1600.034 | 0.4108% |
| 47.6 | 320.75 | V(ve1)-V(ve2), Temperature=47.6 | 1.60181 | 1601.81 | 1604.126406 | 0.144402984 | 3952.227084 | 1597.67  | 0.4025% |
| 47.7 | 320.85 | V(ve1)-V(ve2), Temperature=47.7 | 1.59925 | 1599.25 | 1601.617624 | 0.147827027 | 3937.032929 | 1595.306 | 0.3941% |
| 47.8 | 320.95 | V(ve1)-V(ve2), Temperature=47.8 | 1.59687 | 1596.87 | 1599.10842  | 0.139979228 | 3921.905490 | 1592.942 | 0.3856% |
| 47.9 | 321.05 | V(ve1)-V(ve2), Temperature=47.9 | 1.59448 | 1594.48 | 1596.598813 | 0.132707891 | 3906.844440 | 1590.578 | 0.3771% |
| 48   | 321.15 | V(ve1)-V(ve2), Temperature=48   | 1.5919  | 1591.9  | 1594.088821 | 0.137308601 | 3891.849453 | 1588.214 | 0.3685% |
| 48.1 | 321.25 | V(ve1)-V(ve2), Temperature=48.1 | 1.5895  | 1589.5  | 1591.578464 | 0.130591333 | 3876.920206 | 1585.85  | 0.3599% |
| 48.2 | 321.35 | V(ve1)-V(ve2), Temperature=48.2 | 1.58709 | 1587.09 | 1589.067759 | 0.124460308 | 3862.056378 | 1583.486 | 0.3513% |
| 48.3 | 321.45 | V(ve1)-V(ve2), Temperature=48.3 | 1.58431 | 1584.31 | 1586.556725 | 0.141610107 | 3847.257648 | 1581.122 | 0.3425% |
| 48.4 | 321.55 | V(ve1)-V(ve2), Temperature=48.4 | 1.58171 | 1581.71 | 1584.04538  | 0.147431403 | 3832.523699 | 1578.758 | 0.3338% |
| 48.5 | 321.65 | V(ve1)-V(ve2), Temperature=48.5 | 1.57929 | 1579.29 | 1581.533744 | 0.141871384 | 3817.854213 | 1576.394 | 0.3250% |
| 48.6 | 321.75 | V(ve1)-V(ve2), Temperature=48.6 | 1.57686 | 1576.86 | 1579.021834 | 0.136909688 | 3803.248877 | 1574.03  | 0.3161% |
| 48.7 | 321.85 | V(ve1)-V(ve2), Temperature=48.7 | 1.57424 | 1574.24 | 1576.509669 | 0.143967949 | 3788.707376 | 1571.666 | 0.3072% |
| 48.8 | 321.95 | V(ve1)-V(ve2), Temperature=48.8 | 1.5718  | 1571.8  | 1573.997267 | 0.139597877 | 3774.229400 | 1569.302 | 0.2983% |
| 48.9 | 322.05 | V(ve1)-V(ve2), Temperature=48.9 | 1.56936 | 1569.36 | 1571.484647 | 0.135199961 | 3759.814639 | 1566.938 | 0.2893% |
| 49   | 322.15 | V(ve1)-V(ve2), Temperature=49   | 1.56673 | 1566.73 | 1568.971827 | 0.142885081 | 3745.462785 | 1564.574 | 0.2803% |
| 49.1 | 322.25 | V(ve1)-V(ve2), Temperature=49.1 | 1.5639  | 1563.9  | 1566.458825 | 0.163350936 | 3731.173531 | 1562.21  | 0.2712% |
| 49.2 | 322.35 | V(ve1)-V(ve2), Temperature=49.2 | 1.56181 | 1561.81 | 1563.945661 | 0.136555933 | 3716.946573 | 1559.846 | 0.2621% |
| 49.3 | 322.45 | V(ve1)-V(ve2), Temperature=49.3 | 1.55916 | 1559.16 | 1561.432351 | 0.145529917 | 3702.781607 | 1557.482 | 0.2530% |
| 49.4 | 322.55 | V(ve1)-V(ve2), Temperature=49.4 | 1.55651 | 1556.51 | 1558.918915 | 0.154524731 | 3688.678331 | 1555.118 | 0.2438% |
| 49.5 | 322.65 | V(ve1)-V(ve2), Temperature=49.5 | 1.55422 | 1554.22 | 1556.405371 | 0.140411454 | 3674.636447 | 1552.754 | 0.2346% |
| 49.6 | 322.75 | V(ve1)-V(ve2), Temperature=49.6 | 1.55155 | 1551.55 | 1553.891738 | 0.150701469 | 3660.655655 | 1550.39  | 0.2254% |
| 49.7 | 322.85 | V(ve1)-V(ve2), Temperature=49.7 | 1.54925 | 1549.25 | 1551.378032 | 0.137170466 | 3646.735659 | 1548.026 | 0.2161% |
| 49.8 | 322.95 | V(ve1)-V(ve2), Temperature=49.8 | 1.54658 | 1546.58 | 1548.864274 | 0.147480586 | 3632.876164 | 1545.662 | 0.2067% |
| 49.9 | 323.05 | V(ve1)-V(ve2), Temperature=49.9 | 1.5439  | 1543.9  | 1546.350481 | 0.158468652 | 3619.076876 | 1543.298 | 0.1974% |
| 50   | 323.15 | V(ve1)-V(ve2), Temperature=50   | 1.54176 | 1541.76 | 1543.836671 | 0.134513625 | 3605.337503 | 1540.934 | 0.1880% |
| 50.1 | 323.25 | V(ve1)-V(ve2), Temperature=50.1 | 1.53925 | 1539.25 | 1541.322862 | 0.134485914 | 3591.657755 | 1538.57  | 0.1786% |
| 50.2 | 323.35 | V(ve1)-V(ve2), Temperature=50.2 | 1.53656 | 1536.56 | 1538.809073 | 0.146156738 | 3578.037342 | 1536.206 | 0.1692% |
| 50.3 | 323.45 | V(ve1)-V(ve2), Temperature=50.3 | 1.53404 | 1534.04 | 1536.295322 | 0.146802643 | 3564.475978 | 1533.842 | 0.1597% |
| 50.4 | 323.55 | V(ve1)-V(ve2), Temperature=50.4 | 1.53151 | 1531.51 | 1533.781627 | 0.1481063   | 3550.973377 | 1531.478 | 0.1502% |
| 50.5 | 323.65 | V(ve1)-V(ve2), Temperature=50.5 | 1.52917 | 1529.17 | 1531.268006 | 0.137011041 | 3537.529253 | 1529.114 | 0.1407% |
| 50.6 | 323.75 | V(ve1)-V(ve2), Temperature=50.6 | 1.52663 | 1526.63 | 1528.754478 | 0.138967876 | 3524.143325 | 1526.75  | 0.1311% |
| 50.7 | 323.85 | V(ve1)-V(ve2), Temperature=50.7 | 1.52391 | 1523.91 | 1526.241059 | 0.152732056 | 3510.815311 | 1524.386 | 0.1215% |
| 50.8 | 323.95 | V(ve1)-V(ve2), Temperature=50.8 | 1.52137 | 1521.37 | 1523.72777  | 0.154736934 | 3497.544932 | 1522.022 | 0.1119% |
| 50.9 | 324.05 | V(ve1)-V(ve2), Temperature=50.9 | 1.51882 | 1518.82 | 1521.214626 | 0.157415406 | 3484.331908 | 1519.658 | 0.1023% |
| 51   | 324.15 | V(ve1)-V(ve2), Temperature=51   | 1.51609 | 1516.09 | 1518.701647 | 0.171965796 | 3471.175964 | 1517.294 | 0.0927% |
| 51.1 | 324.25 | V(ve1)-V(ve2), Temperature=51.1 | 1.51407 | 1514.07 | 1516.188851 | 0.139748483 | 3458.076823 | 1514.93  | 0.0830% |
| 51.2 | 324.35 | V(ve1)-V(ve2), Temperature=51.2 | 1.5115  | 1511.5  | 1513.676255 | 0.143772813 | 3445.034212 | 1512.566 | 0.0733% |
| 51.3 | 324.45 | V(ve1)-V(ve2), Temperature=51.3 | 1.50893 | 1508.93 | 1511.163877 | 0.147824963 | 3432.047859 | 1510.202 | 0.0637% |
| 51.4 | 324.55 | V(ve1)-V(ve2), Temperature=51.4 | 1.50618 | 1506.18 | 1508.651736 | 0.163837429 | 3419.117492 | 1507.838 | 0.0539% |
| 51.5 | 324.65 | V(ve1)-V(ve2), Temperature=51.5 | 1.50378 | 1503.78 | 1506.139849 | 0.156681933 | 3406.242841 | 1505.474 | 0.0442% |
| 51.6 | 324.75 | V(ve1)-V(ve2), Temperature=51.6 | 1.50138 | 1501.38 | 1503.628234 | 0.149520616 | 3393.423640 | 1503.11  | 0.0345% |
| 51.7 | 324.85 | V(ve1)-V(ve2), Temperature=51.7 | 1.49861 | 1498.61 | 1501.116909 | 0.167002921 | 3380.659620 | 1500.746 | 0.0247% |
| 51.8 | 324.95 | V(ve1)-V(ve2), Temperature=51.8 | 1.49619 | 1496.19 | 1498.605892 | 0.161209282 | 3367.950517 | 1498.382 | 0.0149% |
| 51.9 | 325.05 | V(ve1)-V(ve2), Temperature=51.9 | 1.49377 | 1493.77 | 1496.0952   | 0.155417909 | 3355.296066 | 1496.018 | 0.0052% |
| 52   | 325.15 | V(ve1)-V(ve2), Temperature=52   | 1.49135 | 1491.35 | 1493.584851 | 0.149630013 | 3342.696005 | 1493.654 | 0.0046% |
| 52.1 | 325.25 | V(ve1)-V(ve2), Temperature=52.1 | 1.48856 | 1488.56 | 1491.074864 | 0.16866113  | 3330.150074 | 1491.29  | 0.0144% |
| 52.2 | 325.35 | V(ve1)-V(ve2), Temperature=52.2 | 1.4863  | 1486.3  | 1488.565255 | 0.152177049 | 3317.658011 | 1488.926 | 0.0242% |
| 52.3 | 325.45 | V(ve1)-V(ve2), Temperature=52.3 | 1.48368 | 1483.68 | 1486.056042 | 0.159889126 | 3305.219559 | 1486.562 | 0.0340% |
| 52.4 | 325.55 | V(ve1)-V(ve2), Temperature=52.4 | 1.48124 | 1481.24 | 1483.547244 | 0.155522084 | 3292.834462 | 1484.198 | 0.0439% |
| 52.5 | 325.65 | V(ve1)-V(ve2), Temperature=52.5 | 1.47844 | 1478.44 | 1481.038877 | 0.175476591 | 3280.502462 | 1481.834 | 0.0537% |
| 52.6 | 325.75 | V(ve1)-V(ve2), Temperature=52.6 | 1.4765  | 1476.5  | 1478.530959 | 0.137363284 | 3268.223306 | 1479.47  | 0.0635% |
| 52.7 | 325.85 | V(ve1)-V(ve2), Temperature=52.7 | 1.47369 | 1473.69 | 1476.023507 | 0.158094187 | 3255.996740 | 1477.106 | 0.0733% |
| 52.8 | 325.95 | V(ve1)-V(ve2), Temperature=52.8 | 1.4714  | 1471.4  | 1473.51654  | 0.143638729 | 3243.822514 | 1474.742 | 0.0832% |
| 52.9 | 326.05 | V(ve1)-V(ve2), Temperature=52.9 | 1.46875 | 1468.75 | 1471.010075 | 0.15364103  | 3231.700376 | 1472.378 | 0.0930% |
| 53   | 326.15 | V(ve1)-V(ve2), Temperature=53   | 1.46627 | 1466.27 | 1468.504129 | 0.152136366 | 3219.630078 | 1470.014 | 0.1028% |
| 53.1 | 326.25 | V(ve1)-V(ve2), Temperature=53.1 | 1.46379 | 1463.79 | 1465.998719 | 0.150663114 | 3207.611373 | 1467.65  | 0.1126% |
| 53.2 | 326.35 | V(ve1)-V(ve2), Temperature=53.2 | 1.46096 | 1460.96 | 1463.493863 | 0.173137959 | 3195.644012 | 1465.286 | 0.1225% |
| 53.3 | 326.45 | V(ve1)-V(ve2), Temperature=53.3 | 1.45881 | 1458.81 | 1460.989579 | 0.14918512  | 3183.727753 | 1462.922 | 0.1323% |

|      |        |                                 |         |         |             |             |             |          |         |
|------|--------|---------------------------------|---------|---------|-------------|-------------|-------------|----------|---------|
| 53.4 | 326.55 | V(ve1)-V(ve2), Temperature=53.4 | 1.45632 | 1456.32 | 1458.485883 | 0.148502164 | 3171.862350 | 1460.558 | 0.1421% |
| 53.5 | 326.65 | V(ve1)-V(ve2), Temperature=53.5 | 1.45347 | 1453.47 | 1455.982793 | 0.172583973 | 3160.047561 | 1458.194 | 0.1519% |
| 53.6 | 326.75 | V(ve1)-V(ve2), Temperature=53.6 | 1.45097 | 1450.97 | 1453.480326 | 0.172711377 | 3148.283145 | 1455.83  | 0.1617% |
| 53.7 | 326.85 | V(ve1)-V(ve2), Temperature=53.7 | 1.44888 | 1448.8  | 1450.978499 | 0.150139985 | 3136.568861 | 1453.466 | 0.1714% |
| 53.8 | 326.95 | V(ve1)-V(ve2), Temperature=53.8 | 1.44628 | 1446.28 | 1448.47733  | 0.15169927  | 3124.904472 | 1451.102 | 0.1812% |
| 53.9 | 327.05 | V(ve1)-V(ve2), Temperature=53.9 | 1.44342 | 1443.42 | 1445.976835 | 0.176824035 | 3113.289739 | 1448.738 | 0.1910% |
| 54   | 327.15 | V(ve1)-V(ve2), Temperature=54   | 1.44107 | 1441.07 | 1443.477031 | 0.166752317 | 3101.724427 | 1446.374 | 0.2007% |
| 54.1 | 327.25 | V(ve1)-V(ve2), Temperature=54.1 | 1.43888 | 1438.88 | 1440.977937 | 0.145591184 | 3090.208300 | 1444.01  | 0.2104% |
| 54.2 | 327.35 | V(ve1)-V(ve2), Temperature=54.2 | 1.43617 | 1436.17 | 1438.479568 | 0.160556196 | 3078.741125 | 1441.646 | 0.2201% |
| 54.3 | 327.45 | V(ve1)-V(ve2), Temperature=54.3 | 1.43397 | 1433.97 | 1435.981942 | 0.140109127 | 3067.322668 | 1439.282 | 0.2298% |
| 54.4 | 327.55 | V(ve1)-V(ve2), Temperature=54.4 | 1.43109 | 1431.09 | 1433.485075 | 0.167080578 | 3055.952699 | 1436.918 | 0.2395% |
| 54.5 | 327.65 | V(ve1)-V(ve2), Temperature=54.5 | 1.42871 | 1428.71 | 1430.988985 | 0.159259432 | 3044.630987 | 1434.554 | 0.2491% |
| 54.6 | 327.75 | V(ve1)-V(ve2), Temperature=54.6 | 1.42616 | 1426.16 | 1428.493688 | 0.163367062 | 3033.357304 | 1432.19  | 0.2588% |
| 54.7 | 327.85 | V(ve1)-V(ve2), Temperature=54.7 | 1.4236  | 1423.6  | 1425.999201 | 0.168247037 | 3022.131421 | 1429.826 | 0.2684% |
| 54.8 | 327.95 | V(ve1)-V(ve2), Temperature=54.8 | 1.42121 | 1421.21 | 1423.505542 | 0.16125977  | 3010.953113 | 1427.462 | 0.2779% |
| 54.9 | 328.05 | V(ve1)-V(ve2), Temperature=54.9 | 1.41881 | 1418.81 | 1421.012726 | 0.15501097  | 2999.822153 | 1425.098 | 0.2875% |
| 55   | 328.15 | V(ve1)-V(ve2), Temperature=55   | 1.41624 | 1416.24 | 1418.52077  | 0.16078509  | 2988.738318 | 1422.734 | 0.2970% |
| 55.1 | 328.25 | V(ve1)-V(ve2), Temperature=55.1 | 1.41367 | 1413.67 | 1416.029691 | 0.166641372 | 2977.701385 | 1420.37  | 0.3065% |
| 55.2 | 328.35 | V(ve1)-V(ve2), Temperature=55.2 | 1.41109 | 1411.09 | 1413.539506 | 0.173288837 | 2966.711131 | 1418.006 | 0.3160% |
| 55.3 | 328.45 | V(ve1)-V(ve2), Temperature=55.3 | 1.40867 | 1408.67 | 1411.050231 | 0.168685081 | 2955.767336 | 1415.642 | 0.3254% |
| 55.4 | 328.55 | V(ve1)-V(ve2), Temperature=55.4 | 1.40625 | 1406.25 | 1408.561883 | 0.164130729 | 2944.869780 | 1413.278 | 0.3348% |
| 55.5 | 328.65 | V(ve1)-V(ve2), Temperature=55.5 | 1.40366 | 1403.66 | 1406.074478 | 0.171717645 | 2934.018246 | 1410.914 | 0.3442% |
| 55.6 | 328.75 | V(ve1)-V(ve2), Temperature=55.6 | 1.40139 | 1401.39 | 1403.588032 | 0.156600959 | 2923.212514 | 1408.55  | 0.3535% |
| 55.7 | 328.85 | V(ve1)-V(ve2), Temperature=55.7 | 1.39879 | 1398.79 | 1401.102563 | 0.165053084 | 2912.452371 | 1406.186 | 0.3628% |
| 55.8 | 328.95 | V(ve1)-V(ve2), Temperature=55.8 | 1.39635 | 1396.35 | 1398.618086 | 0.162166193 | 2901.737599 | 1403.822 | 0.3721% |
| 55.9 | 329.05 | V(ve1)-V(ve2), Temperature=55.9 | 1.39374 | 1393.74 | 1396.134617 | 0.171517642 | 2891.067986 | 1401.458 | 0.3813% |
| 56   | 329.15 | V(ve1)-V(ve2), Temperature=56   | 1.39145 | 1391.45 | 1393.652174 | 0.15801458  | 2880.443319 | 1399.094 | 0.3905% |
| 56.1 | 329.25 | V(ve1)-V(ve2), Temperature=56.1 | 1.389   | 1389    | 1391.170771 | 0.156039161 | 2869.863386 | 1396.73  | 0.3996% |
| 56.2 | 329.35 | V(ve1)-V(ve2), Temperature=56.2 | 1.38638 | 1386.38 | 1388.690426 | 0.166374445 | 2859.327976 | 1394.366 | 0.4087% |
| 56.3 | 329.45 | V(ve1)-V(ve2), Temperature=56.3 | 1.38391 | 1383.91 | 1386.211154 | 0.166003136 | 2848.836879 | 1392.002 | 0.4177% |
| 56.4 | 329.55 | V(ve1)-V(ve2), Temperature=56.4 | 1.38161 | 1381.61 | 1383.732972 | 0.153423503 | 2838.389888 | 1389.638 | 0.4267% |
| 56.5 | 329.65 | V(ve1)-V(ve2), Temperature=56.5 | 1.37897 | 1378.97 | 1381.255895 | 0.165493934 | 2827.986795 | 1387.274 | 0.4357% |
| 56.6 | 329.75 | V(ve1)-V(ve2), Temperature=56.6 | 1.37633 | 1376.33 | 1378.779939 | 0.177688934 | 2817.627394 | 1384.91  | 0.4446% |
| 56.7 | 329.85 | V(ve1)-V(ve2), Temperature=56.7 | 1.37402 | 1374.02 | 1376.305121 | 0.166033047 | 2807.311479 | 1382.546 | 0.4535% |
| 56.8 | 329.95 | V(ve1)-V(ve2), Temperature=56.8 | 1.37153 | 1371.53 | 1373.831456 | 0.167521022 | 2797.038846 | 1380.182 | 0.4623% |
| 56.9 | 330.05 | V(ve1)-V(ve2), Temperature=56.9 | 1.36904 | 1369.04 | 1371.358961 | 0.16909947  | 2786.809292 | 1377.818 | 0.4710% |
| 57   | 330.15 | V(ve1)-V(ve2), Temperature=57   | 1.36671 | 1366.71 | 1368.88765  | 0.159081725 | 2776.622616 | 1375.454 | 0.4797% |
| 57.1 | 330.25 | V(ve1)-V(ve2), Temperature=57.1 | 1.36437 | 1364.37 | 1366.41754  | 0.149847301 | 2766.478615 | 1373.09  | 0.4883% |
| 57.2 | 330.35 | V(ve1)-V(ve2), Temperature=57.2 | 1.36171 | 1361.71 | 1363.948646 | 0.164129791 | 2756.377091 | 1370.726 | 0.4969% |
| 57.3 | 330.45 | V(ve1)-V(ve2), Temperature=57.3 | 1.35904 | 1359.04 | 1361.480984 | 0.17928888  | 2746.317843 | 1368.362 | 0.5054% |
| 57.4 | 330.55 | V(ve1)-V(ve2), Temperature=57.4 | 1.35685 | 1356.85 | 1359.01457  | 0.159274943 | 2736.300675 | 1365.998 | 0.5139% |
| 57.5 | 330.65 | V(ve1)-V(ve2), Temperature=57.5 | 1.35449 | 1354.49 | 1356.549418 | 0.151813    | 2726.325389 | 1363.634 | 0.5223% |
| 57.6 | 330.75 | V(ve1)-V(ve2), Temperature=57.6 | 1.35182 | 1351.82 | 1354.085546 | 0.167311863 | 2716.391790 | 1361.27  | 0.5306% |
| 57.7 | 330.85 | V(ve1)-V(ve2), Temperature=57.7 | 1.34929 | 1349.29 | 1351.622967 | 0.172604867 | 2706.499682 | 1358.906 | 0.5388% |
| 57.8 | 330.95 | V(ve1)-V(ve2), Temperature=57.8 | 1.34692 | 1346.92 | 1349.161698 | 0.166154844 | 2696.648871 | 1356.542 | 0.5470% |
| 57.9 | 331.05 | V(ve1)-V(ve2), Temperature=57.9 | 1.34423 | 1344.23 | 1346.701753 | 0.183541236 | 2686.839166 | 1354.178 | 0.5552% |
| 58   | 331.15 | V(ve1)-V(ve2), Temperature=58   | 1.34186 | 1341.86 | 1344.243148 | 0.177285512 | 2677.070373 | 1351.814 | 0.5632% |
| 58.1 | 331.25 | V(ve1)-V(ve2), Temperature=58.1 | 1.33963 | 1339.63 | 1341.785899 | 0.160673838 | 2667.342302 | 1349.45  | 0.5712% |
| 58.2 | 331.35 | V(ve1)-V(ve2), Temperature=58.2 | 1.33693 | 1336.93 | 1339.33002  | 0.179195573 | 2657.654764 | 1347.086 | 0.5791% |
| 58.3 | 331.45 | V(ve1)-V(ve2), Temperature=58.3 | 1.33454 | 1334.54 | 1336.875527 | 0.174700376 | 2648.007568 | 1344.722 | 0.5869% |
| 58.4 | 331.55 | V(ve1)-V(ve2), Temperature=58.4 | 1.33215 | 1332.15 | 1334.422434 | 0.170293422 | 2638.400527 | 1342.358 | 0.5947% |
| 58.5 | 331.65 | V(ve1)-V(ve2), Temperature=58.5 | 1.32943 | 1329.43 | 1331.970757 | 0.190751675 | 2628.833455 | 1339.994 | 0.6024% |
| 58.6 | 331.75 | V(ve1)-V(ve2), Temperature=58.6 | 1.32734 | 1327.34 | 1329.52051  | 0.16400725  | 2619.306165 | 1337.63  | 0.6100% |
| 58.7 | 331.85 | V(ve1)-V(ve2), Temperature=58.7 | 1.32494 | 1324.94 | 1327.071709 | 0.160632525 | 2609.818471 | 1335.266 | 0.6175% |
| 58.8 | 331.95 | V(ve1)-V(ve2), Temperature=58.8 | 1.32237 | 1322.37 | 1324.624368 | 0.170189228 | 2600.370190 | 1332.902 | 0.6249% |
| 58.9 | 332.05 | V(ve1)-V(ve2), Temperature=58.9 | 1.31995 | 1319.95 | 1322.178502 | 0.168547775 | 2590.961139 | 1330.538 | 0.6323% |
| 59   | 332.15 | V(ve1)-V(ve2), Temperature=59   | 1.31769 | 1317.69 | 1319.734127 | 0.154889286 | 2581.591135 | 1328.174 | 0.6395% |
| 59.1 | 332.25 | V(ve1)-V(ve2), Temperature=59.1 | 1.31526 | 1315.26 | 1317.291256 | 0.154199438 | 2572.259997 | 1325.81  | 0.6467% |
| 59.2 | 332.35 | V(ve1)-V(ve2), Temperature=59.2 | 1.31252 | 1312.52 | 1314.849904 | 0.17719922  | 2562.967544 | 1323.446 | 0.6538% |
| 59.3 | 332.45 | V(ve1)-V(ve2), Temperature=59.3 | 1.31009 | 1310.09 | 1312.410086 | 0.176780562 | 2553.713598 | 1321.082 | 0.6608% |
| 59.4 | 332.55 | V(ve1)-V(ve2), Temperature=59.4 | 1.30766 | 1307.66 | 1309.971816 | 0.176478306 | 2544.497978 | 1318.718 | 0.6677% |
| 59.5 | 332.65 | V(ve1)-V(ve2), Temperature=59.5 | 1.30521 | 1305.21 | 1307.535109 | 0.177823857 | 2535.320509 | 1316.354 | 0.6745% |
| 59.6 | 332.75 | V(ve1)-V(ve2), Temperature=59.6 | 1.30292 | 1302.92 | 1305.09998  | 0.167035472 | 2526.181013 | 1313.99  | 0.6812% |
| 59.7 | 332.85 | V(ve1)-V(ve2), Temperature=59.7 | 1.30047 | 1300.47 | 1302.666442 | 0.168611219 | 2517.079313 | 1311.626 | 0.6878% |
| 59.8 | 332.95 | V(ve1)-V(ve2), Temperature=59.8 | 1.29802 | 1298.02 | 1300.23451  | 0.170316181 | 2508.015236 | 1309.262 | 0.6943% |
| 59.9 | 333.05 | V(ve1)-V(ve2), Temperature=59.9 | 1.29557 | 1295.57 | 1297.804198 | 0.172152175 | 2498.988608 | 1306.898 | 0.7007% |
| 60   | 333.15 | V(ve1)-V(ve2), Temperature=60   | 1.29341 | 1293.41 | 1295.375521 | 0.151733657 | 2489.999254 | 1304.534 | 0.7070% |

|      |        |                                 |         |         |             |             |             |          |         |
|------|--------|---------------------------------|---------|---------|-------------|-------------|-------------|----------|---------|
| 60.1 | 333.25 | V(ve1)-V(ve2), Temperature=60.1 | 1.29064 | 1290.64 | 1292.948491 | 0.178544735 | 2481.047002 | 1302.17  | 0.7132% |
| 60.2 | 333.35 | V(ve1)-V(ve2), Temperature=60.2 | 1.28817 | 1288.17 | 1290.523125 | 0.182338852 | 2472.131682 | 1299.806 | 0.7193% |
| 60.3 | 333.45 | V(ve1)-V(ve2), Temperature=60.3 | 1.2857  | 1285.7  | 1288.099435 | 0.186277181 | 2463.253123 | 1297.442 | 0.7253% |
| 60.4 | 333.55 | V(ve1)-V(ve2), Temperature=60.4 | 1.28353 | 1283.53 | 1285.677436 | 0.167027576 | 2454.411154 | 1295.078 | 0.7312% |
| 60.5 | 333.65 | V(ve1)-V(ve2), Temperature=60.5 | 1.28105 | 1281.05 | 1283.257141 | 0.17199523  | 2445.605608 | 1292.714 | 0.7369% |
| 60.6 | 333.75 | V(ve1)-V(ve2), Temperature=60.6 | 1.27856 | 1278.56 | 1280.838565 | 0.177896309 | 2436.836316 | 1290.35  | 0.7426% |
| 60.7 | 333.85 | V(ve1)-V(ve2), Temperature=60.7 | 1.27622 | 1276.22 | 1278.42172  | 0.172221711 | 2428.103110 | 1287.986 | 0.7481% |
| 60.8 | 333.95 | V(ve1)-V(ve2), Temperature=60.8 | 1.27373 | 1273.73 | 1276.006621 | 0.178417658 | 2419.405826 | 1285.622 | 0.7536% |
| 60.9 | 334.05 | V(ve1)-V(ve2), Temperature=60.9 | 1.27154 | 1271.54 | 1273.593282 | 0.161219588 | 2410.744297 | 1283.258 | 0.7589% |
| 61   | 334.15 | V(ve1)-V(ve2), Temperature=61   | 1.26874 | 1268.74 | 1271.181716 | 0.192082349 | 2402.118359 | 1280.894 | 0.7640% |
| 61.1 | 334.25 | V(ve1)-V(ve2), Temperature=61.1 | 1.26654 | 1266.54 | 1268.771936 | 0.175913109 | 2393.527848 | 1278.53  | 0.7691% |
| 61.2 | 334.35 | V(ve1)-V(ve2), Temperature=61.2 | 1.26403 | 1264.03 | 1266.363957 | 0.184303784 | 2384.972601 | 1276.166 | 0.7740% |
| 61.3 | 334.45 | V(ve1)-V(ve2), Temperature=61.3 | 1.26152 | 1261.52 | 1263.957791 | 0.192869656 | 2376.452457 | 1273.802 | 0.7788% |
| 61.4 | 334.55 | V(ve1)-V(ve2), Temperature=61.4 | 1.2593  | 1259.3  | 1261.553452 | 0.178625177 | 2367.967253 | 1271.438 | 0.7835% |
| 61.5 | 334.65 | V(ve1)-V(ve2), Temperature=61.5 | 1.25679 | 1256.79 | 1259.150953 | 0.187503592 | 2359.516829 | 1269.074 | 0.7881% |
| 61.6 | 334.75 | V(ve1)-V(ve2), Temperature=61.6 | 1.25456 | 1254.56 | 1256.750308 | 0.174283467 | 2351.101026 | 1266.71  | 0.7925% |
| 61.7 | 334.85 | V(ve1)-V(ve2), Temperature=61.7 | 1.25204 | 1252.04 | 1254.351529 | 0.184280834 | 2342.719685 | 1264.346 | 0.7968% |
| 61.8 | 334.95 | V(ve1)-V(ve2), Temperature=61.8 | 1.2498  | 1249.8  | 1251.954631 | 0.172101345 | 2334.372648 | 1261.982 | 0.8009% |
| 61.9 | 335.05 | V(ve1)-V(ve2), Temperature=61.9 | 1.24727 | 1247.27 | 1249.559625 | 0.183234533 | 2326.059757 | 1259.618 | 0.8050% |
| 62   | 335.15 | V(ve1)-V(ve2), Temperature=62   | 1.24503 | 1245.03 | 1247.166525 | 0.171310288 | 2317.780856 | 1257.254 | 0.8088% |
| 62.1 | 335.25 | V(ve1)-V(ve2), Temperature=62.1 | 1.24249 | 1242.49 | 1244.775343 | 0.183594844 | 2309.535790 | 1254.89  | 0.8126% |
| 62.2 | 335.35 | V(ve1)-V(ve2), Temperature=62.2 | 1.24024 | 1240.24 | 1242.386094 | 0.172739694 | 2301.324404 | 1252.526 | 0.8162% |
| 62.3 | 335.45 | V(ve1)-V(ve2), Temperature=62.3 | 1.2377  | 1237.7  | 1239.998789 | 0.185386382 | 2293.146543 | 1250.162 | 0.8196% |
| 62.4 | 335.55 | V(ve1)-V(ve2), Temperature=62.4 | 1.23544 | 1235.44 | 1237.613441 | 0.175615527 | 2285.002055 | 1247.798 | 0.8229% |
| 62.5 | 335.65 | V(ve1)-V(ve2), Temperature=62.5 | 1.23303 | 1233.03 | 1235.230064 | 0.178109624 | 2276.890786 | 1245.434 | 0.8261% |
| 62.6 | 335.75 | V(ve1)-V(ve2), Temperature=62.6 | 1.23076 | 1230.76 | 1232.848669 | 0.169418077 | 2268.812585 | 1243.07  | 0.8291% |
| 62.7 | 335.85 | V(ve1)-V(ve2), Temperature=62.7 | 1.2282  | 1228.2  | 1230.469269 | 0.184423024 | 2260.767301 | 1240.706 | 0.8319% |
| 62.8 | 335.95 | V(ve1)-V(ve2), Temperature=62.8 | 1.22593 | 1225.93 | 1228.091876 | 0.176035385 | 2252.754783 | 1238.342 | 0.8346% |
| 62.9 | 336.05 | V(ve1)-V(ve2), Temperature=62.9 | 1.22365 | 1223.65 | 1225.716504 | 0.168595616 | 2244.774883 | 1235.978 | 0.8372% |
| 63   | 336.15 | V(ve1)-V(ve2), Temperature=63   | 1.22137 | 1221.37 | 1223.343165 | 0.161292824 | 2236.827451 | 1233.614 | 0.8396% |
| 63.1 | 336.25 | V(ve1)-V(ve2), Temperature=63.1 | 1.2188  | 1218.8  | 1220.97187  | 0.177880448 | 2228.912340 | 1231.25  | 0.8418% |
| 63.2 | 336.35 | V(ve1)-V(ve2), Temperature=63.2 | 1.21637 | 1216.37 | 1218.602633 | 0.183212538 | 2221.029402 | 1228.886 | 0.8439% |
| 63.3 | 336.45 | V(ve1)-V(ve2), Temperature=63.3 | 1.21407 | 1214.07 | 1216.235465 | 0.178046511 | 2213.178491 | 1226.522 | 0.8458% |
| 63.4 | 336.55 | V(ve1)-V(ve2), Temperature=63.4 | 1.21178 | 1211.78 | 1213.870378 | 0.17220768  | 2205.359460 | 1224.158 | 0.8475% |
| 63.5 | 336.65 | V(ve1)-V(ve2), Temperature=63.5 | 1.20948 | 1209.48 | 1211.507385 | 0.167344015 | 2197.572166 | 1221.794 | 0.8491% |
| 63.6 | 336.75 | V(ve1)-V(ve2), Temperature=63.6 | 1.20717 | 1207.17 | 1209.146498 | 0.16346221  | 2189.816463 | 1219.43  | 0.8505% |
| 63.7 | 336.85 | V(ve1)-V(ve2), Temperature=63.7 | 1.20458 | 1204.58 | 1206.787728 | 0.182942496 | 2182.092208 | 1217.066 | 0.8517% |
| 63.8 | 336.95 | V(ve1)-V(ve2), Temperature=63.8 | 1.20227 | 1202.27 | 1204.431087 | 0.179428029 | 2174.399258 | 1214.702 | 0.8528% |
| 63.9 | 337.05 | V(ve1)-V(ve2), Temperature=63.9 | 1.19995 | 1199.95 | 1202.076588 | 0.17690949  | 2166.737471 | 1212.338 | 0.8536% |
| 64   | 337.15 | V(ve1)-V(ve2), Temperature=64   | 1.19763 | 1197.63 | 1199.724241 | 0.174560196 | 2159.106705 | 1209.974 | 0.8543% |
| 64.1 | 337.25 | V(ve1)-V(ve2), Temperature=64.1 | 1.19531 | 1195.31 | 1197.374059 | 0.172382175 | 2151.506821 | 1207.61  | 0.8549% |
| 64.2 | 337.35 | V(ve1)-V(ve2), Temperature=64.2 | 1.19298 | 1192.98 | 1195.026054 | 0.171214174 | 2143.937677 | 1205.246 | 0.8552% |
| 64.3 | 337.45 | V(ve1)-V(ve2), Temperature=64.3 | 1.19065 | 1190.65 | 1192.680236 | 0.170224692 | 2136.399134 | 1202.882 | 0.8554% |
| 64.4 | 337.55 | V(ve1)-V(ve2), Temperature=64.4 | 1.18832 | 1188.32 | 1190.336618 | 0.169415773 | 2128.891054 | 1200.518 | 0.8553% |
| 64.5 | 337.65 | V(ve1)-V(ve2), Temperature=64.5 | 1.18584 | 1185.84 | 1187.995211 | 0.181415761 | 2121.413299 | 1198.154 | 0.8551% |
| 64.6 | 337.75 | V(ve1)-V(ve2), Temperature=64.6 | 1.18336 | 1183.36 | 1185.656025 | 0.193650215 | 2113.965732 | 1195.79  | 0.8547% |
| 64.7 | 337.85 | V(ve1)-V(ve2), Temperature=64.7 | 1.18102 | 1181.02 | 1183.319074 | 0.194290269 | 2106.548216 | 1193.426 | 0.8541% |
| 64.8 | 337.95 | V(ve1)-V(ve2), Temperature=64.8 | 1.17895 | 1178.95 | 1180.984367 | 0.172260246 | 2099.160614 | 1191.062 | 0.8533% |
| 64.9 | 338.05 | V(ve1)-V(ve2), Temperature=64.9 | 1.1766  | 1176.6  | 1178.651916 | 0.17409004  | 2091.802793 | 1188.698 | 0.8523% |
| 65   | 338.15 | V(ve1)-V(ve2), Temperature=65   | 1.17424 | 1174.24 | 1176.321731 | 0.176969544 | 2084.474616 | 1186.334 | 0.8512% |
| 65.1 | 338.25 | V(ve1)-V(ve2), Temperature=65.1 | 1.17188 | 1171.88 | 1173.993825 | 0.180054189 | 2077.175951 | 1183.97  | 0.8498% |
| 65.2 | 338.35 | V(ve1)-V(ve2), Temperature=65.2 | 1.16938 | 1169.38 | 1171.668208 | 0.195294896 | 2069.906665 | 1181.606 | 0.8482% |
| 65.3 | 338.45 | V(ve1)-V(ve2), Temperature=65.3 | 1.16702 | 1167.02 | 1169.344891 | 0.198819965 | 2062.666624 | 1179.242 | 0.8464% |
| 65.4 | 338.55 | V(ve1)-V(ve2), Temperature=65.4 | 1.16492 | 1164.92 | 1167.023885 | 0.180277791 | 2055.455697 | 1176.878 | 0.8444% |
| 65.5 | 338.65 | V(ve1)-V(ve2), Temperature=65.5 | 1.16255 | 1162.55 | 1164.7052   | 0.18504254  | 2048.273752 | 1174.514 | 0.8422% |
| 65.6 | 338.75 | V(ve1)-V(ve2), Temperature=65.6 | 1.16018 | 1160.18 | 1162.388848 | 0.190026596 | 2041.120660 | 1172.15  | 0.8397% |
| 65.7 | 338.85 | V(ve1)-V(ve2), Temperature=65.7 | 1.15807 | 1158.07 | 1160.074839 | 0.172819781 | 2033.996290 | 1169.786 | 0.8371% |
| 65.8 | 338.95 | V(ve1)-V(ve2), Temperature=65.8 | 1.15569 | 1155.69 | 1157.763183 | 0.179067966 | 2026.900512 | 1167.422 | 0.8343% |
| 65.9 | 339.05 | V(ve1)-V(ve2), Temperature=65.9 | 1.15344 | 1153.44 | 1155.453891 | 0.174294383 | 2019.833198 | 1165.058 | 0.8312% |
| 66   | 339.15 | V(ve1)-V(ve2), Temperature=66   | 1.15105 | 1151.05 | 1153.146974 | 0.181847969 | 2012.794221 | 1162.694 | 0.8279% |
| 66.1 | 339.25 | V(ve1)-V(ve2), Temperature=66.1 | 1.14879 | 1148.79 | 1150.842442 | 0.178342589 | 2005.783452 | 1160.33  | 0.8244% |
| 66.2 | 339.35 | V(ve1)-V(ve2), Temperature=66.2 | 1.1464  | 1146.4  | 1148.540306 | 0.186350073 | 1998.800766 | 1157.966 | 0.8207% |
| 66.3 | 339.45 | V(ve1)-V(ve2), Temperature=66.3 | 1.144   | 1144    | 1146.240574 | 0.195471572 | 1991.846035 | 1155.602 | 0.8167% |
| 66.4 | 339.55 | V(ve1)-V(ve2), Temperature=66.4 | 1.14187 | 1141.87 | 1143.943259 | 0.181237942 | 1984.919135 | 1153.238 | 0.8125% |
| 66.5 | 339.65 | V(ve1)-V(ve2), Temperature=66.5 | 1.1396  | 1139.6  | 1141.64837  | 0.179422121 | 1978.019940 | 1150.874 | 0.8081% |
| 66.6 | 339.75 | V(ve1)-V(ve2), Temperature=66.6 | 1.13719 | 1137.19 | 1139.355916 | 0.190100073 | 1971.148327 | 1148.51  | 0.8034% |
| 66.7 | 339.85 | V(ve1)-V(ve2), Temperature=66.7 | 1.13505 | 1135.05 | 1137.065909 | 0.17729042  | 1964.304171 | 1146.146 | 0.7986% |

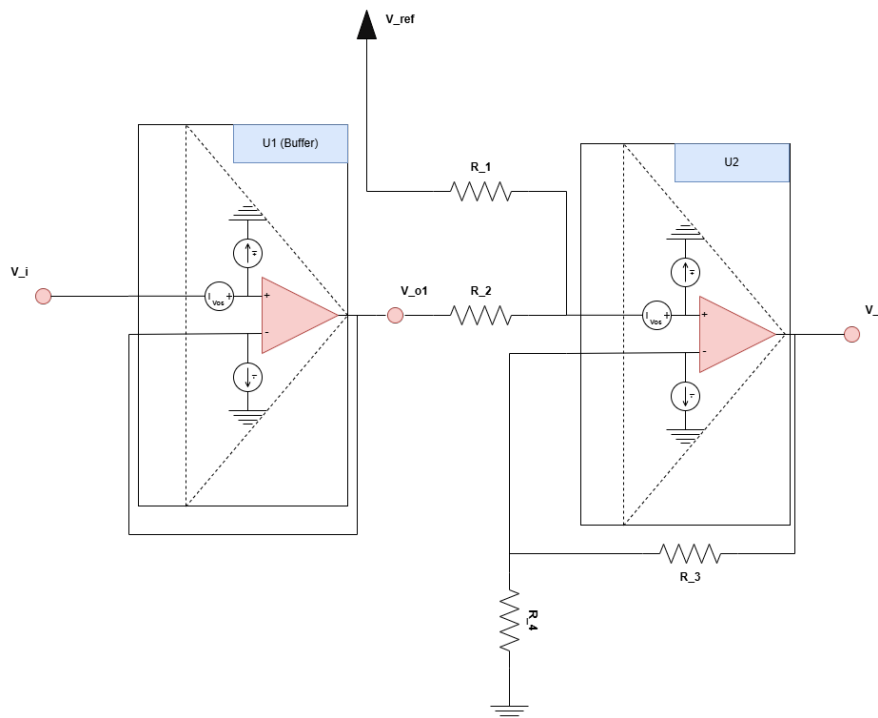
|      |        |                                |            |           |             |             |             |          |         |
|------|--------|--------------------------------|------------|-----------|-------------|-------------|-------------|----------|---------|
| 66.8 | 339.95 | V(ve1)-V(v2), Temperature=66.8 | 1.13263    | 1132.63   | 1134.778357 | 0.189319567 | 1957.487350 | 1143.782 | 0.7934% |
| 66.9 | 340.05 | V(v1)-V(v2), Temperature=66.9  | 1.13048    | 1130.48   | 1132.493272 | 0.177773427 | 1950.697742 | 1141.418 | 0.7881% |
| 67   | 340.15 | V(v1)-V(v2), Temperature=67    | 1.12806    | 1128.06   | 1130.210662 | 0.190288585 | 1943.935223 | 1139.054 | 0.7825% |
| 67.1 | 340.25 | V(v1)-V(v2), Temperature=67.1  | 1.12577    | 1125.77   | 1127.930537 | 0.191548789 | 1937.199674 | 1136.69  | 0.7766% |
| 67.2 | 340.35 | V(v1)-V(v2), Temperature=67.2  | 1.12348    | 1123.48   | 1125.652908 | 0.193035337 | 1930.490974 | 1134.326 | 0.7705% |
| 67.3 | 340.45 | V(v1)-V(v2), Temperature=67.3  | 1.12131    | 1121.31   | 1123.377783 | 0.184068349 | 1923.809002 | 1131.962 | 0.7641% |
| 67.4 | 340.55 | V(v1)-V(v2), Temperature=67.4  | 1.11915    | 1119.15   | 1121.105172 | 0.174396854 | 1917.153639 | 1129.598 | 0.7575% |
| 67.5 | 340.65 | V(v1)-V(v2), Temperature=67.5  | 1.11684    | 1116.84   | 1118.835085 | 0.178318084 | 1910.524767 | 1127.234 | 0.7507% |
| 67.6 | 340.75 | V(v1)-V(v2), Temperature=67.6  | 1.1144     | 1114.4    | 1116.567532 | 0.194124533 | 1903.922267 | 1124.87  | 0.7436% |
| 67.7 | 340.85 | V(v1)-V(v2), Temperature=67.7  | 1.11249    | 1112.49   | 1114.30252  | 0.162659602 | 1897.346021 | 1122.506 | 0.7362% |
| 67.8 | 340.95 | V(v1)-V(v2), Temperature=67.8  | 1.11004    | 1110.04   | 1112.040061 | 0.179855082 | 1890.795913 | 1120.142 | 0.7286% |
| 67.9 | 341.05 | V(v1)-V(v2), Temperature=67.9  | 1.10773    | 1107.73   | 1109.780162 | 0.184735845 | 1884.271825 | 1117.778 | 0.7207% |
| 68   | 341.15 | V(v1)-V(v2), Temperature=68    | 1.10541    | 1105.41   | 1107.522833 | 0.190771067 | 1877.773642 | 1115.414 | 0.7125% |
| 68.1 | 341.25 | V(v1)-V(v2), Temperature=68.1  | 1.10321    | 1103.21   | 1105.268084 | 0.186206759 | 1871.301249 | 1113.05  | 0.7041% |
| 68.2 | 341.35 | V(v1)-V(v2), Temperature=68.2  | 1.10102    | 1101.02   | 1103.015923 | 0.180951387 | 1864.854530 | 1110.686 | 0.6954% |
| 68.3 | 341.45 | V(v1)-V(v2), Temperature=68.3  | 1.09882    | 1098.82   | 1100.766339 | 0.176818515 | 1858.433372 | 1108.322 | 0.6864% |
| 68.4 | 341.55 | V(v1)-V(v2), Temperature=68.4  | 1.09649    | 1096.49   | 1098.519401 | 0.184739628 | 1852.037660 | 1105.958 | 0.6771% |
| 68.5 | 341.65 | V(v1)-V(v2), Temperature=68.5  | 1.09428    | 1094.28   | 1096.275058 | 0.181985163 | 1845.667283 | 1103.594 | 0.6676% |
| 68.6 | 341.75 | V(v1)-V(v2), Temperature=68.6  | 1.09194    | 1091.94   | 1094.033339 | 0.191341384 | 1839.322126 | 1101.23  | 0.6578% |
| 68.7 | 341.85 | V(v1)-V(v2), Temperature=68.7  | 1.08973    | 1089.73   | 1091.794251 | 0.18906964  | 1833.002078 | 1098.866 | 0.6477% |
| 68.8 | 341.95 | V(v1)-V(v2), Temperature=68.8  | 1.08751    | 1087.51   | 1089.557806 | 0.187948312 | 1826.707028 | 1096.502 | 0.6373% |
| 68.9 | 342.05 | V(v1)-V(v2), Temperature=68.9  | 1.08542    | 1085.42   | 1087.324009 | 0.175109646 | 1820.436865 | 1094.138 | 0.6267% |
| 69   | 342.15 | V(v1)-V(v2), Temperature=69    | 1.0832     | 1083.2    | 1085.092871 | 0.174443215 | 1814.191478 | 1091.774 | 0.6157% |
| 69.1 | 342.25 | V(v1)-V(v2), Temperature=69.1  | 1.08097    | 1080.97   | 1082.864399 | 0.174943314 | 1807.970757 | 1089.41  | 0.6045% |
| 69.2 | 342.35 | V(v1)-V(v2), Temperature=69.2  | 1.07874    | 1078.74   | 1080.638602 | 0.175692595 | 1801.774594 | 1087.046 | 0.5929% |
| 69.3 | 342.45 | V(v1)-V(v2), Temperature=69.3  | 1.07651    | 1076.51   | 1078.415488 | 0.176693289 | 1795.602878 | 1084.682 | 0.5811% |
| 69.4 | 342.55 | V(v1)-V(v2), Temperature=69.4  | 1.07428    | 1074.28   | 1076.195065 | 0.177947784 | 1789.455503 | 1082.318 | 0.5689% |
| 69.5 | 342.65 | V(v1)-V(v2), Temperature=69.5  | 1.07204    | 1072.04   | 1073.977343 | 0.180389515 | 1783.332361 | 1079.954 | 0.5565% |
| 69.6 | 342.75 | V(v1)-V(v2), Temperature=69.6  | 1.06992    | 1069.92   | 1071.762327 | 0.171896971 | 1777.233343 | 1077.59  | 0.5437% |
| 69.7 | 342.85 | V(v1)-V(v2), Temperature=69.7  | 1.06755    | 1067.55   | 1069.550028 | 0.18699711  | 1771.158345 | 1075.226 | 0.5307% |
| 69.8 | 342.95 | V(v1)-V(v2), Temperature=69.8  | 1.0653     | 1065.3    | 1067.340451 | 0.191171567 | 1765.107258 | 1072.862 | 0.5173% |
| 69.9 | 343.05 | V(v1)-V(v2), Temperature=69.9  | 1.06318    | 1063.18   | 1065.133607 | 0.183414296 | 1759.079979 | 1070.498 | 0.5036% |
| 70   | 343.15 | V(v1)-V(v2), Temperature=70    | 1.06092    | 1060.92   | 1062.929502 | 0.189053208 | 1753.076401 | 1068.134 | 0.4896% |
| 70.1 | 343.25 | V(v1)-V(v2), Temperature=70.1  | 1.05892    | 1058.92   | 1060.728145 | 0.170462574 | 1747.096420 | 1065.77  | 0.4753% |
| 70.2 | 343.35 | V(v1)-V(v2), Temperature=70.2  | 1.05653    | 1056.53   | 1058.529542 | 0.188898079 | 1741.139933 | 1063.406 | 0.4607% |
| 70.3 | 343.45 | V(v1)-V(v2), Temperature=70.3  | 1.05439    | 1054.39   | 1056.333701 | 0.184004494 | 1735.206834 | 1061.042 | 0.4457% |
| 70.4 | 343.55 | V(v1)-V(v2), Temperature=70.4  | 1.05237    | 1052.37   | 1054.140632 | 0.167969217 | 1729.297023 | 1058.678 | 0.4304% |
| 70.5 | 343.65 | V(v1)-V(v2), Temperature=70.5  | 1.05011    | 1050.11   | 1051.950339 | 0.174945468 | 1723.410395 | 1056.314 | 0.4148% |
| 70.6 | 343.75 | V(v1)-V(v2), Temperature=70.6  | 1.04783    | 1047.83   | 1049.762832 | 0.184120825 | 1717.546848 | 1053.95  | 0.3989% |
| 70.7 | 343.85 | V(v1)-V(v2), Temperature=70.7  | 1.04556    | 1045.56   | 1047.578117 | 0.192646005 | 1711.706282 | 1051.586 | 0.3826% |
| 70.8 | 343.95 | V(v1)-V(v2), Temperature=70.8  | 1.04353    | 1043.53   | 1045.396203 | 0.178516302 | 1705.888595 | 1049.222 | 0.3660% |
| 70.9 | 344.05 | V(v1)-V(v2), Temperature=70.9  | 1.04137    | 1041.37   | 1043.217095 | 0.177057572 | 1700.093686 | 1046.858 | 0.3490% |
| 71   | 344.15 | V(v1)-V(v2), Temperature=71    | 1.03909    | 1039.09   | 1041.040802 | 0.187389546 | 1694.321456 | 1044.494 | 0.3317% |
| 71.1 | 344.25 | V(v1)-V(v2), Temperature=71.1  | 1.03692    | 1036.92   | 1038.86733  | 0.187447381 | 1688.571804 | 1042.13  | 0.3141% |
| 71.2 | 344.35 | V(v1)-V(v2), Temperature=71.2  | 1.03476    | 1034.76   | 1036.696686 | 0.186813201 | 1682.844632 | 1039.766 | 0.2961% |
| 71.3 | 344.45 | V(v1)-V(v2), Temperature=71.3  | 1.03271    | 1032.71   | 1034.528878 | 0.175817082 | 1677.139841 | 1037.402 | 0.2777% |
| 71.4 | 344.55 | V(v1)-V(v2), Temperature=71.4  | 1.03041    | 1030.41   | 1032.363913 | 0.189265943 | 1671.457333 | 1035.038 | 0.2590% |
| 71.5 | 344.65 | V(v1)-V(v2), Temperature=71.5  | 1.02812    | 1028.12   | 1030.201797 | 0.202076629 | 1665.797009 | 1032.674 | 0.2400% |
| 71.6 | 344.75 | V(v1)-V(v2), Temperature=71.6  | 1.02606    | 1026.06   | 1028.042537 | 0.19284586  | 1660.158774 | 1030.31  | 0.2206% |
| 71.7 | 344.85 | V(v1)-V(v2), Temperature=71.7  | 1.02388    | 1023.88   | 1025.88614  | 0.195551955 | 1654.542529 | 1027.946 | 0.2008% |
| 71.8 | 344.95 | V(v1)-V(v2), Temperature=71.8  | 1.02169    | 1021.69   | 1023.732613 | 0.199526068 | 1648.948180 | 1025.582 | 0.1807% |
| 71.9 | 345.05 | V(v1)-V(v2), Temperature=71.9  | 1.01962    | 1019.62   | 1021.581962 | 0.192051389 | 1643.375629 | 1023.218 | 0.1601% |
| 72   | 345.15 | V(v1)-V(v2), Temperature=72    | 1.01767    | 1017.67   | 1019.434194 | 0.173056178 | 1637.824781 | 1020.854 | 0.1393% |
| 72.1 | 345.25 | V(v1)-V(v2), Temperature=72.1  | 1.01548    | 1015.48   | 1017.289315 | 0.177856501 | 1632.295543 | 1018.49  | 0.1180% |
| 72.2 | 345.35 | V(v1)-V(v2), Temperature=72.2  | 1.01328    | 1013.28   | 1015.147332 | 0.183946873 | 1626.787818 | 1016.126 | 0.0964% |
| 72.3 | 345.45 | V(v1)-V(v2), Temperature=72.3  | 1.0112     | 1011.2    | 1013.008251 | 0.178503066 | 1621.301514 | 1013.762 | 0.0744% |
| 72.4 | 345.55 | V(v1)-V(v2), Temperature=72.4  | 1.00911    | 1009.11   | 1010.872078 | 0.174312653 | 1615.836536 | 1011.398 | 0.0520% |
| 72.5 | 345.65 | V(v1)-V(v2), Temperature=72.5  | 1.00679    | 1006.79   | 1008.73882  | 0.193193707 | 1610.392792 | 1009.034 | 0.0293% |
| 72.6 | 345.75 | V(v1)-V(v2), Temperature=72.6  | 1.00494    | 1004.94   | 1006.608483 | 0.165752901 | 1604.970189 | 1006.67  | 0.0061% |
| 72.7 | 345.85 | V(v1)-V(v2), Temperature=72.7  | 1.00261    | 1002.61   | 1004.481072 | 0.186272527 | 1599.568634 | 1004.306 | 0.0174% |
| 72.8 | 345.95 | V(v1)-V(v2), Temperature=72.8  | 1.00051    | 1000.51   | 1002.356595 | 0.184225375 | 1594.188037 | 1001.942 | 0.0414% |
| 72.9 | 346.05 | V(v1)-V(v2), Temperature=72.9  | 0.99841249 | 998.41249 | 1000.235057 | 0.182213855 | 1588.828305 | 999.578  | 0.0657% |
| 73   | 346.15 | V(v1)-V(v2), Temperature=73    | 0.99619325 | 996.19325 | 998.1164635 | 0.192684281 | 1583.489348 | 997.214  | 0.0904% |
| 73.1 | 346.25 | V(v1)-V(v2), Temperature=73.1  | 0.99408822 | 994.08822 | 996.0008209 | 0.192028041 | 1578.171075 | 994.85   | 0.1155% |
| 73.2 | 346.35 | V(v1)-V(v2), Temperature=73.2  | 0.99198022 | 991.98022 | 993.8881351 | 0.191964769 | 1572.873397 | 992.486  | 0.1411% |
| 73.3 | 346.45 | V(v1)-V(v2), Temperature=73.3  | 0.98986925 | 989.86925 | 991.7784113 | 0.192498775 | 1567.596223 | 990.122  | 0.1670% |
| 73.4 | 346.55 | V(v1)-V(v2), Temperature=73.4  | 0.98775538 | 987.75538 | 989.6716557 | 0.193627421 | 1562.339465 | 987.758  | 0.1934% |

|      |        |                                |            |           |             |             |             |         |         |
|------|--------|--------------------------------|------------|-----------|-------------|-------------|-------------|---------|---------|
| 73.5 | 346.65 | V(ve1)-V(v2), Temperature=73.5 | 0.98587001 | 985.87001 | 987.5678738 | 0.171923757 | 1557.103034 | 985.394 | 0.2201% |
| 73.6 | 346.75 | V(v1)-V(v2), Temperature=73.6  | 0.98374994 | 983.74994 | 985.4670712 | 0.17424542  | 1551.886842 | 983.03  | 0.2473% |
| 73.7 | 346.85 | V(v1)-V(v2), Temperature=73.7  | 0.98162701 | 981.62701 | 983.3692531 | 0.177170789 | 1546.690800 | 980.666 | 0.2749% |
| 73.8 | 346.95 | V(v1)-V(v2), Temperature=73.8  | 0.97950129 | 979.50129 | 981.2744253 | 0.18069719  | 1541.514822 | 978.302 | 0.3029% |
| 73.9 | 347.05 | V(v1)-V(v2), Temperature=73.9  | 0.97725811 | 977.25811 | 979.1825929 | 0.196539742 | 1536.358820 | 975.938 | 0.3314% |
| 74   | 347.15 | V(v1)-V(v2), Temperature=74    | 0.97524159 | 975.24159 | 977.0937615 | 0.189559241 | 1531.222708 | 973.574 | 0.3602% |
| 74.1 | 347.25 | V(v1)-V(v2), Temperature=74.1  | 0.97322189 | 973.22189 | 975.007936  | 0.183182712 | 1526.106399 | 971.21  | 0.3895% |
| 74.2 | 347.35 | V(v1)-V(v2), Temperature=74.2  | 0.97131296 | 971.31296 | 972.9251218 | 0.165702557 | 1521.009808 | 968.846 | 0.4193% |
| 74.3 | 347.45 | V(v1)-V(v2), Temperature=74.3  | 0.96905928 | 969.05928 | 970.8453243 | 0.183967956 | 1515.932850 | 966.482 | 0.4494% |
| 74.4 | 347.55 | V(v1)-V(v2), Temperature=74.4  | 0.96703039 | 967.03039 | 968.7685481 | 0.179419337 | 1510.875439 | 964.118 | 0.4800% |
| 74.5 | 347.65 | V(v1)-V(v2), Temperature=74.5  | 0.96488524 | 964.88524 | 966.6947985 | 0.187190255 | 1505.837491 | 961.754 | 0.5111% |
| 74.6 | 347.75 | V(v1)-V(v2), Temperature=74.6  | 0.9627375  | 962.7375  | 964.6240799 | 0.195576701 | 1500.818921 | 959.39  | 0.5426% |
| 74.7 | 347.85 | V(v1)-V(v2), Temperature=74.7  | 0.96081274 | 960.81274 | 962.5563982 | 0.181148679 | 1495.819647 | 957.026 | 0.5746% |
| 74.8 | 347.95 | V(v1)-V(v2), Temperature=74.8  | 0.95865949 | 958.65949 | 960.4917573 | 0.190763464 | 1490.839584 | 954.662 | 0.6070% |
| 74.9 | 348.05 | V(v1)-V(v2), Temperature=74.9  | 0.95672829 | 956.72829 | 958.4301626 | 0.177568761 | 1485.878650 | 952.298 | 0.6398% |
| 75   | 348.15 | V(v1)-V(v2), Temperature=75    | 0.95468166 | 954.68166 | 956.3716184 | 0.176705199 | 1480.936762 | 949.934 | 0.6731% |
| 75.1 | 348.25 | V(v1)-V(v2), Temperature=75.1  | 0.95263211 | 952.63211 | 954.3161295 | 0.176463485 | 1476.013838 | 947.57  | 0.7069% |
| 75.2 | 348.35 | V(v1)-V(v2), Temperature=75.2  | 0.95057972 | 950.57972 | 952.2637004 | 0.176839716 | 1471.109796 | 945.206 | 0.7411% |
| 75.3 | 348.45 | V(v1)-V(v2), Temperature=75.3  | 0.94852451 | 948.52451 | 950.2143358 | 0.177836282 | 1466.224555 | 942.842 | 0.7759% |
| 75.4 | 348.55 | V(v1)-V(v2), Temperature=75.4  | 0.94646647 | 946.46647 | 948.1680402 | 0.17945872  | 1461.358034 | 940.478 | 0.8110% |
| 75.5 | 348.65 | V(v1)-V(v2), Temperature=75.5  | 0.9445165  | 944.5165  | 946.1248178 | 0.169990027 | 1456.510152 | 938.114 | 0.8467% |
| 75.6 | 348.75 | V(v1)-V(v2), Temperature=75.6  | 0.94234213 | 942.34213 | 944.0846731 | 0.184574875 | 1451.680829 | 935.75  | 0.8828% |
| 75.7 | 348.85 | V(v1)-V(v2), Temperature=75.7  | 0.94027588 | 940.27588 | 942.0476103 | 0.188072265 | 1446.869985 | 933.386 | 0.9194% |
| 75.8 | 348.95 | V(v1)-V(v2), Temperature=75.8  | 0.9382069  | 938.2069  | 940.0136334 | 0.192202888 | 1442.077540 | 931.022 | 0.9565% |
| 75.9 | 349.05 | V(v1)-V(v2), Temperature=75.9  | 0.93613529 | 936.13529 | 937.9827474 | 0.196960701 | 1437.303417 | 928.658 | 0.9941% |
| 76   | 349.15 | V(v1)-V(v2), Temperature=76    | 0.93428022 | 934.28022 | 935.9549556 | 0.178933351 | 1432.547535 | 926.294 | 1.0322% |
| 76.1 | 349.25 | V(v1)-V(v2), Temperature=76.1  | 0.93220288 | 932.20288 | 933.9302625 | 0.184958402 | 1427.809817 | 923.93  | 1.0708% |
| 76.2 | 349.35 | V(v1)-V(v2), Temperature=76.2  | 0.93023212 | 930.23212 | 931.9086717 | 0.179905154 | 1423.090184 | 921.566 | 1.1098% |
| 76.3 | 349.45 | V(v1)-V(v2), Temperature=76.3  | 0.92804055 | 928.04055 | 929.8901879 | 0.198909285 | 1418.388560 | 919.202 | 1.1494% |
| 76.4 | 349.55 | V(v1)-V(v2), Temperature=76.4  | 0.9262815  | 926.2815  | 927.8748142 | 0.171716504 | 1413.704866 | 916.838 | 1.1895% |
| 76.5 | 349.65 | V(v1)-V(v2), Temperature=76.5  | 0.92408496 | 924.08496 | 925.8625548 | 0.191993374 | 1409.039026 | 914.474 | 1.2300% |
| 76.6 | 349.75 | V(v1)-V(v2), Temperature=76.6  | 0.92210276 | 922.10276 | 923.8534132 | 0.18949469  | 1404.390963 | 912.11  | 1.2711% |
| 76.7 | 349.85 | V(v1)-V(v2), Temperature=76.7  | 0.92011762 | 920.11762 | 921.8473936 | 0.187642079 | 1399.760602 | 909.746 | 1.3127% |
| 76.8 | 349.95 | V(v1)-V(v2), Temperature=76.8  | 0.91823736 | 918.23736 | 919.8444993 | 0.174718581 | 1395.147866 | 907.382 | 1.3548% |
| 76.9 | 350.05 | V(v1)-V(v2), Temperature=76.9  | 0.91613886 | 916.13886 | 917.8447339 | 0.185856476 | 1390.552680 | 905.018 | 1.3975% |
| 77   | 350.15 | V(v1)-V(v2), Temperature=77    | 0.91425242 | 914.25242 | 915.8481015 | 0.174229928 | 1385.974970 | 902.654 | 1.4406% |
| 77.1 | 350.25 | V(v1)-V(v2), Temperature=77.1  | 0.91236277 | 912.36277 | 913.8546046 | 0.163246381 | 1381.414659 | 900.29  | 1.4843% |
| 77.2 | 350.35 | V(v1)-V(v2), Temperature=77.2  | 0.91025637 | 910.25637 | 911.8642476 | 0.176328616 | 1376.871675 | 897.926 | 1.5285% |
| 77.3 | 350.45 | V(v1)-V(v2), Temperature=77.3  | 0.90825426 | 908.25426 | 909.8770331 | 0.178350813 | 1372.345942 | 895.562 | 1.5733% |
| 77.4 | 350.55 | V(v1)-V(v2), Temperature=77.4  | 0.90624941 | 906.24941 | 907.8929646 | 0.181029559 | 1367.837387 | 893.198 | 1.6186% |
| 77.5 | 350.65 | V(v1)-V(v2), Temperature=77.5  | 0.90424194 | 904.24194 | 905.9120461 | 0.184356316 | 1363.345938 | 890.834 | 1.6644% |
| 77.6 | 350.75 | V(v1)-V(v2), Temperature=77.6  | 0.90244344 | 902.44344 | 903.9342802 | 0.16492794  | 1358.871521 | 888.47  | 1.7108% |
| 77.7 | 350.85 | V(v1)-V(v2), Temperature=77.7  | 0.90021909 | 900.21909 | 901.9596699 | 0.192977581 | 1354.414063 | 886.106 | 1.7577% |
| 77.8 | 350.95 | V(v1)-V(v2), Temperature=77.8  | 0.89830911 | 898.30911 | 899.9882186 | 0.186570063 | 1349.973492 | 883.742 | 1.8052% |
| 77.9 | 351.05 | V(v1)-V(v2), Temperature=77.9  | 0.89639613 | 896.39613 | 898.0199291 | 0.180819943 | 1345.549736 | 881.378 | 1.8532% |
| 78   | 351.15 | V(v1)-V(v2), Temperature=78    | 0.89427039 | 894.27039 | 896.0548048 | 0.199141259 | 1341.142724 | 879.014 | 1.9018% |
| 78.1 | 351.25 | V(v1)-V(v2), Temperature=78.1  | 0.89245658 | 892.45658 | 894.0928483 | 0.183008765 | 1336.752384 | 876.65  | 1.9509% |
| 78.2 | 351.35 | V(v1)-V(v2), Temperature=78.2  | 0.89063934 | 890.63934 | 892.1340624 | 0.167544589 | 1332.378645 | 874.286 | 2.0066% |
| 78.3 | 351.45 | V(v1)-V(v2), Temperature=78.3  | 0.88840206 | 888.40206 | 890.1784506 | 0.199554436 | 1328.021438 | 871.922 | 2.0509% |
| 78.4 | 351.55 | V(v1)-V(v2), Temperature=78.4  | 0.8865791  | 886.5791  | 888.2260145 | 0.185416155 | 1323.680690 | 869.558 | 2.1017% |
| 78.5 | 351.65 | V(v1)-V(v2), Temperature=78.5  | 0.88475281 | 884.75281 | 886.2767582 | 0.171949473 | 1319.356334 | 867.194 | 2.1531% |
| 78.6 | 351.75 | V(v1)-V(v2), Temperature=78.6  | 0.88281978 | 882.81978 | 884.3306832 | 0.170852744 | 1315.048298 | 864.83  | 2.2051% |
| 78.7 | 351.85 | V(v1)-V(v2), Temperature=78.7  | 0.88088396 | 880.88396 | 882.3877932 | 0.170427702 | 1310.756515 | 862.466 | 2.2577% |
| 78.8 | 351.95 | V(v1)-V(v2), Temperature=78.8  | 0.87894538 | 878.94538 | 880.4480899 | 0.170675578 | 1306.480914 | 860.102 | 2.3109% |
| 78.9 | 352.05 | V(v1)-V(v2), Temperature=78.9  | 0.87690126 | 876.90126 | 878.5115759 | 0.183300475 | 1302.221427 | 857.738 | 2.3646% |
| 79   | 352.15 | V(v1)-V(v2), Temperature=79    | 0.87506009 | 875.06009 | 876.5782539 | 0.173192059 | 1297.977986 | 855.374 | 2.4190% |
| 79.1 | 352.25 | V(v1)-V(v2), Temperature=79.1  | 0.87321577 | 873.21577 | 874.6481259 | 0.163763674 | 1293.750522 | 853.01  | 2.4739% |
| 79.2 | 352.35 | V(v1)-V(v2), Temperature=79.2  | 0.87116413 | 871.16413 | 872.7211949 | 0.178414934 | 1289.538969 | 850.646 | 2.5295% |
| 79.3 | 352.45 | V(v1)-V(v2), Temperature=79.3  | 0.86931414 | 869.31414 | 870.797463  | 0.170340763 | 1285.343259 | 848.282 | 2.5856% |
| 79.4 | 352.55 | V(v1)-V(v2), Temperature=79.4  | 0.86746108 | 867.46108 | 868.8769325 | 0.162952015 | 1281.163325 | 845.918 | 2.6424% |
| 79.5 | 352.65 | V(v1)-V(v2), Temperature=79.5  | 0.86550359 | 865.50359 | 866.9596053 | 0.167944999 | 1276.999100 | 843.554 | 2.6997% |
| 79.6 | 352.75 | V(v1)-V(v2), Temperature=79.6  | 0.86344238 | 863.44238 | 865.0454834 | 0.185320127 | 1272.850517 | 841.19  | 2.7577% |
| 79.7 | 352.85 | V(v1)-V(v2), Temperature=79.7  | 0.86148009 | 861.48009 | 863.1345693 | 0.191682665 | 1268.717511 | 838.826 | 2.8163% |
| 79.8 | 352.95 | V(v1)-V(v2), Temperature=79.8  | 0.85917689 | 859.17689 | 861.2268651 | 0.17532838  | 1264.600016 | 836.462 | 2.8755% |
| 79.9 | 353.05 | V(v1)-V(v2), Temperature=79.9  | 0.8576487  | 857.6487  | 859.3223724 | 0.194766539 | 1260.497966 | 834.098 | 2.9354% |
| 80   | 353.15 | V(v1)-V(v2), Temperature=80    | 0.85597986 | 855.97986 | 857.4210934 | 0.168089329 | 1256.411296 | 831.734 | 2.9959% |

## Chapter 3. Lựa chọn OPAMP sử dụng, mô phỏng OPAMP theo datasheet

### 3.1 Tại tầng khuếch đại

Ta xét ở tầng khuếch đại:



1. Xét tại OPAMP đầu tiên có chức năng là một Buffer:

$$V_{o1} = V_i \pm V_{os}$$

2. Xét tại OPAMP thứ hai có chức năng là thực hiện hàm  $V_o = V_i + V_{ref}$ :

$$V_o = \left(1 + \frac{R_3}{R_4}\right) \left(V_{ref} \frac{R_2}{R_1 + R_2} + V_{o1} \frac{R_1}{R_1 + R_2}\right) \pm V_{os} \left(1 + \frac{R_3}{R_4}\right) \mp I_- R_3$$

Chọn  $R_1 = R_2 = 1 \text{ k}\Omega$

Chọn  $R_3 = R_4 = 1 \text{ k}\Omega$

Ta xét 2 con OPAMP trên là cùng một Package, ta có:

1. Xét tại OPAMP đầu tiên có chức năng là một Buffer với ngõ điện áp ngõ ra sai số 5% giá trị thì tại đây nhóm em chọn sai số phải rơi vào dưới 1%:

$$\pm V_{os} < 0.08 \text{ V}$$

$\Rightarrow V_{os}$  phải có giá trị ở dưới 10 lần ngưỡng trên là cỡ 8 mV.

2. Xét tại OPAMP thứ hai phải có ngõ ra cũng rơi vào dưới 1%:

$$\pm V_{os} \left( 1 + \frac{R_3}{R_4} \right) \mp I_- R_3 < 0.05 \text{ V}$$

$\Rightarrow$  Với  $V_{os} < 10 \text{ mV}$  thì  $I_- > 70 \mu\text{A}$ .

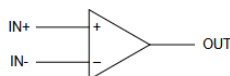
$$\Rightarrow \begin{cases} I_{os} = |I_+ - I_-| < 30 \mu\text{A} \\ I_{ib} = \frac{I_+ + I_-}{2} < 15 \mu\text{A} \end{cases}$$

Từ đó ở tầng khuếch đại nhóm em sử dụng **LMV358IDGKR**, với thông số sau:

| Device Information |               |                        |
|--------------------|---------------|------------------------|
| PART NUMBER        | CHANNEL COUNT | PACKAGE <sup>(1)</sup> |
| TL081x             | Single        | P (PDIP, 8)            |
|                    |               | DCK (SC70, 5)          |
|                    |               | PS (SO, 8)             |
|                    |               | D (SOIC, 8)            |
|                    |               | DBV (SOT-23, 5)        |
| TL082x             | Dual          | P (PDIP, 8)            |
|                    |               | PS (SO, 8)             |
|                    |               | D (SOIC, 8)            |
|                    |               | DDF (SOT-23, 8)        |
|                    |               | PW (TSSOP, 8)          |
| TL082M             | Dual          | JG (CDIP, 8)           |
|                    |               | FK (LCCC, 20)          |
| TL084x             | Quad          | N (PDIP, 14)           |
|                    |               | D (SOIC, 14)           |
|                    |               | DYY (SOT-23, 14)       |
|                    |               | PW (TSSOP, 14)         |
| TL084M             | Quad          | J (CDIP, 14)           |
|                    |               | FK (LCCC, 20)          |

(1) For more information, see [Section 11](#).

TL081 (Each Amplifier)  
TL082 (Each Amplifier)  
TL084 (Each Amplifier)



**5.7 Electrical Characteristics: TL08xH**

For  $V_S = (V_{CC+} - V_{CC-}) = 4.5\text{ V to }40\text{ V}$  ( $\pm 2.25\text{ V to } \pm 20\text{ V}$ ) at  $T_A = 25^\circ\text{C}$ ,  $R_L = 10\text{ k}\Omega$  connected to  $V_S / 2$ ,  $V_{CM} = V_S / 2$ , and  $V_{out} = V_S / 2$ , unless otherwise noted.

| PARAMETER            |                                          | TEST CONDITIONS                                                 |                                      | MIN | TYP  | MAX  | UNIT  |
|----------------------|------------------------------------------|-----------------------------------------------------------------|--------------------------------------|-----|------|------|-------|
| OFFSET VOLTAGE       |                                          |                                                                 |                                      |     |      |      |       |
| V <sub>OS</sub>      | Input offset voltage                     |                                                                 | T <sub>A</sub> = −40°C to +125°C     |     | ±1   | ±4   | mV    |
|                      |                                          |                                                                 |                                      |     |      | ±5   |       |
| dV <sub>OS</sub> /dT | Input offset voltage drift               |                                                                 | T <sub>A</sub> = −40°C to +125°C     |     | ±2   |      | µV/°C |
| PSRR                 | Input offset voltage versus power supply | V <sub>S</sub> = 5V to 40V, V <sub>CM</sub> = V <sub>S</sub> /2 | T <sub>A</sub> = −40°C to +125°C     |     | ±1   | ±10  | µV/V  |
|                      | Channel separation                       | f = 0 Hz                                                        |                                      |     | 10   |      | µV/V  |
| INPUT BIAS CURRENT   |                                          |                                                                 |                                      |     |      |      |       |
| I <sub>B</sub>       | Input bias current                       |                                                                 |                                      |     | ±1   | ±120 | pA    |
|                      |                                          |                                                                 | DCK and DBV packages                 |     | ±1   | ±300 | pA    |
|                      |                                          |                                                                 | T <sub>A</sub> = −40°C to +125°C (1) |     |      | ±5   | nA    |
| I <sub>OS</sub>      | Input offset current                     |                                                                 |                                      |     | ±0.5 | ±120 | pA    |
|                      |                                          |                                                                 | DCK and DBV packages                 |     | ±0.5 | ±250 | pA    |
|                      |                                          |                                                                 | T <sub>A</sub> = −40°C to +125°C (1) |     |      | ±5   | nA    |

## Chapter 4. Thực hiện mạch đọc điện áp cho ngõ ra ADC

### 4.1 Tổng quan về ADC

Theo chuẩn ngõ ra điện áp đọc từ Sensor là từ  $0 \rightarrow 5\text{ V}$ , với ngõ vào là từ  $20^\circ\text{C} \rightarrow 80^\circ\text{C}$  với bước nhảy là  $0.1^\circ\text{C}$ . Thực hiện tính giá trị của một bước nhảy nhỏ nhất là:

$$\frac{5 - 0}{\frac{80 - 20}{0.1} + 1} \approx 8.3195\text{ mV}$$

Vậy có nghĩa là  $\Delta_{min} \approx 8.3195\text{ mV}$ , từ đó ta có:

$$2^Q = \frac{V_{ref+} - V_{ref-}}{\Delta_{min}}$$

Giả sử, chọn  $V_{ref+} = 5\text{ V}$  và  $V_{ref-} = 0\text{ V}$ . Ta có  $Q \approx 9.23\text{ bit}$

Tuy nhiên, trong thị trường không có ADC 9.23 bit cho nên nhóm em sẽ chọn con có trên thị trường là **ADS1115IDGSR**.

### 4.2 Electrical Characteristics của ADS1115

- Ultra-small packages:
  - + X2QFN:  $2\text{ mm} \times 1.5\text{ mm} \times 0.4\text{ mm}$
  - + SOT:  $2.9\text{ mm} \times 2.8\text{ mm} \times 0.6\text{ mm}$
- Wide supply range:  $2.0\text{ V}$  to  $5.5\text{ V}$
- Low current consumption:  $150\text{ }\mu\text{A}$  (continuous-conversion mode)
- Programmable data rate: 8SPS to 860SPS
- Single-cycle settling
- Internal low-drift voltage reference
- Internal oscillator
- I2C interface: four pin-selectable addresses

- Operating temperature range:  $-40^{\circ}\text{C}$  to  $+125^{\circ}\text{C}$
- Family of devices:
  - + ADS1115: four single-ended or two differential inputs with comparator and PGA

Từ những thông tin, ADC ADS1115 có tầm hoạt động ở dải nhiệt độ phù hợp trong khoảng  $20^{\circ}\text{C} \rightarrow 80^{\circ}\text{C}$ , nhóm em sử dụng nguồn cấp là  $V_{DD} = 5.5\text{V}$  và cho phép nguồn sai số trong khoảng từ  $[5.3, 5.5]\text{V}$  phù hợp với thông số dưới đây:

| DIGITAL INPUT/OUTPUT |                          |                       |         |         |               |
|----------------------|--------------------------|-----------------------|---------|---------|---------------|
| $V_{IH}$             | High-level input voltage |                       | 0.7 VDD | 5.5     | V             |
| $V_{IL}$             | Low-level input voltage  |                       | GND     | 0.3 VDD | V             |
| $V_{OL}$             | Low-level output voltage | $I_{OL} = 3\text{mA}$ | GND     | 0.15    | 0.4 V         |
|                      | Input leakage current    | $GND < V_{DIG} < VDD$ | -10     | 10      | $\mu\text{A}$ |

Hình 4.1: Digital Input/Output của ADS1115.

Như vậy, với  $GND \approx 0\text{V} < V_{IN} < V_{DD} = 5.5\text{V}$  thì có thể phù hợp cho tầm đo từ  $0 \rightarrow 5\text{V}$  từ mạch đọc giá trị của sensor.

Tiếp theo, ta cần quan tâm đến giá trị như sau:

| SYSTEM PERFORMANCE |                                            |                                                        |                                               |                  |
|--------------------|--------------------------------------------|--------------------------------------------------------|-----------------------------------------------|------------------|
|                    | Resolution (no missing codes)              |                                                        | 16                                            | Bits             |
| DR                 | Data rate                                  |                                                        | 8, 16, 32, 64, 128, 250, 475, 860             | SPS              |
|                    | Data rate variation                        | All data rates                                         | -10% 10%                                      |                  |
|                    | Output noise                               |                                                        | See <a href="#">Noise Performance</a> section |                  |
| INL                | Integral nonlinearity                      | DR = 8SPS, FSR = $\pm 2.048V^{(2)}$                    | 1                                             | LSB              |
|                    | Offset error                               | FSR = $\pm 2.048V$ , differential inputs               | 3                                             | LSB              |
|                    |                                            | FSR = $\pm 2.048V$ , single-ended inputs               | 3                                             | LSB              |
|                    | Offset drift over temperature              | FSR = $\pm 2.048V$                                     | 0.005                                         | LSB/ $^{\circ}C$ |
|                    | Long-term Offset drift                     | FSR = $\pm 2.048V$ , $T_A = 125^{\circ}C$ , 1000 hours | $\pm 1$                                       | LSB              |
|                    | Offset power-supply rejection              | FSR = $\pm 2.048V$ , DC supply variation               | 1                                             | LSB/V            |
|                    | Offset channel match                       | Match between any two inputs                           | 3                                             | LSB              |
|                    | Gain error <sup>(3)</sup>                  | FSR = $\pm 2.048V$ , $T_A = 25^{\circ}C$               | 0.01% 0.15%                                   |                  |
|                    | Gain drift over temperature <sup>(3)</sup> | FSR = $\pm 0.256V$                                     | 7                                             | ppm/ $^{\circ}C$ |
|                    |                                            | FSR = $\pm 2.048V$                                     | 5 40                                          |                  |
|                    |                                            | FSR = $\pm 6.144V^{(1)}$                               | 5                                             |                  |
|                    | Long-term gain drift <sup>(3)</sup>        | FSR = $\pm 2.048V$ , $T_A = 125^{\circ}C$ , 1000 hours | $\pm 0.05\%$                                  |                  |
|                    | Gain power-supply rejection                |                                                        | 80                                            | ppm/V            |
|                    | Gain match <sup>(3)</sup>                  | Match between any two gains                            | 0.02% 0.1%                                    |                  |
|                    | Gain channel match                         | Match between any two inputs                           | 0.05% 0.1%                                    |                  |
| CMRR               | Common-mode rejection ratio                | At DC, FSR = $\pm 0.256V$                              | 105                                           | dB               |
|                    |                                            | At DC, FSR = $\pm 2.048V$                              | 100                                           |                  |
|                    |                                            | At DC, FSR = $\pm 6.144V^{(1)}$                        | 90                                            |                  |
|                    |                                            | $f_{CM} = 60Hz$ , DR = 8SPS                            | 105                                           |                  |
|                    |                                            | $f_{CM} = 50Hz$ , DR = 8SPS                            | 105                                           |                  |

Hình 4.2: Electrical Characteristics của ADS1115.

- Data rate (DR) là giá trị dữ liệu Digital được đổi theo đơn vị thời gian. Với DR = 128 Sample per second là giá trị default của ADS1115, với thời gian cách nhau đọc dữ liệu khoảng tầm 8 ms.
- Integral Nonlinearity (INL) là thước đo độ lệch của đặc tuyến chuyển đổi thực tế so với đặc tuyến lý tưởng trong các bộ chuyển đổi ADC/DAC, sau khi đã loại bỏ sai số offset và gain. Với INL = 1 LSB.
- Offset error là sai số khi sử dụng chế độ Single-Ended với giá trị là  $\pm 3$  LSB.
- Gain drift over temperature là sai số của phép đo khi có ảnh hưởng của nhiệt độ với tại FSR =  $\pm 6.144V$  thì Gain drift =  $5ppm/^{\circ}C$  tức là mỗi  $^{\circ}C$  sẽ gây sai số là 5 phần ngàn.
- Common-mode Rejection Ratio (CMRR) là thước đo khả năng của một bộ khuếch đại vi sai trong việc loại bỏ tín hiệu chung xuất hiện đồng thời và cùng pha trên cả hai đầu vào. Với CMRR = 90 dB.

### 4.3 Typical Characteristics của ADS1115

- Single-Ended Offset Error vs Temperature:

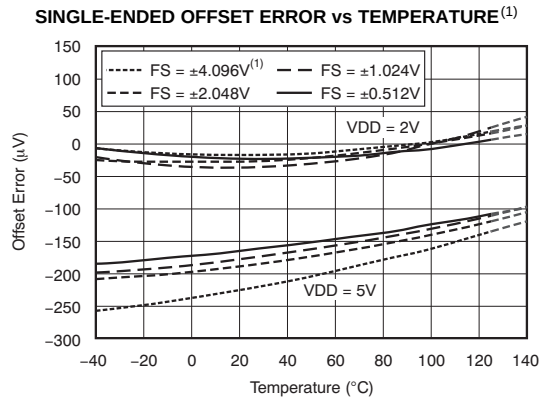


Figure 4.

+ Tại  $T = 20^{\circ}\text{C}$ : Offset Error =  $-225\mu\text{V}$

+ Tại  $T = 80^{\circ}\text{C}$ : Offset Error =  $-120\mu\text{V}$

- Gain Error vs Temperature:

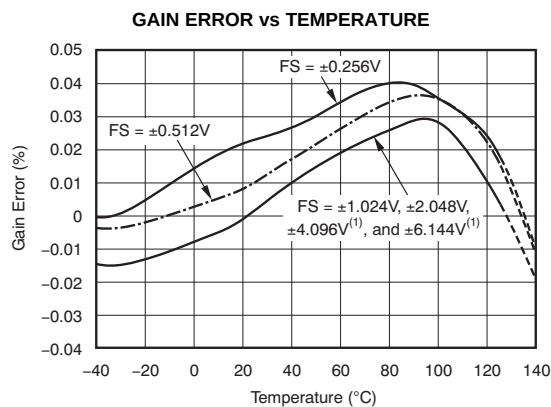


Figure 6.

+ Tại  $T = 20^{\circ}\text{C}$ : Gain Error = 0.00 %

+ Tại  $T = 80^{\circ}\text{C}$ : Gain Error = 0.03 %

- Gain Error vs Supply:

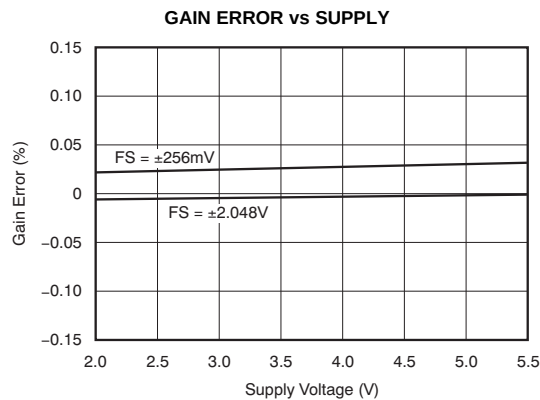


Figure 7.

+ Tại  $V_{DD} = 5.5\text{V}$ : Gain Error = 0.00 %

- INL vs Supply Voltage:

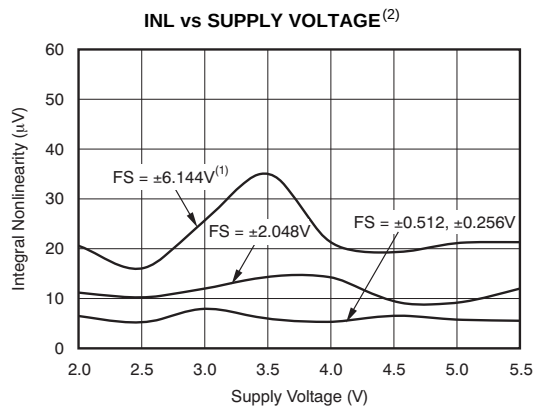


Figure 8.

+ Tại  $V_{DD} = 5.5V$ : INL vs Voltage =  $20\mu V$

- INL vs Temperature:

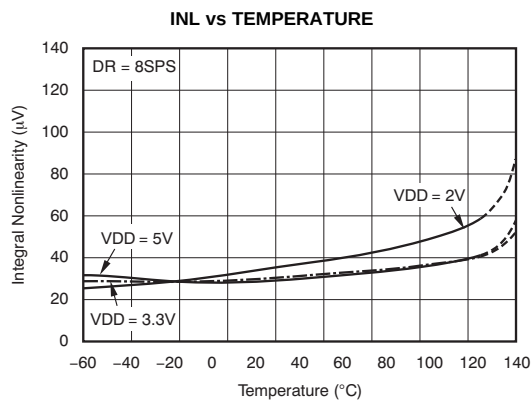


Figure 13.

+ Tại  $T = 20^{\circ}C$ : INL vs Temperature =  $22\mu V$

+ Tại  $T = 80^{\circ}C$ : INL vs Temperature =  $38\mu V$

- Data Rate vs Temperature:

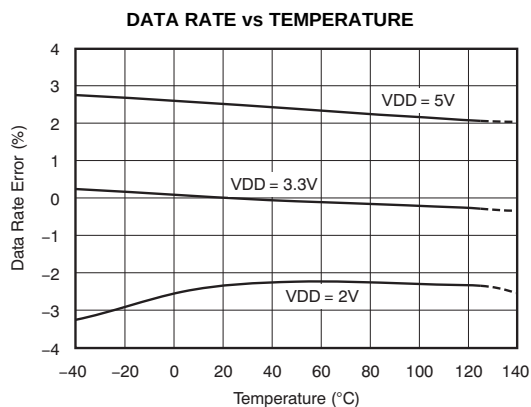


Figure 20.

+ Tại  $T = 20^{\circ}C$ : Data Rate Error =  $2.8\%$

+ Tại  $T = 80^{\circ}C$ : Data Rate Error =  $1.2\%$

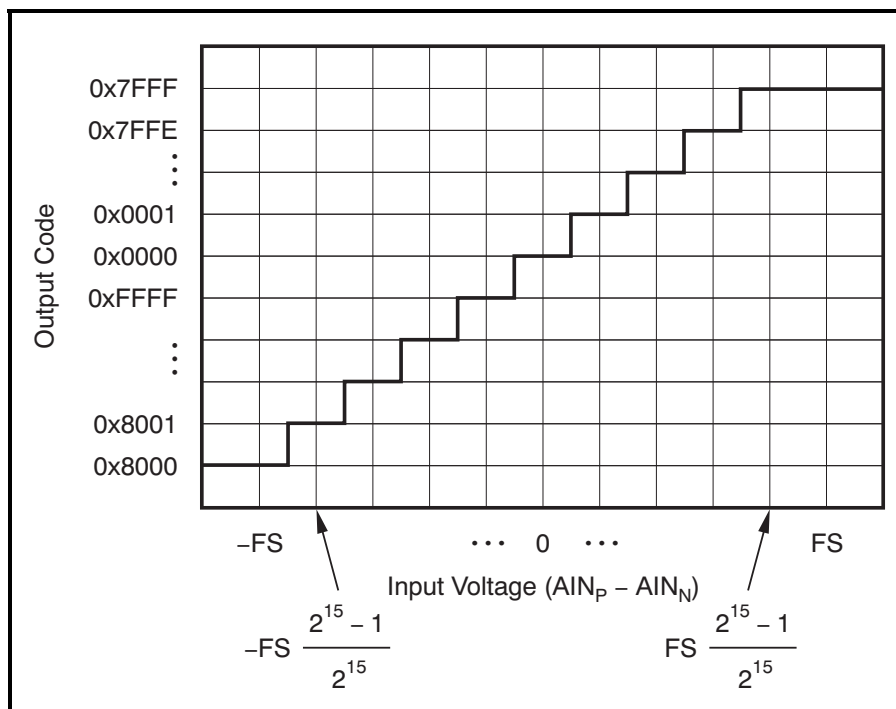
#### 4.4 Data Format

Với ADS1115 là một ADC 16-bit với ngõ ra là một chuỗi bit có dấu theo giá trị như sau:

**Table 4. Input Signal versus Ideal Output Code**

| INPUT SIGNAL, $V_{IN}$<br>( $A_{INP} - A_{INN}$ ) | IDEAL OUTPUT CODE <sup>(1)</sup> |
|---------------------------------------------------|----------------------------------|
| $\geq FS (2^{15} - 1)/2^{15}$                     | 7FFFh                            |
| $+FS/2^{15}$                                      | 0001h                            |
| 0                                                 | 0                                |
| $-FS/2^{15}$                                      | FFFFh                            |
| $\leq -FS$                                        | 8000h                            |

1. Excludes the effects of noise, INL, offset, and gain errors.



**Figure 27. ADS1113/4/5 Code Transition Diagram**

Hình 4.3: ADS1115 Code Transition Diagram.

Theo hình 4.3, với  $FS = 6.144$  như ở trên thì giá trị lớn nhất mà ADC có thể đọc được

là  $V_{ref} \times \frac{2^{15} - 1}{2^{15}} \approx 6.1438 \text{ V}$  vẫn phù hợp với tầm đo là từ  $0 \rightarrow 5 \text{ V}$ . Từ đó ta có công thức suy ngược từ  $D_{out}$  nhận được từ ADC là:

$$V_{out} = \frac{V_{ref}}{2^{15}} \times D_{out} = \frac{6.1438}{32768} \times D_{out}$$

## 4.5 Kết quả của việc tính toán ADC

Ta có bảng tính toán giá bên dưới, trong bảng giá trị có các giá trị như sau:

- **T\_s** : là giá trị nhiệt độ chuẩn cần đo.
- **V\_i** : là giá trị điện áp đo được từ sensor trong khoảng từ  $0 \rightarrow 5 \text{ V}$  với sai số là 5%.
- **Offset\_Error** : là giá trị sai số của ADS1115 do nhiệt độ tạo nên.
- **Gain\_Error** : là giá trị của độ lợi của ADS1115 do nhiệt độ tạo nên.
- **V\_i\_ADC** : là giá trị mà ADC thực sự nhận tại chân IN0.
- **D\_out** : là giá trị tính toán từ giá trị điện đọc từ chân IN0 với công thức là  $\frac{V_{i\_ADC} \times 2^{15}}{FS}$ .
- **D\_cal\_Error** : là giá trị mà thực sự ADS1115 cho ra ở ngõ ra với các giá trị sai số cộng thêm do chính ADC và nhiệt độ.
- **Cal\_V** : là giá trị được tính trong MCU chuyển đổi từ giá trị Digital của ADS1115 gửi về cho MCU.
- **Cal\_Temp** : là giá trị chuyển đổi từ điện áp tính toán được **Cal\_V** về giá trị nhiệt độ theo công thức chuẩn là  $T = \frac{V_{in} + \frac{5}{3}}{\frac{1}{12}}$ .

| Requirement |    |  |  |
|-------------|----|--|--|
| ERROR       | 5% |  |  |

| ADC                          |                |            |             |        |
|------------------------------|----------------|------------|-------------|--------|
|                              | Min            | Typ        | Max         | Unit   |
| VDD                          | 5.2            |            | 5.5         | 5.5 V  |
| GND                          | -0.3           |            | 0           | 0.3 V  |
| Resolution                   |                |            | 16          | Bit    |
| Analog_In_V_to_GND           |                |            |             | V      |
| Temperature                  | -60            |            | 150         | dó C   |
| PGA (program gain amplifier) | 0.66666667     | 0.66686667 |             | V      |
| FS (Full-Scale)              | -6.142157353   | 6.1438125  | 6.142157353 | V      |
| Common-Mode Input Impedance  |                | 6000000    |             | Ohm    |
| DR (Data Rate)               |                | 128        |             | SPS    |
| Range_Temperature            | 20             |            | 80          | dó C   |
| Offset Voltage               | -0.000225      |            | -0.00012    | V      |
| Gain Error                   | 0%             |            | 0.03%       | %      |
| Gain Drift                   |                | 0.000500%  |             | %/dó C |
|                              | Tai FS = 6.144 | 0.030000%  |             |        |
| Common Mode Rejection        | Tai FS = 6.144 | 90         |             | dB     |
| INL_vs_VoltageSupply         |                | 0.000020   |             | V      |
| INL_vs_Temperature           | 0.000022       |            | 0.000038    | V      |
| Digital Input/Output         |                |            |             |        |
| V_IH                         | 3.85           |            | 5.5         | V      |
| V_IL                         | -0.5           |            | 1.65        | V      |

| Calibration |                |                   |       |
|-------------|----------------|-------------------|-------|
|             | Value_Standard | Value_Measurement | Error |
| raw_zero_25 | 0              | 0                 | 0     |
| raw_full_25 | 5              | 5                 | 0     |
| raw_zero_20 | 0              | 0                 | 0     |
| raw_full_20 | 5              | 5                 | 0     |
| raw_zero_50 | 0              | 0                 | 0     |
| raw_full_50 | 5              | 5                 | 0     |
| raw_zero_80 | 0              | 0                 | 0     |
| raw_full_80 | 5              | 5                 | 0     |

| Digital output                |                      |               |          |
|-------------------------------|----------------------|---------------|----------|
| Input Signal (V_i)            | >= FS(2^15-1)/(2^15) |               |          |
| Offset Error                  | -3                   | 6.141969909 V | 3 LSB    |
| Offset Drift                  |                      | 0.005         | LSB/dó C |
|                               | FS = 6.144V          | 0.3           | LSB      |
| Offset power supply rejection |                      | 1             | LSB/V    |
|                               | VDD = 5.5V           | 0.3           | LSB      |
| INL                           |                      | 1             | LSB      |

| s    | V           | Offset_Vol | Gain_Error | G_E       | V_I_ADC     | Error_V_I_ADC | D_out     | D_cat_error | BIN                | Cal_V       | Cal_Temp   | Error_re | Error_ab  |
|------|-------------|------------|------------|-----------|-------------|---------------|-----------|-------------|--------------------|-------------|------------|----------|-----------|
| 20   | 0.017       | -0.000225  | 0.000000%  | -0.001500 | 0.017257201 | 1.5129%       | 92.0688   | 94          | 0000000001011110   | 0.017624462 | 20.2114935 | 1%       | 3.6989%   |
| 20.1 | 0.019954682 | -0.000224  | 0.000100%  | -0.001469 | 0.020207083 | 1.2649%       | 107.8067  | 111         | 0000000001011111   | 0.020811865 | 20.2497424 | 1%       | 0.0055%   |
| 20.2 | 0.025642402 | -0.000224  | 0.000100%  | -0.001439 | 0.025887233 | 0.9548%       | 138.1109  | 137         | 0000000010001001   | 0.025686716 | 20.3082406 | 1%       | 0.10821%  |
| 20.3 | 0.032330742 | -0.000223  | 0.000200%  | -0.001408 | 0.032656971 | 0.7277%       | 173.7426  | 175         | 0000000010101111   | 0.032811499 | 20.393738  | 0%       | 0.09373%  |
| 20.4 | 0.038074328 | -0.000223  | 0.000200%  | -0.001378 | 0.038303627 | 0.5996%       | 204.3482  | 202         | 0000000011001010   | 0.037873844 | 20.4544861 | 0%       | 0.05446%  |
| 20.5 | 0.044816929 | -0.000222  | 0.000300%  | -0.001347 | 0.045036343 | 0.4896%       | 240.2732  | 240         | 0000000011110000   | 0.044998627 | 20.5399835 | 0%       | 0.03998%  |
| 20.6 | 0.05158625  | -0.000222  | 0.000300%  | -0.001317 | 0.051798108 | 0.4107%       | 276.3479  | 280         | 0000000010001100   | 0.052498398 | 20.6299808 | 0%       | 0.029981% |
| 20.7 | 0.058382283 | -0.000221  | 0.000400%  | -0.001286 | 0.058586016 | 0.3490%       | 312.562   | 313         | 0000000010011001   | 0.058685709 | 20.7042285 | 0%       | 0.004229% |
| 20.8 | 0.065205052 | -0.000220  | 0.000400%  | -0.001256 | 0.065400949 | 0.3005%       | 348.9204  | 347         | 0000000010101011   | 0.065060514 | 20.7807262 | 0%       | 0.01927%  |
| 20.9 | 0.072054452 | -0.000220  | 0.000500%  | -0.001225 | 0.072244024 | 0.2631%       | 385.4288  | 386         | 0000000011000010   | 0.072372791 | 20.8684735 | 0%       | 0.031527% |
| 21   | 0.078930568 | -0.000219  | 0.000500%  | -0.001195 | 0.079113097 | 0.2313%       | 422.076   | 423         | 0000000011010011   | 0.07931008  | 20.951721  | 0%       | 0.048479% |
| 21.1 | 0.084867512 | -0.000219  | 0.000600%  | -0.001164 | 0.085045587 | 0.2098%       | 453.7264  | 454         | 0000000011000110   | 0.085122402 | 21.0214688 | 0%       | 0.078531% |
| 21.2 | 0.091797781 | -0.000218  | 0.000600%  | -0.001134 | 0.091969553 | 0.1871%       | 490.6664  | 489         | 00000000111101001  | 0.091684702 | 21.1002164 | 0%       | 0.099784% |
| 21.3 | 0.098754705 | -0.000218  | 0.000700%  | -0.001103 | 0.098921658 | 0.1691%       | 527.7566  | 532         | 000000001000010100 | 0.099746956 | 21.1969635 | 0%       | 0.103037% |
| 21.4 | 0.105738272 | -0.000217  | 0.000700%  | -0.001073 | 0.105899704 | 0.1527%       | 564.9851  | 570         | 000000001000111010 | 0.106871738 | 21.2824609 | 1%       | 0.178249% |
| 21.5 | 0.112748471 | -0.000216  | 0.000800%  | -0.001042 | 0.112904885 | 0.1387%       | 602.3584  | 605         | 000000001001011011 | 0.113434038 | 21.3612085 | 1%       | 0.138792% |
| 21.6 | 0.118823527 | -0.000216  | 0.000800%  | -0.001012 | 0.118977181 | 0.1293%       | 634.7547  | 639         | 000000001011111111 | 0.119808844 | 21.4377061 | 1%       | 0.16229%  |
| 21.7 | 0.125887774 | -0.000215  | 0.000900%  | -0.000981 | 0.126037188 | 0.1187%       | 672.4205  | 672         | 000000001010100000 | 0.125996155 | 21.5119539 | 1%       | 0.188046% |
| 21.8 | 0.132978615 | -0.000215  | 0.000900%  | -0.000951 | 0.133125068 | 0.1101%       | 710.235   | 709         | 000000001011000101 | 0.132933443 | 21.5952013 | 1%       | 0.204799% |
| 21.9 | 0.140096038 | -0.000214  | 0.001000%  | -0.000920 | 0.140239071 | 0.1021%       | 748.1889  | 749         | 000000001011101101 | 0.140433214 | 21.6851986 | 1%       | 0.214801% |
| 22   | 0.146281565 | -0.000214  | 0.001000%  | -0.000890 | 0.1464233   | 0.0969%       | 781.1824  | 785         | 000000001100010001 | 0.147183008 | 21.7661961 | 1%       | 0.23380%  |
| 22.1 | 0.153452935 | -0.000213  | 0.001100%  | -0.000859 | 0.153592049 | 0.0907%       | 819.4283  | 819         | 000000001100110011 | 0.153557814 | 21.8426938 | 1%       | 0.257306% |
| 22.2 | 0.160650843 | -0.000212  | 0.001100%  | -0.000829 | 0.160787598 | 0.0851%       | 857.8173  | 857         | 000000001010101001 | 0.160682596 | 21.9281912 | 1%       | 0.271809% |
| 22.3 | 0.166919306 | -0.000212  | 0.001200%  | -0.000798 | 0.167056042 | 0.0819%       | 891.26    | 890         | 000000001101111010 | 0.166869907 | 22.0024389 | 1%       | 0.297561% |
| 22.4 | 0.174717107 | -0.000211  | 0.001200%  | -0.000768 | 0.174306255 | 0.0776%       | 929.9406  | 934         | 000000001101001010 | 0.175119656 | 22.1014359 | 1%       | 0.29865%  |
| 22.5 | 0.181449339 | -0.000211  | 0.001300%  | -0.000737 | 0.181584554 | 0.0745%       | 968.7711  | 970         | 000000001110010100 | 0.18186945  | 22.1824334 | 1%       | 0.317567% |
| 22.6 | 0.187800619 | -0.000210  | 0.001300%  | -0.000707 | 0.187935789 | 0.0720%       | 1002.6555 | 1004        | 000000001111101100 | 0.188244255 | 22.2589311 | 2%       | 0.341069% |
| 22.7 | 0.195132651 | -0.000209  | 0.001400%  | -0.000676 | 0.195267689 | 0.0692%       | 1041.7719 | 1040        | 000000100000010000 | 0.194994049 | 22.3399286 | 2%       | 0.360071% |
| 22.8 | 0.202491122 | -0.000209  | 0.001400%  | -0.000646 | 0.202627262 | 0.0672%       | 1081.0359 | 1083        | 000000100000110111 | 0.203056303 | 22.4366756 | 2%       | 0.38332%  |
| 22.9 | 0.208925084 | -0.000208  | 0.001500%  | -0.000615 | 0.209062546 | 0.0658%       | 1115.3688 | 1115        | 000000100010101101 | 0.20905612  | 22.5086734 | 2%       | 0.391327% |
| 23   | 0.216337223 | -0.000208  | 0.001500%  | -0.000585 | 0.216476618 | 0.0644%       | 1154.9236 | 1154        | 000000100100000100 | 0.216368397 | 22.5964208 | 2%       | 0.403579% |
| 23.1 | 0.222826509 | -0.000207  | 0.001600%  | -0.000554 | 0.222968017 | 0.0635%       | 1189.5558 | 1189        | 00000010010100101  | 0.222930696 | 22.6751684 | 2%       | 0.424832% |
| 23.2 | 0.230292244 | -0.000207  | 0.001600%  | -0.000524 | 0.230436526 | 0.0627%       | 1229.401  | 1228        | 000000100110011000 | 0.230242973 | 22.7629157 | 2%       | 0.43708%  |
| 23.3 | 0.236836788 | -0.000206  | 0.001700%  | -0.000493 | 0.236983983 | 0.0622%       | 1264.3323 | 1265        | 000000100111000101 | 0.237180262 | 22.8461631 | 2%       | 0.453837% |
| 23.4 | 0.244356043 | -0.000205  | 0.001700%  | -0.000463 | 0.244505864 | 0.0613%       | 1304.4623 | 1308        | 000000101000011100 | 0.245242516 | 22.9429102 | 2%       | 0.457090% |
| 23.5 | 0.251901602 | -0.000205  | 0.001800%  | -0.000432 | 0.252055739 | 0.0612%       | 1344.7416 | 1347        | 000000101010000101 | 0.252554792 | 23.0306575 | 2%       | 0.469342% |
| 23.6 | 0.258528469 | -0.000204  | 0.001800%  | -0.000402 | 0.2586685   | 0.0611%       | 1380.1173 | 1379        | 000000101011000111 | 0.258554609 | 23.1026553 | 2%       | 0.497345% |
| 23.7 | 0.266127425 | -0.000204  | 0.001900%  | -0.000371 | 0.266290652 | 0.0613%       | 1420.6862 | 1424        | 000000101001000000 | 0.266991852 | 23.2039022 | 2%       | 0.496098% |
| 23.8 | 0.272809362 | -0.000203  | 0.001900%  | -0.000341 | 0.272977297 | 0.0616%       | 1456.3601 | 1460        | 000000101011010100 | 0.273741646 | 23.2848997 | 2%       | 0.515100% |
| 23.9 | 0.28046163  | -0.000203  | 0.002000%  | -0.000310 | 0.280635651 | 0.0620%       | 1497.2182 | 1497        | 000000101101101001 | 0.280678934 | 23.3681472 | 2%       | 0.531853% |
| 24   | 0.287198558 | -0.000202  | 0.002000%  | -0.000280 | 0.287378108 | 0.0625%       | 1533.1898 | 1533        | 000000101111111011 | 0.287428728 | 23.4491447 | 2%       | 0.550855% |
| 24.1 | 0.293963383 | -0.000201  | 0.002100%  | -0.000249 | 0.294149154 | 0.0632%       | 1569.314  | 1574        | 000000100001001010 | 0.295115993 | 23.5413919 | 2%       | 0.586808% |
| 24.2 | 0.300756084 | -0.000201  | 0.002100%  | -0.000219 | 0.300949189 | 0.0642%       | 1605.5929 | 1609        | 000000100100100100 | 0.301678293 | 23.6201395 | 2%       | 0.579860% |
| 24.3 | 0.307576642 | -0.000200  | 0.002200%  | -0.000188 | 0.307776791 | 0.0651%       | 1642.0188 | 1644        | 000000100110110100 | 0.308240593 | 23.6988871 | 2%       | 0.601113% |
| 24.4 | 0.315363091 | -0.000200  | 0.002200%  | -0.000158 | 0.31557124  | 0.0660%       | 1683.6029 | 1688        | 000000101000110000 | 0.316490341 | 23.7978841 | 2%       | 0.602116% |
| 24.5 | 0.32238425  | -0.000199  | 0.002300%  | -0.000127 | 0.32245448  | 0.0670%       | 1720.3257 | 1721        | 000000101010110001 | 0.322677652 | 23.8721318 | 3%       | 0.627868% |
| 24.6 | 0.329141548 | -0.000199  | 0.002300%  | -0.000097 | 0.329366605 | 0.0684%       | 1757.2025 | 1760        | 000000110111000000 | 0.329989929 | 23.9598792 | 3%       | 0.640121% |
| 24.7 | 0.337007873 | -0.000198  | 0.002400%  | -0.000066 | 0.337214619 | 0.0694%       | 1799.2165 | 1802        | 000000110000010100 | 0.337864689 | 24.0543763 | 3%       | 0.64562%  |
| 24.8 | 0.34395653  | -0.000197  | 0.002400%  | -0.000036 | 0.344208241 | 0.0705%       | 1836.3841 | 1838        | 000000110010110110 | 0.344614483 | 24.1353738 | 3%       | 0.66462%  |
| 24.9 | 0.350951106 | -0.000197  | 0.002500%  | -0.000005 | 0.351204351 | 0.0722%       | 1873.709  | 1877        | 000000110101010101 | 0.35192676  | 24.2231211 | 3%       | 0.676879% |
| 25   | 0.358897075 | -0.000196  | 0.002500%  | 0.000025  | 0.359160052 | 0.0733%       | 1916.1534 | 1921        | 000000111100000001 | 0.360176508 | 24.3221181 | 3%       | 0.677882% |
| 25.1 | 0.365937033 | -0.000196  | 0.002600%  | 0.000056  | 0.366211538 | 0.0750%       | 1953.7738 | 1954        | 000000111010000100 | 0.366363819 | 24.3963658 | 3%       | 0.70363%  |
| 25.2 | 0.373004635 | -0.000195  | 0.002600%  | 0.000086  | 0.373289733 | 0.0764%       | 1991.5366 | 1993        | 000000111100010001 | 0.373676096 | 24.4841132 | 3%       | 0.715887% |
| 25.3 | 0.381030001 | -0.000195  | 0.002700%  | 0.000117  | 0.381327609 | 0.0781%       | 2034.4195 | 2035        | 000000111111000111 | 0.381550856 | 24.5786103 | 3%       | 0.721390% |
| 25.4 | 0.388151924 | -0.000194  | 0.002700%  | 0.000147  | 0.388461019 | 0.0796%       | 2072.4769 | 2071        | 000001000000010111 | 0.38830065  | 24.6596078 | 3%       | 0.740392% |
| 25.5 | 0.395301412 | -0.000193  | 0.002800%  | 0.000178  | 0.395622822 | 0.0813%       | 2110.6858 | 2115        | 000001000010000011 | 0.396550398 | 24.7586048 | 3%       | 0.741395% |
| 25.6 | 0.402478435 | -0.000193  | 0.002800%  | 0.000208  | 0.402813208 | 0.0832%       | 2149.0472 | 2148        | 000001000011001000 | 0.402737709 | 24.8328525 | 3%       | 0.767147% |
| 25.7 | 0.410609553 | -0.000192  | 0.002900%  | 0.000239  | 0.410957758 | 0.0848%       | 2192.4992 | 2191        | 000001000010001111 | 0.410799963 | 24.9295996 | 3%       | 0.770400% |
| 25.8 | 0.417840671 | -0.000192  | 0.002900%  | 0.000269  | 0.418203152 | 0.0868%       | 2231.154  | 2230        | 000001000010101100 | 0.41811224  | 25.0173469 | 3%       | 0.782653% |
| 25.9 | 0.42509924  | -0.000191  | 0.003000%  | 0.000300  | 0.425457866 | 0.0886%       | 2269.9546 | 2275        | 000001000011000011 | 0.426549482 | 25.1185938 | 3%       | 0.781406% |
| 26   | 0.432385229 | -0.000191  | 0.003000%  | 0.000330  | 0.432777706 | 0.0906%       | 2308.907  | 2310        | 000000100100000100 | 0.433111782 | 25.1973414 | 3%       | 0.802659% |
| 26.1 | 0.43969861  | -0.000190  | 0.003100%  | 0.000361  | 0.440105467 | 0.0925%       | 2348.005  | 2349        | 000000100101001010 | 0.440236565 | 25.2828388 | 3%       | 0.817161% |
| 26.2 | 0.447961449 | -0.000189  | 0.003100%  | 0.000391  | 0.448383744 | 0.0943%       | 2392.1704 | 2395        | 000000100101010111 | 0.449048796 | 25.3858585 | 3%       | 0.81414%  |
| 26.3 | 0.455328618 | -0.000189  | 0.003200%  | 0.000422  | 0.455767927 | 0.0965%       | 2431.5657 | 2434        | 000001001001000010 | 0.456361073 | 25.4763239 | 3%       | 0.823667% |
| 26.4 | 0.462732084 | -0.000188  | 0.003200%  | 0.000452  | 0.463178415 | 0.0984%       | 2471.1014 | 2474        | 000001001010101010 | 0.463860844 | 25.5663001 | 3%       | 0.83607%  |
| 26.5 | 0.470144    |            |            |           |             |               |           |             |                    |             |            |          |           |

|  |      |             |           |           |          |             |         |           |      |                    |             |            |    |          |  |
|--|------|-------------|-----------|-----------|----------|-------------|---------|-----------|------|--------------------|-------------|------------|----|----------|--|
|  | 28.1 | 0.591668042 | -0.000178 | 0.004100% | 0.000971 | 0.592479283 | 0.1371% | 3160.9339 | 3163 | 0000110001011011   | 0.593044401 | 27.1165328 | 3% | 0.983467 |  |
|  | 28.2 | 0.598645126 | -0.000178 | 0.004100% | 0.001001 | 0.599481148 | 0.1397% | 3198.2896 | 3199 | 0000110001111111   | 0.599794195 | 27.1975303 | 4% | 1.002470 |  |
|  | 28.3 | 0.606553484 | -0.000177 | 0.004200% | 0.001032 | 0.607415277 | 0.1421% | 3240.6189 | 3240 | 0000110010101000   | 0.607481461 | 27.2897775 | 4% | 1.010222 |  |
|  | 28.4 | 0.614848409 | -0.000177 | 0.004200% | 0.001062 | 0.615376878 | 0.1446% | 3283.0948 | 3287 | 0000110011010111   | 0.616293692 | 27.3955423 | 4% | 1.004476 |  |
|  | 28.5 | 0.622449859 | -0.000176 | 0.004300% | 0.001093 | 0.623365134 | 0.1470% | 3325.7129 | 3327 | 0000110011111111   | 0.623793463 | 27.4855216 | 4% | 1.014478 |  |
|  | 28.6 | 0.630437792 | -0.000176 | 0.004300% | 0.001123 | 0.631380767 | 0.1496% | 3368.4771 | 3373 | 0000110100101101   | 0.6324182   | 27.5890184 | 4% | 1.010982 |  |
|  | 28.7 | 0.637553119 | -0.000175 | 0.004400% | 0.001154 | 0.638522908 | 0.1521% | 3406.5811 | 3408 | 0000110101010000   | 0.638980499 | 27.667766  | 4% | 1.032234 |  |
|  | 28.8 | 0.645594828 | -0.000174 | 0.004400% | 0.001184 | 0.646592325 | 0.1545% | 3449.6322 | 3449 | 0000110101111001   | 0.646667765 | 27.7600132 | 4% | 1.039987 |  |
|  | 28.9 | 0.653662889 | -0.000174 | 0.004500% | 0.001215 | 0.654690268 | 0.1572% | 3492.8355 | 3494 | 0000110110100110   | 0.655105007 | 27.8612601 | 4% | 1.038740 |  |
|  | 29   | 0.661757259 | -0.000173 | 0.004500% | 0.001245 | 0.662813339 | 0.1596% | 3536.1732 | 3538 | 0000110111010010   | 0.663354755 | 27.9602571 | 4% | 1.039743 |  |
|  | 29.1 | 0.668962628 | -0.000173 | 0.004600% | 0.001276 | 0.670068561 | 0.1623% | 3574.8802 | 3575 | 0000110111110111   | 0.670292044 | 28.0435045 | 4% | 1.056495 |  |
|  | 29.2 | 0.677130426 | -0.000172 | 0.004600% | 0.001306 | 0.67824614  | 0.1648% | 3618.5084 | 3622 | 0000111000100110   | 0.679104275 | 28.1492513 | 4% | 1.050749 |  |
|  | 29.3 | 0.685304397 | -0.000172 | 0.004700% | 0.001337 | 0.686452106 | 0.1675% | 3662.288  | 3661 | 0000111001001101   | 0.686416552 | 28.2369986 | 4% | 1.063001 |  |
|  | 29.4 | 0.69261208  | -0.000171 | 0.004700% | 0.001367 | 0.693789411 | 0.1700% | 3701.4332 | 3702 | 0000111001110110   | 0.694103817 | 28.3292458 | 4% | 1.070754 |  |
|  | 29.5 | 0.700839208 | -0.000170 | 0.004800% | 0.001398 | 0.702048591 | 0.1726% | 3745.4967 | 3748 | 0000111010100100   | 0.702728554 | 28.4327426 | 4% | 1.067257 |  |
|  | 29.6 | 0.709092369 | -0.000170 | 0.004800% | 0.001428 | 0.710334644 | 0.1752% | 3789.7036 | 3789 | 0000111011001101   | 0.710415819 | 28.5249898 | 4% | 1.075010 |  |
|  | 29.7 | 0.716481964 | -0.000169 | 0.004900% | 0.001459 | 0.717756086 | 0.1778% | 3829.2977 | 3829 | 0000111011110101   | 0.717915559 | 28.6149871 | 4% | 1.085013 |  |
|  | 29.8 | 0.724788002 | -0.000169 | 0.004900% | 0.001489 | 0.726096073 | 0.1805% | 3873.7924 | 3873 | 0000111100100001   | 0.726165339 | 28.7139841 | 4% | 1.086016 |  |
|  | 29.9 | 0.732232296 | -0.000168 | 0.005000% | 0.001520 | 0.733573239 | 0.1831% | 3913.6838 | 3914 | 0000111101010010   | 0.733852604 | 28.8062312 | 4% | 1.093769 |  |
|  | 30   | 0.740591019 | -0.000168 | 0.005000% | 0.001550 | 0.741966978 | 0.1858% | 3956.4652 | 3963 | 0000111101110111   | 0.743039824 | 28.9164779 | 4% | 1.083522 |  |
|  | 30.1 | 0.748089823 | -0.000167 | 0.005100% | 0.001581 | 0.74949969  | 0.1885% | 3999.6529 | 4000 | 0000111110100000   | 0.749977112 | 28.9997253 | 4% | 1.100275 |  |
|  | 30.2 | 0.755616293 | -0.000166 | 0.005100% | 0.001611 | 0.757059823 | 0.1910% | 4038.9869 | 4039 | 0000111111000011   | 0.757289389 | 29.0874727 | 4% | 1.112527 |  |
|  | 30.3 | 0.76405416  | -0.000166 | 0.005200% | 0.001642 | 0.765535073 | 0.1938% | 4084.2032 | 4085 | 0000111111110101   | 0.765914125 | 29.1909695 | 4% | 1.109030 |  |
|  | 30.4 | 0.771634854 | -0.000165 | 0.005200% | 0.001672 | 0.773150464 | 0.1964% | 4124.8321 | 4123 | 0001000000001011   | 0.773038908 | 29.2764669 | 4% | 1.123533 |  |
|  | 30.5 | 0.779243069 | -0.000165 | 0.005300% | 0.001703 | 0.780795666 | 0.1992% | 4165.6199 | 4166 | 0001000001000110   | 0.781101162 | 29.3732139 | 4% | 1.126786 |  |
|  | 30.6 | 0.787759631 | -0.000164 | 0.005300% | 0.001733 | 0.789349473 | 0.2018% | 4211.2553 | 4210 | 0001000001110010   | 0.78935091  | 29.4722109 | 4% | 1.127789 |  |
|  | 30.7 | 0.795421774 | -0.000163 | 0.005400% | 0.001764 | 0.797048665 | 0.2045% | 4252.3313 | 4251 | 0001000001001101   | 0.797038176 | 29.5644581 | 4% | 1.135542 |  |
|  | 30.8 | 0.803990219 | -0.000163 | 0.005400% | 0.001794 | 0.805656463 | 0.2072% | 4298.2547 | 4302 | 0001000001100110   | 0.806600384 | 29.6792046 | 4% | 1.120795 |  |
|  | 30.9 | 0.811706085 | -0.000162 | 0.005500% | 0.001825 | 0.813410453 | 0.2100% | 4339.623  | 4338 | 0001000001110010   | 0.813350178 | 29.7602021 | 4% | 1.139798 |  |
|  | 31   | 0.820326202 | -0.000162 | 0.005500% | 0.001855 | 0.822071036 | 0.2127% | 4385.828  | 4388 | 0001000100100100   | 0.822724892 | 29.8726987 | 4% | 1.127301 |  |
|  | 31.1 | 0.828995583 | -0.000161 | 0.005600% | 0.001886 | 0.829879626 | 0.2154% | 4427.4876 | 4432 | 0001000101010000   | 0.83097464  | 29.9716957 | 4% | 1.128304 |  |
|  | 31.2 | 0.835892131 | -0.000161 | 0.005600% | 0.001916 | 0.837716084 | 0.2182% | 4469.2959 | 4468 | 0001000101110100   | 0.837724434 | 30.0526932 | 4% | 1.147307 |  |
|  | 31.3 | 0.844589884 | -0.000160 | 0.005700% | 0.001947 | 0.846455777 | 0.2209% | 4515.923  | 4517 | 0001000110100101   | 0.846911654 | 30.1629398 | 4% | 1.137060 |  |
|  | 31.4 | 0.852439586 | -0.000159 | 0.005700% | 0.001977 | 0.854345512 | 0.2236% | 4558.0154 | 4559 | 0001000111001111   | 0.854786413 | 30.257437  | 4% | 1.142563 |  |
|  | 31.5 | 0.860316339 | -0.000159 | 0.005800% | 0.002008 | 0.86226465  | 0.2265% | 4600.2648 | 4599 | 0001000111110111   | 0.862286184 | 30.3474342 | 4% | 1.152566 |  |
|  | 31.6 | 0.869692248 | -0.000158 | 0.005800% | 0.002038 | 0.871955175 | 0.2291% | 4651.9647 | 4657 | 00010001000110001  | 0.873160852 | 30.4779302 | 4% | 1.122070 |  |
|  | 31.7 | 0.877890886 | -0.000158 | 0.005900% | 0.002069 | 0.879927336 | 0.2320% | 4694.4969 | 4697 | 00010001001011001  | 0.880660624 | 30.5679275 | 4% | 1.132073 |  |
|  | 31.8 | 0.885846367 | -0.000157 | 0.005900% | 0.002099 | 0.887925    | 0.2346% | 4737.1652 | 4741 | 00010001010000101  | 0.888910372 | 30.6669245 | 4% | 1.133076 |  |
|  | 31.9 | 0.893828637 | -0.000157 | 0.006000% | 0.002130 | 0.895951891 | 0.2375% | 4779.9895 | 4783 | 0001000101010111   | 0.896785131 | 30.7614216 | 4% | 1.138578 |  |
|  | 32   | 0.902704792 | -0.000156 | 0.006000% | 0.002160 | 0.904873193 | 0.2402% | 4827.5855 | 4828 | 00010001011011100  | 0.905222374 | 30.8626685 | 4% | 1.137332 |  |
|  | 32.1 | 0.910739479 | -0.000155 | 0.006100% | 0.002191 | 0.912952631 | 0.2430% | 4870.6901 | 4873 | 000100011000001001 | 0.913659616 | 30.9639154 | 4% | 1.136085 |  |
|  | 32.2 | 0.918000785 | -0.000155 | 0.006100% | 0.002221 | 0.921059329 | 0.2458% | 4913.9401 | 4915 | 00010001100110011  | 0.921534376 | 31.0584125 | 4% | 1.141587 |  |
|  | 32.3 | 0.926888652 | -0.000154 | 0.006200% | 0.002252 | 0.929193064 | 0.2486% | 4957.3343 | 4958 | 0001000110101110   | 0.92959663  | 31.1551596 | 4% | 1.144840 |  |
|  | 32.4 | 0.935003022 | -0.000154 | 0.006200% | 0.002282 | 0.937353931 | 0.2514% | 5000.8733 | 5002 | 00010001110001010  | 0.937846378 | 31.2541565 | 4% | 1.145843 |  |
|  | 32.5 | 0.944060625 | -0.000153 | 0.006300% | 0.002313 | 0.946405927 | 0.2542% | 5049.1666 | 5050 | 00010001101101010  | 0.946846104 | 31.3621532 | 4% | 1.137847 |  |
|  | 32.6 | 0.952172229 | -0.000153 | 0.006300% | 0.002343 | 0.954619768 | 0.2570% | 5092.9882 | 5095 | 00010001111100111  | 0.955283346 | 31.4634002 | 3% | 1.136600 |  |
|  | 32.7 | 0.959504569 | -0.000152 | 0.006400% | 0.002374 | 0.961998215 | 0.2599% | 5132.353  | 5134 | 0001010000000110   | 0.962595623 | 31.5511475 | 4% | 1.148853 |  |
|  | 32.8 | 0.967724366 | -0.000151 | 0.006400% | 0.002404 | 0.970265746 | 0.2626% | 5176.461  | 5179 | 0001010000011011   | 0.971032866 | 31.6523944 | 3% | 1.147606 |  |
|  | 32.9 | 0.975970371 | -0.000151 | 0.006500% | 0.002435 | 0.978562028 | 0.2655% | 5220.7225 | 5223 | 0001010001100111   | 0.979282614 | 31.7513914 | 3% | 1.148609 |  |
|  | 33   | 0.984242523 | -0.000150 | 0.006500% | 0.002465 | 0.986883047 | 0.2683% | 5265.1159 | 5269 | 0001010010010101   | 0.987907351 | 31.8548882 | 3% | 1.145112 |  |
|  | 33.1 | 0.992540761 | -0.000150 | 0.006600% | 0.002496 | 0.995232717 | 0.2712% | 5309.6622 | 5308 | 0001010010111100   | 0.995219627 | 31.9426355 | 3% | 1.157364 |  |
|  | 33.2 | 1.000865022 | -0.000149 | 0.006600% | 0.002526 | 1.003606987 | 0.2740% | 5354.3398 | 5358 | 0001010011101110   | 1.004594341 | 32.0551321 | 3% | 1.144868 |  |
|  | 33.3 | 1.009215245 | -0.000149 | 0.006700% | 0.002557 | 1.012009806 | 0.2769% | 5399.1696 | 5401 | 00010101000011001  | 1.012656595 | 32.1518791 | 3% | 1.148121 |  |
|  | 33.4 | 1.017591368 | -0.000148 | 0.006700% | 0.002587 | 1.020437089 | 0.2797% | 5444.13   | 5445 | 00010101010000101  | 1.020906343 | 32.2508761 | 3% | 1.149124 |  |
|  | 33.5 | 1.025993327 | -0.000147 | 0.006800% | 0.002618 | 1.028891814 | 0.2825% | 5489.2368 | 5489 | 0001010101110001   | 1.029156092 | 32.3498731 | 3% | 1.150127 |  |
|  | 33.6 | 1.034421058 | -0.000147 | 0.006800% | 0.002648 | 1.037372868 | 0.2854% | 5534.484  | 5535 | 00010101100110111  | 1.037780828 | 32.4533699 | 3% | 1.146630 |  |
|  | 33.7 | 1.042874498 | -0.000146 | 0.006900% | 0.002679 | 1.045880256 | 0.2882% | 5579.8717 | 5584 | 0001010111010000   | 1.046968048 | 32.5636166 | 3% | 1.136383 |  |
|  | 33.8 | 1.051353582 | -0.000146 | 0.006900% | 0.002709 | 1.054413832 | 0.2911% | 5625.3992 | 5625 | 00010101111111001  | 1.054655313 | 32.6558638 | 3% | 1.144136 |  |
|  | 33.9 | 1.059858245 | -0.000145 | 0.007000% | 0.002740 | 1.062973634 | 0.2939% | 5671.0665 | 5673 | 00010101000101001  | 1.063655039 | 32.7638605 | 3% | 1.136140 |  |
|  | 34   | 1.067541328 | -0.000145 | 0.007000% | 0.002770 | 1.070710035 | 0.2968% | 5712.341  | 5717 | 0001011001010101   | 1.071904787 | 32.8628574 | 3% | 1.137143 |  |
|  | 34.1 | 1.076097968 | -0.000144 | 0.007100% | 0.002801 | 1.079322989 | 0.2997% | 5758.2919 | 5761 | 00010110100000001  | 1.080154535 | 32.9618544 | 3% | 1.138146 |  |
|  | 34.2 | 1.084679989 | -0.000143 | 0.007100% | 0.002831 | 1.087960842 | 0.3025% | 5804.3757 | 5803 | 0001011010101011   | 1.088029295 | 33.0563515 | 3% | 1.143648 |  |
|  | 34.3 | 1.093287325 | -0.000143 | 0.007200% | 0.002862 | 1.096626704 | 0.3054% | 5850.6089 | 5854 | 0001011010110110   | 1.097591503 | 33.17      |    |          |  |

|  |      |             |           |           |          |             |         |           |      |                   |             |            |    |          |  |
|--|------|-------------|-----------|-----------|----------|-------------|---------|-----------|------|-------------------|-------------|------------|----|----------|--|
|  | 36.3 | 1.263232066 | -0.000131 | 0.008200% | 0.003472 | 1.267822745 | 0.3634% | 6763.9562 | 6768 | 0001101001110000  | 1.268961273 | 35.2275353 | 3% | 1.072465 |  |
|  | 36.4 | 1.272364096 | -0.000131 | 0.008200% | 0.003502 | 1.277025034 | 0.3663% | 6811.0513 | 6817 | 0001101010100001  | 1.278148493 | 35.3377819 | 3% | 1.062218 |  |
|  | 36.5 | 1.280698525 | -0.000130 | 0.008300% | 0.003533 | 1.285427736 | 0.3693% | 6857.8806 | 6856 | 0001101011001000  | 1.28546077  | 35.4255292 | 3% | 1.074471 |  |
|  | 36.6 | 1.289058699 | -0.000130 | 0.008300% | 0.003563 | 1.293856503 | 0.3722% | 6902.8488 | 6901 | 0001101011110101  | 1.293898012 | 35.5267761 | 3% | 1.073224 |  |
|  | 36.7 | 1.298263789 | -0.000129 | 0.008400% | 0.003594 | 1.303134044 | 0.3751% | 6952.3454 | 6952 | 0001101100101000  | 1.30346022  | 35.6415226 | 3% | 1.058477 |  |
|  | 36.8 | 1.306674182 | -0.000128 | 0.008400% | 0.003624 | 1.311613258 | 0.3780% | 6997.5828 | 6998 | 0001101101010110  | 1.312084957 | 35.7450195 | 3% | 1.054981 |  |
|  | 36.9 | 1.315110094 | -0.000128 | 0.008500% | 0.003655 | 1.320120924 | 0.3810% | 7042.9721 | 7042 | 0001101110000010  | 1.320334705 | 35.8440165 | 3% | 1.055984 |  |
|  | 37   | 1.324387549 | -0.000127 | 0.008500% | 0.003685 | 1.329471437 | 0.3839% | 7092.8579 | 7092 | 0001101110110100  | 1.329709419 | 35.956513  | 3% | 1.043487 |  |
|  | 37.1 | 1.332873218 | -0.000127 | 0.008600% | 0.003716 | 1.338030123 | 0.3869% | 7138.5194 | 7140 | 0001101111100100  | 1.338709145 | 36.0645097 | 3% | 1.035490 |  |
|  | 37.2 | 1.340570175 | -0.000126 | 0.008600% | 0.003746 | 1.345795307 | 0.3898% | 7179.9474 | 7182 | 0001110000001110  | 1.346583904 | 36.1590069 | 3% | 1.040993 |  |
|  | 37.3 | 1.349107395 | -0.000126 | 0.008700% | 0.003777 | 1.35440677  | 0.3928% | 7225.8903 | 7229 | 0001110000111101  | 1.355396135 | 36.2647536 | 3% | 1.035246 |  |
|  | 37.4 | 1.358481641 | -0.000125 | 0.008700% | 0.003807 | 1.363856623 | 0.3957% | 7276.3062 | 7278 | 0001110001101110  | 1.364583355 | 36.3750003 | 3% | 1.025000 |  |
|  | 37.5 | 1.367067994 | -0.000124 | 0.008800% | 0.003838 | 1.372517493 | 0.3986% | 7322.5128 | 7324 | 0001110010011100  | 1.373208092 | 36.4784971 | 3% | 1.021503 |  |
|  | 37.6 | 1.375679321 | -0.000124 | 0.008800% | 0.003868 | 1.381203592 | 0.4016% | 7368.854  | 7368 | 0001110011001000  | 1.38145784  | 36.5774941 | 3% | 1.022506 |  |
|  | 37.7 | 1.384315544 | -0.000123 | 0.008900% | 0.003899 | 1.389915599 | 0.4045% | 7415.3334 | 7416 | 0001110011111000  | 1.390457565 | 36.6854908 | 3% | 1.014509 |  |
|  | 37.8 | 1.393784257 | -0.000123 | 0.008900% | 0.003929 | 1.399463522 | 0.4075% | 7466.2724 | 7470 | 0001110100101110  | 1.400582256 | 36.8069871 | 3% | 0.993013 |  |
|  | 37.9 | 1.402468973 | -0.000122 | 0.009000% | 0.003960 | 1.408225316 | 0.4104% | 7513.0175 | 7513 | 0001110101011001  | 1.40864451  | 36.9037341 | 3% | 0.996266 |  |
|  | 38   | 1.411178344 | -0.000122 | 0.009000% | 0.003990 | 1.417011199 | 0.4134% | 7559.8952 | 7562 | 0001110110001010  | 1.41783173  | 37.0139808 | 3% | 0.986019 |  |
|  | 38.1 | 1.419912286 | -0.000121 | 0.009100% | 0.004021 | 1.425824293 | 0.4164% | 7606.9097 | 7608 | 0001110110111000  | 1.426456467 | 37.1174776 | 3% | 0.982522 |  |
|  | 38.2 | 1.429474163 | -0.000120 | 0.009100% | 0.004051 | 1.435467005 | 0.4192% | 7656.3545 | 7662 | 0001110111101110  | 1.436581158 | 37.2389739 | 3% | 0.961026 |  |
|  | 38.3 | 1.438255946 | -0.000120 | 0.009200% | 0.004082 | 1.444329462 | 0.4223% | 7705.6365 | 7707 | 0001110000101011  | 1.4450184   | 37.3402208 | 3% | 0.959779 |  |
|  | 38.4 | 1.447062055 | -0.000119 | 0.009200% | 0.004112 | 1.453214435 | 0.4252% | 7753.0387 | 7754 | 0001110010010100  | 1.453830631 | 37.4459676 | 2% | 0.954032 |  |
|  | 38.5 | 1.455892409 | -0.000119 | 0.009300% | 0.004143 | 1.46212676  | 0.4282% | 7800.5868 | 7805 | 0001110011111101  | 1.463392839 | 37.5607141 | 2% | 0.939286 |  |
|  | 38.6 | 1.464746923 | -0.000118 | 0.009300% | 0.004173 | 1.47106142  | 0.4311% | 7848.2541 | 7853 | 0001111010101011  | 1.472392565 | 37.6687108 | 2% | 0.931289 |  |
|  | 38.7 | 1.472827387 | -0.000117 | 0.009400% | 0.004204 | 1.479220788 | 0.4341% | 7891.7851 | 7891 | 0001111011010011  | 1.479517347 | 37.7542082 | 2% | 0.945792 |  |
|  | 38.8 | 1.481731037 | -0.000117 | 0.009400% | 0.004234 | 1.488206859 | 0.4370% | 7939.7267 | 7938 | 0001111100000010  | 1.488329578 | 37.8599459 | 2% | 0.940045 |  |
|  | 38.9 | 1.490658597 | -0.000116 | 0.009500% | 0.004265 | 1.497217984 | 0.4400% | 7987.8019 | 7991 | 0001111100110111  | 1.498266775 | 37.9792013 | 2% | 0.920799 |  |
|  | 39   | 1.499609982 | -0.000116 | 0.009500% | 0.004295 | 1.50625309  | 0.4430% | 8036.0051 | 8036 | 0001111101100100  | 1.506704018 | 38.0804482 | 2% | 0.919552 |  |
|  | 39.1 | 1.508585107 | -0.000115 | 0.009600% | 0.004326 | 1.515313101 | 0.4460% | 8084.3411 | 8089 | 0001111110011001  | 1.516641214 | 38.1999646 | 2% | 0.900305 |  |
|  | 39.2 | 1.516791096 | -0.000115 | 0.009600% | 0.004356 | 1.523606046 | 0.4489% | 8128.5559 | 8131 | 0001111111000011  | 1.524515974 | 38.2941917 | 2% | 0.905808 |  |
|  | 39.3 | 1.525814514 | -0.000114 | 0.009700% | 0.004387 | 1.532710259 | 0.4519% | 8177.1566 | 8179 | 0001111111110011  | 1.533515699 | 38.4021884 | 2% | 0.897812 |  |
|  | 39.4 | 1.534070765 | -0.000113 | 0.009700% | 0.004417 | 1.54104832  | 0.4548% | 8221.641  | 8223 | 0010000000001111  | 1.541765448 | 38.5011854 | 2% | 0.898815 |  |
|  | 39.5 | 1.543142133 | -0.000113 | 0.009800% | 0.004448 | 1.550208201 | 0.4579% | 8270.5098 | 8271 | 0010000001001111  | 1.550765173 | 38.6091821 | 2% | 0.890818 |  |
|  | 39.6 | 1.55223681  | -0.000112 | 0.009800% | 0.004478 | 1.559389496 | 0.4608% | 8319.4929 | 8320 | 0010000001000000  | 1.559952393 | 38.7194287 | 2% | 0.880571 |  |
|  | 39.7 | 1.56135471  | -0.000112 | 0.009900% | 0.004509 | 1.568597253 | 0.4639% | 8368.6172 | 8367 | 0010000001010111  | 1.568764624 | 38.8251755 | 2% | 0.874825 |  |
|  | 39.8 | 1.569709386 | -0.000111 | 0.009900% | 0.004539 | 1.577036291 | 0.4668% | 8413.6402 | 8417 | 0010000001100001  | 1.578139338 | 38.9376721 | 2% | 0.862328 |  |
|  | 39.9 | 1.57887454  | -0.000111 | 0.010000% | 0.004570 | 1.586292633 | 0.4698% | 8463.0237 | 8468 | 0010000100001000  | 1.587701546 | 39.0524185 | 2% | 0.847581 |  |
|  | 40   | 1.588062653 | -0.000110 | 0.010000% | 0.004600 | 1.595570009 | 0.4727% | 8512.5194 | 8516 | 0010000010000100  | 1.596701271 | 39.1604153 | 2% | 0.839585 |  |
|  | 40.1 | 1.596490504 | -0.000109 | 0.010100% | 0.004631 | 1.604085757 | 0.4757% | 8557.9517 | 8557 | 0010000010110101  | 1.604388536 | 39.2526624 | 2% | 0.847338 |  |
|  | 40.2 | 1.605725342 | -0.000109 | 0.010100% | 0.004661 | 1.613412186 | 0.4787% | 8607.7091 | 8606 | 0010000110011110  | 1.613575756 | 39.3629091 | 2% | 0.837091 |  |
|  | 40.3 | 1.614982872 | -0.000108 | 0.010200% | 0.004692 | 1.622762602 | 0.4817% | 8657.5945 | 8656 | 0010000111010000  | 1.62295047  | 39.4754056 | 2% | 0.824584 |  |
|  | 40.4 | 1.623483103 | -0.000108 | 0.010200% | 0.004722 | 1.631352073 | 0.4847% | 8703.4202 | 8708 | 00100010000000100 | 1.632700172 | 39.5924021 | 2% | 0.807598 |  |
|  | 40.5 | 1.632786822 | -0.000107 | 0.010300% | 0.004753 | 1.640750031 | 0.4877% | 8753.5592 | 8754 | 0010001000110010  | 1.641324909 | 39.6958989 | 2% | 0.804101 |  |
|  | 40.6 | 1.641335221 | -0.000107 | 0.010300% | 0.004783 | 1.649388971 | 0.4907% | 8799.6487 | 8798 | 0010001001011110  | 1.649574657 | 39.7948959 | 2% | 0.805104 |  |
|  | 40.7 | 1.650684768 | -0.000106 | 0.010400% | 0.004814 | 1.658834117 | 0.4937% | 8850.0395 | 8851 | 0010001010001011  | 1.659511854 | 39.9141423 | 2% | 0.785858 |  |
|  | 40.8 | 1.659280977 | -0.000105 | 0.010400% | 0.004844 | 1.667521168 | 0.4966% | 8896.3858 | 8901 | 0010001011000101  | 1.668886568 | 40.0266388 | 2% | 0.773361 |  |
|  | 40.9 | 1.668675989 | -0.000105 | 0.010500% | 0.004875 | 1.677014151 | 0.4997% | 8947.0317 | 8947 | 0010001011100111  | 1.677511305 | 40.1301357 | 2% | 0.769864 |  |
|  | 41   | 1.677319644 | -0.000104 | 0.010500% | 0.004905 | 1.685749963 | 0.5026% | 8993.6381 | 8996 | 0010001100100100  | 1.686698524 | 40.2403823 | 2% | 0.759618 |  |
|  | 41.1 | 1.686759755 | -0.000104 | 0.010600% | 0.004936 | 1.695289416 | 0.5057% | 9044.532  | 9043 | 0010001101010011  | 1.695510756 | 40.3461291 | 2% | 0.753871 |  |
|  | 41.2 | 1.694679247 | -0.000103 | 0.010600% | 0.004966 | 1.70329854  | 0.5086% | 9087.2615 | 9087 | 0010001101111111  | 1.703760504 | 40.4445126 | 2% | 0.754874 |  |
|  | 41.3 | 1.703395014 | -0.000103 | 0.010700% | 0.004997 | 1.712111143 | 0.5117% | 9134.2776 | 9139 | 0010001101100111  | 1.713510206 | 40.5621225 | 2% | 0.737878 |  |
|  | 41.4 | 1.712903706 | -0.000102 | 0.010700% | 0.005027 | 1.721718493 | 0.5146% | 9185.5337 | 9190 | 0010001111100110  | 1.723072414 | 40.676869  | 2% | 0.723131 |  |
|  | 41.5 | 1.721666002 | -0.000101 | 0.010800% | 0.005058 | 1.730577972 | 0.5176% | 9232.7999 | 9234 | 0010010000010010  | 1.731322163 | 40.775866  | 2% | 0.724134 |  |
|  | 41.6 | 1.730451967 | -0.000101 | 0.010800% | 0.005088 | 1.73946105  | 0.5206% | 9280.1919 | 9279 | 0010010000111111  | 1.739759405 | 40.8771129 | 2% | 0.722887 |  |
|  | 41.7 | 1.740027319 | -0.000100 | 0.010900% | 0.005119 | 1.749138864 | 0.5236% | 9331.824  | 9333 | 0010010001110101  | 1.749884096 | 40.9986092 | 2% | 0.701391 |  |
|  | 41.8 | 1.748859255 | -0.000100 | 0.010900% | 0.005149 | 1.758069255 | 0.5266% | 9379.4685 | 9382 | 0010010010100110  | 1.759071316 | 41.1088558 | 2% | 0.691144 |  |
|  | 41.9 | 1.758478215 | -0.000099 | 0.011000% | 0.005180 | 1.767792077 | 0.5297% | 9431.3407 | 9432 | 0010010011011000  | 1.76844603  | 41.2213524 | 2% | 0.678648 |  |
|  | 42   | 1.767355744 | -0.000099 | 0.011000% | 0.005210 | 1.77676941  | 0.5326% | 9479.2356 | 9479 | 0010010100000111  | 1.777258261 | 41.3270991 | 2% | 0.672901 |  |
|  | 42.1 | 1.776256472 | -0.000098 | 0.011100% | 0.005241 | 1.785771396 | 0.5357% | 9527.2621 | 9526 | 0010010100101100  | 1.786070492 | 41.4328459 | 2% | 0.667154 |  |
|  | 42.2 | 1.785940674 | -0.000097 | 0.011100% | 0.005271 | 1.795559764 | 0.5386% | 9579.484  | 9584 | 0010010101110000  | 1.79694516  | 41.5633419 | 2% | 0.636658 |  |
|  | 42.3 | 1.794886427 | -0.000097 | 0.011200% | 0.005302 | 1.804609157 | 0.5417% | 9627.7634 | 9632 | 0010010110100000  | 1.805944885 | 41.6713386 | 1% | 0.628661 |  |
|  | 42.4 | 1.803855094 | -0.000096 | 0.011200% | 0.005332 | 1.813679323 | 0.5446% | 9676.1536 | 9675 | 0010010111001011  | 1.814007139 | 41.7680857 | 1% | 0.631914 |  |
|  | 42.5 | 1.812846581 | -0.000096 | 0.011300% | 0.005363 | 1.822775817 | 0.5477% | 9724.6842 | 9728 | 0010011000000000  | 1.823944336 | 41.887332  | 1% | 0.61     |  |

|  |      |             |           |           |          |                     |         |            |       |                    |             |            |    |          |
|--|------|-------------|-----------|-----------|----------|---------------------|---------|------------|-------|--------------------|-------------|------------|----|----------|
|  | 44.5 | 1.994377615 | -0.000084 | 0.012300% | 0.005973 | <b>2.006504117</b>  | 0.6080% | 10704.8924 | 10708 | 0010100111010100   | 2.007688728 | 44.0922647 | 1% | 0.407735 |
|  | 44.6 | 2.003096018 | -0.000084 | 0.012300% | 0.006003 | <b>2.01533573</b>   | 0.6110% | 10752.0099 | 10756 | 0010101000000100   | 2.016688454 | 44.2002614 | 1% | 0.399739 |
|  | 44.7 | 2.011837264 | -0.000083 | 0.012400% | 0.006034 | <b>2.024191888</b>  | 0.6141% | 10799.2583 | 10800 | 0010101000110000   | 2.024938202 | 44.2992584 | 1% | 0.400742 |
|  | 44.8 | 2.020601253 | -0.000082 | 0.012400% | 0.006064 | <b>2.033069435</b>  | 0.6171% | 10846.6209 | 10847 | 0010101001011111   | 2.033750433 | 44.4050052 | 1% | 0.394995 |
|  | 44.9 | 2.030118609 | -0.000082 | 0.012500% | 0.006095 | <b>2.042708564</b>  | 0.6202% | 10896.0466 | 10902 | 0010101010010110   | 2.044062618 | 44.5287514 | 1% | 0.371249 |
|  | 45   | 2.038926672 | -0.000081 | 0.012500% | 0.006125 | <b>2.05163156</b>   | 0.6231% | 10945.6516 | 10945 | 0010101011000001   | 2.052124872 | 44.6254985 | 1% | 0.374502 |
|  | 45.1 | 2.049214211 | -0.000081 | 0.012600% | 0.006156 | <b>2.062046814</b>  | 0.6262% | 11001.218  | 11005 | 00101010111111101  | 2.063374529 | 44.7604943 | 1% | 0.395906 |
|  | 45.2 | 2.058064832 | -0.000080 | 0.012600% | 0.006186 | <b>2.071013765</b>  | 0.6292% | 11049.0576 | 11054 | 00101010110010110  | 2.072561749 | 44.8707471 | 1% | 0.329259 |
|  | 45.3 | 2.067663996 | -0.000080 | 0.012700% | 0.006217 | <b>2.080737581</b>  | 0.6323% | 11100.9351 | 11105 | 001010101101100001 | 2.082123957 | 44.9854875 | 1% | 0.314513 |
|  | 45.4 | 2.076557867 | -0.000079 | 0.012700% | 0.006247 | <b>2.089749168</b>  | 0.6352% | 11149.0127 | 11152 | 00101010110010000  | 2.090936188 | 45.0912343 | 1% | 0.308766 |
|  | 45.5 | 2.085473762 | -0.000078 | 0.012800% | 0.006278 | <b>2.098785573</b>  | 0.6383% | 11197.2228 | 11202 | 0010101111000010   | 2.100310902 | 45.2037308 | 1% | 0.296269 |
|  | 45.6 | 2.095134571 | -0.000078 | 0.012800% | 0.006308 | <b>2.108571072</b>  | 0.6413% | 11249.4294 | 11248 | 00101011111110000  | 2.108935638 | 45.3072277 | 1% | 0.292772 |
|  | 45.7 | 2.104093093 | -0.000077 | 0.012900% | 0.006339 | <b>2.117651519</b>  | 0.6444% | 11297.8745 | 11296 | 0010110000100000   | 2.117935364 | 45.4152244 | 1% | 0.284776 |
|  | 45.8 | 2.113073326 | -0.000077 | 0.012900% | 0.006369 | <b>2.126753248</b>  | 0.6474% | 11346.4331 | 11350 | 0010110001010110   | 2.128060055 | 45.5367207 | 1% | 0.263279 |
|  | 45.9 | 2.122794843 | -0.000076 | 0.013000% | 0.006400 | <b>2.136602729</b>  | 0.6505% | 11398.981  | 11403 | 0010110010001011   | 2.137997252 | 45.655967  | 1% | 0.244033 |
|  | 46   | 2.131817076 | -0.000076 | 0.013000% | 0.006430 | <b>2.145747862</b>  | 0.6535% | 11447.7712 | 11448 | 00101100101110000  | 2.146434494 | 45.7572139 | 1% | 0.242786 |
|  | 46.1 | 2.140860705 | -0.000075 | 0.013100% | 0.006461 | <b>2.1549416245</b> | 0.6565% | 11496.6854 | 11495 | 0010110011100111   | 2.155246725 | 45.8629607 | 1% | 0.237039 |
|  | 46.2 | 2.149925626 | -0.000074 | 0.013100% | 0.006491 | <b>2.164104452</b>  | 0.6595% | 11545.7053 | 11548 | 0010110100001100   | 2.165183922 | 45.9822071 | 0% | 0.217793 |
|  | 46.3 | 2.159011732 | -0.000074 | 0.013200% | 0.006522 | <b>2.173317732</b>  | 0.6626% | 11594.8591 | 11599 | 0010110101001111   | 2.17474613  | 46.0969536 | 0% | 0.203046 |
|  | 46.4 | 2.167404779 | -0.000073 | 0.013200% | 0.006552 | <b>2.181830754</b>  | 0.6656% | 11640.2769 | 11639 | 0010110101101111   | 2.182245901 | 46.1869508 | 0% | 0.213049 |
|  | 46.5 | 2.177247078 | -0.000073 | 0.013300% | 0.006583 | <b>2.191806358</b>  | 0.6687% | 11693.4977 | 11697 | 0010110110110001   | 2.193120569 | 46.3174468 | 0% | 0.182553 |
|  | 46.6 | 2.186396106 | -0.000072 | 0.013300% | 0.006613 | <b>2.201081474</b>  | 0.6717% | 11742.9813 | 11743 | 0010110111011111   | 2.201745306 | 46.4209437 | 0% | 0.179056 |
|  | 46.7 | 2.195565895 | -0.000071 | 0.013400% | 0.006644 | <b>2.210380277</b>  | 0.6747% | 11792.5913 | 11797 | 001011000010101    | 2.211869997 | 46.54244   | 0% | 0.157560 |
|  | 46.8 | 2.20475634  | -0.000071 | 0.013400% | 0.006674 | <b>2.219699225</b>  | 0.6778% | 11842.3088 | 11847 | 0010110010001111   | 2.221244711 | 46.6549365 | 0% | 0.145063 |
|  | 46.9 | 2.213967332 | -0.000070 | 0.013500% | 0.006705 | <b>2.22904066</b>   | 0.6808% | 11892.1462 | 11897 | 0010110011111001   | 2.230619425 | 46.7674331 | 0% | 0.132567 |
|  | 47   | 2.223198765 | -0.000070 | 0.013500% | 0.006735 | <b>2.238402012</b>  | 0.6838% | 11942.0899 | 11946 | 0010111010101010   | 2.239806644 | 46.8776797 | 0% | 0.122320 |
|  | 47.1 | 2.232450531 | -0.000069 | 0.013600% | 0.006766 | <b>2.247785657</b>  | 0.6869% | 11992.1526 | 11997 | 0010111011011011   | 2.249368853 | 46.9924262 | 0% | 0.107574 |
|  | 47.2 | 2.241722524 | -0.000069 | 0.013600% | 0.006796 | <b>2.257188986</b>  | 0.6899% | 12042.3203 | 12047 | 0010111100001111   | 2.258743567 | 47.1049228 | 0% | 0.095077 |
|  | 47.3 | 2.250310476 | -0.000068 | 0.013700% | 0.006827 | <b>2.266905416</b>  | 0.6930% | 12088.8233 | 12088 | 0010111100111000   | 2.266430832 | 47.19717   | 0% | 0.102830 |
|  | 47.4 | 2.259623707 | -0.000067 | 0.013700% | 0.006857 | <b>2.275350387</b>  | 0.6960% | 12139.2131 | 12139 | 0010111101101011   | 2.27599304  | 47.3119165 | 0% | 0.088084 |
|  | 47.5 | 2.268956841 | -0.000067 | 0.013800% | 0.006888 | <b>2.284819277</b>  | 0.6991% | 12189.7305 | 12190 | 0010111110011110   | 2.285555248 | 47.4226663 | 0% | 0.073337 |
|  | 47.6 | 2.278309769 | -0.000066 | 0.013800% | 0.006918 | <b>2.294305373</b>  | 0.7021% | 12240.3398 | 12244 | 0010111111010100   | 2.295679939 | 47.5481593 | 0% | 0.051841 |
|  | 47.7 | 2.288382104 | -0.000066 | 0.013900% | 0.006949 | <b>2.30451981</b>   | 0.7052% | 12294.8348 | 12295 | 0011000000000111   | 2.305242147 | 47.6629058 | 0% | 0.037094 |
|  | 47.8 | 2.297074574 | -0.000065 | 0.013900% | 0.006979 | <b>2.313341983</b>  | 0.7082% | 12341.9019 | 12341 | 0011000000110101   | 2.313866884 | 47.7664026 | 0% | 0.033597 |
|  | 47.9 | 2.306486233 | -0.000065 | 0.014000% | 0.007010 | <b>2.322892301</b>  | 0.7113% | 12392.8538 | 12392 | 0011000001101000   | 2.323429092 | 47.8811491 | 0% | 0.018851 |
|  | 48   | 2.315917251 | -0.000064 | 0.014000% | 0.007040 | <b>2.332459357</b>  | 0.7143% | 12443.8949 | 12446 | 0011000010011110   | 2.333553783 | 48.0026454 | 0% | 0.002645 |
|  | 48.1 | 2.32467224  | -0.000063 | 0.014100% | 0.007071 | <b>2.341348505</b>  | 0.7174% | 12491.3194 | 12494 | 0011000011001110   | 2.342553509 | 48.1106421 | 0% | 0.010642 |
|  | 48.2 | 2.333444859 | -0.000063 | 0.014100% | 0.007101 | <b>2.350258363</b>  | 0.7204% | 12538.8543 | 12538 | 0011000011111010   | 2.350803257 | 48.2096391 | 0% | 0.009639 |
|  | 48.3 | 2.34293925  | -0.000062 | 0.014200% | 0.007132 | <b>2.359898958</b>  | 0.7235% | 12590.2377 | 12595 | 00110000100110011  | 2.361490431 | 48.3378852 | 0% | 0.037885 |
|  | 48.4 | 2.351756885 | -0.000062 | 0.014200% | 0.007162 | <b>2.368842116</b>  | 0.7265% | 12636.0004 | 12641 | 00110000101100001  | 2.370115168 | 48.441382  | 0% | 0.041382 |
|  | 48.5 | 2.361286386 | -0.000061 | 0.014300% | 0.007193 | <b>2.378513618</b>  | 0.7296% | 12688.5988 | 12688 | 00110000110010000  | 2.378927399 | 48.5471288 | 0% | 0.047129 |
|  | 48.6 | 2.370144865 | -0.000061 | 0.014300% | 0.007223 | <b>2.38750842</b>   | 0.7326% | 12737.587  | 12736 | 00110000111000000  | 2.387927124 | 48.6551255 | 0% | 0.055125 |
|  | 48.7 | 2.379712769 | -0.000060 | 0.014400% | 0.007254 | <b>2.397127971</b>  | 0.7357% | 12789.398  | 12788 | 0011000011110100   | 2.397676826 | 48.7721219 | 0% | 0.072122 |
|  | 48.8 | 2.388611651 | -0.000059 | 0.014400% | 0.007284 | <b>2.406255393</b>  | 0.7387% | 12837.6039 | 12836 | 0011001000100100   | 2.406676552 | 48.8801186 | 0% | 0.080119 |
|  | 48.9 | 2.398217515 | -0.000059 | 0.014500% | 0.007315 | <b>2.416007184</b>  | 0.7418% | 12889.6306 | 12891 | 0011001001010111   | 2.416988737 | 49.0038648 | 0% | 0.103865 |
|  | 49   | 2.407156358 | -0.000058 | 0.014500% | 0.007345 | <b>2.425084176</b>  | 0.7448% | 12938.0572 | 12941 | 00110010100010101  | 2.426363451 | 49.1163614 | 0% | 0.116361 |
|  | 49.1 | 2.416799739 | -0.000058 | 0.014600% | 0.007376 | <b>2.434874949</b>  | 0.7479% | 12990.292  | 12990 | 0011001010111110   | 2.435550671 | 49.226608  | 0% | 0.126608 |
|  | 49.2 | 2.425778099 | -0.000057 | 0.014600% | 0.007406 | <b>2.443992881</b>  | 0.7509% | 13038.937  | 13044 | 0011001011110100   | 2.445675362 | 49.3481043 | 0% | 0.148104 |
|  | 49.3 | 2.434776606 | -0.000057 | 0.014700% | 0.007437 | <b>2.453135141</b>  | 0.7540% | 13087.7118 | 13086 | 00110010110001110  | 2.453550121 | 49.4426015 | 0% | 0.142601 |
|  | 49.4 | 2.444475984 | -0.000056 | 0.014700% | 0.007467 | <b>2.462980628</b>  | 0.7570% | 13140.2385 | 13139 | 00110010101010011  | 2.463487318 | 49.5618478 | 0% | 0.161848 |
|  | 49.5 | 2.45351334  | -0.000055 | 0.014800% | 0.007498 | <b>2.472162177</b>  | 0.7601% | 13189.2229 | 13192 | 00110011100010000  | 2.473424515 | 49.6810942 | 0% | 0.181094 |
|  | 49.6 | 2.462570509 | -0.000055 | 0.014800% | 0.007528 | <b>2.481362771</b>  | 0.7631% | 13238.309  | 13238 | 0011001110110110   | 2.482049252 | 49.784591  | 0% | 0.184591 |
|  | 49.7 | 2.470969877 | -0.000054 | 0.014900% | 0.007559 | <b>2.489902613</b>  | 0.7662% | 13283.8698 | 13287 | 0011001111100111   | 2.491236471 | 49.8948377 | 0% | 0.194838 |
|  | 49.8 | 2.480743835 | -0.000054 | 0.014900% | 0.007589 | <b>2.498826576</b>  | 0.7692% | 13336.8151 | 13337 | 0011010000011001   | 2.500611185 | 50.0073342 | 0% | 0.207334 |
|  | 49.9 | 2.490535048 | -0.000053 | 0.015000% | 0.007620 | <b>2.509770054</b>  | 0.7723% | 13389.8645 | 13394 | 0011010001010010   | 2.511298359 | 50.1355803 | 0% | 0.235580 |
|  | 50   | 2.499669234 | -0.000053 | 0.015000% | 0.007650 | <b>2.519050526</b>  | 0.7754% | 13439.3768 | 13444 | 0011010010000100   | 2.520673073 | 50.2480769 | 0% | 0.248077 |
|  | 50.1 | 2.508822671 | -0.000052 | 0.015100% | 0.007681 | <b>2.528352501</b>  | 0.7784% | 13489.0037 | 13494 | 001101001010110    | 2.530047787 | 50.3605734 | 1% | 0.260573 |
|  | 50.2 | 2.517995247 | -0.000051 | 0.015100% | 0.007711 | <b>2.537671786</b>  | 0.7814% | 13538.723  | 13543 | 0011010011100111   | 2.539235006 | 50.4708201 | 1% | 0.270820 |
|  | 50.3 | 2.527186848 | -0.000051 | 0.015200% | 0.007742 | <b>2.547014384</b>  | 0.7846% | 13588.5666 | 13589 | 00110101000010101  | 2.547859743 | 50.5743169 | 1% | 0.274317 |
|  | 50.4 | 2.536397362 | -0.000050 | 0.015200% | 0.007772 | <b>2.556373043</b>  | 0.7876% | 13638.496  | 13643 | 00110101010010011  | 2.557984434 | 50.6958132 | 1% | 0.295813 |
|  | 50.5 | 2.545626676 | -0.000050 | 0.015300% | 0.007803 | <b>2.565754808</b>  | 0.7907% | 13688.5486 | 13689 | 0011010101111001   | 2.566609171 | 50.7993101 | 1% | 0.299310 |
|  | 50.6 | 2.554874677 | -0.000049 | 0.015300% | 0.007833 | <b>2.575152391</b>  | 0.7937% | 13738.6856 | 13737 | 00110101010101001  | 2.575608896 | 50.9073068 | 1% | 0.307307 |
|  | 50.7 | 2.56414125  | -0.000048 |           |          |                     |         |            |       |                    |             |            |    |          |

|  |      |             |           |           |          |             |         |            |       |                   |             |            |    |          |  |
|--|------|-------------|-----------|-----------|----------|-------------|---------|------------|-------|-------------------|-------------|------------|----|----------|--|
|  | 52.7 | 2.748043725 | -0.000037 | 0.016400% | 0.008474 | 2.771624983 | 0.8581% | 14796.8858 | 14789 | 0011100111000101  | 2.772852877 | 53.2742345 | 1% | 0.574235 |  |
|  | 52.8 | 2.757039689 | -0.000036 | 0.016400% | 0.008504 | 2.780780957 | 0.8611% | 14833.7338 | 14840 | 0011100111111000  | 2.782415085 | 53.388981  | 1% | 0.588981 |  |
|  | 52.9 | 2.766053919 | -0.000036 | 0.016500% | 0.008535 | 2.789959573 | 0.8643% | 14884.7034 | 14885 | 0011101000100101  | 2.790852327 | 53.4902279 | 1% | 0.590228 |  |
|  | 53   | 2.775086295 | -0.000035 | 0.016500% | 0.008565 | 2.799153548 | 0.8673% | 14933.7533 | 14935 | 0011101001010111  | 2.800227041 | 53.6027245 | 1% | 0.602724 |  |
|  | 53.1 | 2.784136703 | -0.000035 | 0.016600% | 0.008596 | 2.808369952 | 0.8704% | 14982.9237 | 14986 | 0011101010001010  | 2.809789249 | 53.717471  | 1% | 0.617471 |  |
|  | 53.2 | 2.793843728 | -0.000034 | 0.016600% | 0.008626 | 2.818245412 | 0.8734% | 15035.6102 | 15035 | 0011101010111011  | 2.818976469 | 53.8271716 | 1% | 0.627718 |  |
|  | 53.3 | 2.802928744 | -0.000034 | 0.016700% | 0.008657 | 2.827497891 | 0.8766% | 15084.9731 | 15086 | 0011101011110110  | 2.828538677 | 53.9424641 | 1% | 0.642464 |  |
|  | 53.4 | 2.812031443 | -0.000033 | 0.016700% | 0.008687 | 2.836764916 | 0.8796% | 15134.4136 | 15133 | 0011101100001101  | 2.837350908 | 54.0482109 | 1% | 0.648211 |  |
|  | 53.5 | 2.8217871   | -0.000032 | 0.016800% | 0.008718 | 2.846694073 | 0.8827% | 15187.3866 | 15191 | 0011101101010111  | 2.848225576 | 54.1787069 | 1% | 0.678707 |  |
|  | 53.6 | 2.83092371  | -0.000032 | 0.016800% | 0.008748 | 2.855997469 | 0.8857% | 15237.0211 | 15238 | 0011101110000110  | 2.857037807 | 54.2844537 | 1% | 0.684454 |  |
|  | 53.7 | 2.840077654 | -0.000031 | 0.016900% | 0.008779 | 2.865320796 | 0.8888% | 15286.7619 | 15285 | 0011101110110101  | 2.865850039 | 54.3902005 | 1% | 0.690200 |  |
|  | 53.8 | 2.849248813 | -0.000031 | 0.016900% | 0.008809 | 2.874660184 | 0.8919% | 15336.5885 | 15336 | 0011101111110100  | 2.875412247 | 54.504947  | 1% | 0.704947 |  |
|  | 53.9 | 2.858437073 | -0.000030 | 0.017000% | 0.008840 | 2.884019291 | 0.8950% | 15386.5202 | 15385 | 0011110000011001  | 2.884599466 | 54.6151936 | 1% | 0.715194 |  |
|  | 54   | 2.867642315 | -0.000030 | 0.017000% | 0.008870 | 2.893394207 | 0.8980% | 15436.5363 | 15439 | 0011110001001111  | 2.894724157 | 54.7366899 | 1% | 0.736690 |  |
|  | 54.1 | 2.876864423 | -0.000029 | 0.017100% | 0.008901 | 2.902788628 | 0.9011% | 15486.6564 | 15487 | 0011110001111111  | 2.903723883 | 54.8446866 | 1% | 0.744687 |  |
|  | 54.2 | 2.88610328  | -0.000028 | 0.017100% | 0.008931 | 2.912197595 | 0.9041% | 15536.8542 | 15538 | 0011110010110010  | 2.913286091 | 54.9594331 | 1% | 0.759433 |  |
|  | 54.3 | 2.894105618 | -0.000028 | 0.017200% | 0.008962 | 2.920363395 | 0.9073% | 15580.4195 | 15583 | 0011110011011111  | 2.921723333 | 55.06068   | 1% | 0.760680 |  |
|  | 54.4 | 2.903379823 | -0.000027 | 0.017200% | 0.008992 | 2.929809145 | 0.9103% | 15630.8135 | 15629 | 0011110100000101  | 2.93034807  | 55.1641768 | 1% | 0.764177 |  |
|  | 54.5 | 2.912670425 | -0.000027 | 0.017300% | 0.009023 | 2.939275989 | 0.9134% | 15681.32   | 15682 | 0011110101000010  | 2.940285267 | 55.2834232 | 1% | 0.783423 |  |
|  | 54.6 | 2.921977309 | -0.000026 | 0.017300% | 0.009053 | 2.94875587  | 0.9165% | 15731.8961 | 15731 | 0011110101110011  | 2.949472486 | 55.3936698 | 1% | 0.793670 |  |
|  | 54.7 | 2.930678179 | -0.000025 | 0.017400% | 0.009084 | 2.957627742 | 0.9196% | 15779.2284 | 15784 | 0011110110101000  | 2.959409683 | 55.5129162 | 1% | 0.812916 |  |
|  | 54.8 | 2.940018371 | -0.000025 | 0.017400% | 0.009114 | 2.967143387 | 0.9226% | 15829.9953 | 15834 | 00111101110101010 | 2.968784397 | 55.6254128 | 2% | 0.825413 |  |
|  | 54.9 | 2.949374493 | -0.000024 | 0.017500% | 0.009145 | 2.976677768 | 0.9257% | 15880.8616 | 15881 | 0011110000001001  | 2.977596628 | 55.7311595 | 2% | 0.831160 |  |
|  | 55   | 2.958746427 | -0.000024 | 0.017500% | 0.009175 | 2.986226523 | 0.9288% | 15931.8056 | 15932 | 0011110000111100  | 2.987158836 | 55.845906  | 2% | 0.845906 |  |
|  | 55.1 | 2.968134057 | -0.000023 | 0.017600% | 0.009206 | 2.995793801 | 0.9319% | 15982.848  | 15981 | 0011110011001101  | 2.996346056 | 55.9561527 | 2% | 0.856153 |  |
|  | 55.2 | 2.976920574 | -0.000023 | 0.017600% | 0.009236 | 3.004752937 | 0.9349% | 16030.6458 | 16033 | 0011111010100001  | 3.006095758 | 56.0731491 | 2% | 0.873149 |  |
|  | 55.3 | 2.98634034  | -0.000022 | 0.017700% | 0.009267 | 3.01435382  | 0.9381% | 16081.8675 | 16085 | 0011111011010101  | 3.015845461 | 56.1901455 | 2% | 0.890146 |  |
|  | 55.4 | 2.995160951 | -0.000021 | 0.017700% | 0.009297 | 3.023347473 | 0.9411% | 16129.8495 | 16128 | 0011111100000000  | 3.023907715 | 56.2868926 | 2% | 0.886893 |  |
|  | 55.5 | 3.004612383 | -0.000021 | 0.017800% | 0.009328 | 3.032982503 | 0.9442% | 16181.2533 | 16180 | 0011111100110100  | 3.033657417 | 56.403889  | 2% | 0.903889 |  |
|  | 55.6 | 3.013466616 | -0.000020 | 0.017800% | 0.009358 | 3.042011214 | 0.9472% | 16229.4223 | 16234 | 0011111101101010  | 3.043782108 | 56.5253853 | 2% | 0.925385 |  |
|  | 55.7 | 3.022949244 | -0.000020 | 0.017900% | 0.009389 | 3.051678913 | 0.9504% | 16281.0004 | 16283 | 0011111110010111  | 3.052969328 | 56.6356319 | 2% | 0.935632 |  |
|  | 55.8 | 3.031226505 | -0.000019 | 0.017900% | 0.009419 | 3.060126288 | 0.9534% | 16326.068  | 16331 | 0011111111000101  | 3.061969053 | 56.7436286 | 2% | 0.943629 |  |
|  | 55.9 | 3.040131925 | -0.000019 | 0.018000% | 0.009450 | 3.069212435 | 0.9566% | 16374.5434 | 16375 | 0011111111110111  | 3.070218801 | 56.8426256 | 2% | 0.942626 |  |
|  | 56   | 3.049054171 | -0.000018 | 0.018000% | 0.009480 | 3.078312019 | 0.9596% | 16423.0906 | 16426 | 01000000000101010 | 3.07978101  | 56.9573721 | 2% | 0.957372 |  |
|  | 56.1 | 3.058599965 | -0.000017 | 0.018100% | 0.009511 | 3.088044808 | 0.9627% | 16475.016  | 16480 | 0100000001100000  | 3.089905701 | 57.0788684 | 2% | 0.978868 |  |
|  | 56.2 | 3.067554415 | -0.000017 | 0.018100% | 0.009541 | 3.097179046 | 0.9657% | 16523.748  | 16523 | 01000000010001011 | 3.097967955 | 57.1756155 | 2% | 0.975615 |  |
|  | 56.3 | 3.077129995 | -0.000016 | 0.018200% | 0.009572 | 3.106943099 | 0.9689% | 16575.8402 | 16580 | 01000000011000100 | 3.108655128 | 57.3038615 | 2% | 1.003862 |  |
|  | 56.4 | 3.086116176 | -0.000016 | 0.018200% | 0.009602 | 3.116110512 | 0.9719% | 16624.7492 | 16624 | 01000000011100000 | 3.116904877 | 57.4028585 | 2% | 1.002859 |  |
|  | 56.5 | 3.095118592 | -0.000015 | 0.018300% | 0.009633 | 3.125297019 | 0.9750% | 16673.7601 | 16676 | 01000000100100100 | 3.126654579 | 57.5198549 | 2% | 1.019855 |  |
|  | 56.6 | 3.104137124 | -0.000015 | 0.018300% | 0.009663 | 3.134498219 | 0.9781% | 16722.8494 | 16725 | 01000000101010101 | 3.135841799 | 57.6301016 | 2% | 1.030102 |  |
|  | 56.7 | 3.113171653 | -0.000014 | 0.018400% | 0.009694 | 3.143718296 | 0.9812% | 16772.0394 | 16776 | 01000000100010000 | 3.145404007 | 57.7448481 | 2% | 1.044848 |  |
|  | 56.8 | 3.122222061 | -0.000013 | 0.018400% | 0.009724 | 3.1529518   | 0.9842% | 16821.301  | 16825 | 01000000110111001 | 3.154591227 | 57.8550947 | 2% | 1.050509 |  |
|  | 56.9 | 3.13128823  | -0.000013 | 0.018500% | 0.009755 | 3.162205984 | 0.9874% | 16870.673  | 16871 | 01000000111100111 | 3.163215963 | 57.9585916 | 2% | 1.058592 |  |
|  | 57   | 3.140370042 | -0.000012 | 0.018500% | 0.009785 | 3.171472331 | 0.9904% | 16920.1098 | 16921 | 01000010000011001 | 3.172590677 | 58.0710881 | 2% | 1.071088 |  |
|  | 57.1 | 3.149467378 | -0.000012 | 0.018600% | 0.009816 | 3.178075914 | 0.9936% | 16969.6558 | 16968 | 01000010010010000 | 3.181402908 | 58.1768349 | 2% | 1.076835 |  |
|  | 57.2 | 3.15858012  | -0.000011 | 0.018600% | 0.009846 | 3.190057859 | 0.9966% | 17019.2654 | 17021 | 0100001001111101  | 3.191340105 | 58.2960813 | 2% | 1.096081 |  |
|  | 57.3 | 3.167114368 | -0.000011 | 0.018700% | 0.009877 | 3.198777117 | 0.9997% | 17065.7835 | 17065 | 0100001010101001  | 3.199589853 | 58.3950782 | 2% | 1.095078 |  |
|  | 57.4 | 3.176258652 | -0.000010 | 0.018700% | 0.009907 | 3.208108811 | 1.0028% | 17115.569  | 17119 | 0100001011011111  | 3.209714544 | 58.5165745 | 2% | 1.116575 |  |
|  | 57.5 | 3.184826374 | -0.000009 | 0.018800% | 0.009938 | 3.216861972 | 1.0059% | 17162.2679 | 17167 | 0100001100001111  | 3.21871427  | 58.6245712 | 2% | 1.124571 |  |
|  | 57.6 | 3.194592256 | -0.000009 | 0.018800% | 0.009968 | 3.226823656 | 1.0089% | 17215.4145 | 17220 | 01000011010000100 | 3.228651466 | 58.7438176 | 2% | 1.143818 |  |
|  | 57.7 | 3.203781339 | -0.000008 | 0.018900% | 0.009999 | 3.236205579 | 1.0121% | 17265.4679 | 17266 | 0100001101110010  | 3.237276203 | 58.8473144 | 2% | 1.147314 |  |
|  | 57.8 | 3.212396753 | -0.000008 | 0.018900% | 0.010029 | 3.24500634  | 1.0151% | 17312.4208 | 17316 | 01000011010010010 | 3.246650917 | 58.959811  | 2% | 1.159811 |  |
|  | 57.9 | 3.221616194 | -0.000007 | 0.019000% | 0.010060 | 3.254420077 | 1.0182% | 17362.644  | 17366 | 0100001111010110  | 3.256025631 | 59.0723076 | 2% | 1.172308 |  |
|  | 58   | 3.230850096 | -0.000007 | 0.019000% | 0.010090 | 3.263846724 | 1.0213% | 17412.9361 | 17417 | 01000100000001001 | 3.265587839 | 59.1870541 | 2% | 1.187054 |  |
|  | 58.1 | 3.239513216 | -0.000006 | 0.019100% | 0.010121 | 3.272699622 | 1.0244% | 17460.1671 | 17463 | 0100010000101111  | 3.274212576 | 59.2905509 | 2% | 1.190551 |  |
|  | 58.2 | 3.248776766 | -0.000005 | 0.019100% | 0.010151 | 3.282156346 | 1.0275% | 17510.6197 | 17510 | 0100010001100110  | 3.283024807 | 59.3962977 | 2% | 1.196298 |  |
|  | 58.3 | 3.257471456 | -0.000005 | 0.019200% | 0.010182 | 3.291043268 | 1.0306% | 17558.0323 | 17559 | 0100010010010111  | 3.292212027 | 59.5065443 | 2% | 1.206544 |  |
|  | 58.4 | 3.266182291 | -0.000004 | 0.019200% | 0.010212 | 3.299942714 | 1.0336% | 17605.5117 | 17609 | 0100010011001001  | 3.30158674  | 59.6190409 | 2% | 1.219041 |  |
|  | 58.5 | 3.275489964 | -0.000004 | 0.019300% | 0.010243 | 3.309450066 | 1.0368% | 17656.2343 | 17658 | 0100010011111010  | 3.31077396  | 59.7292875 | 2% | 1.229288 |  |
|  | 58.6 | 3.284231654 | -0.000003 | 0.019300% | 0.010273 | 3.318381795 | 1.0398% | 17703.8859 | 17707 | 0100010100101011  | 3.31996118  | 59.8395342 | 2% | 1.239534 |  |
|  | 58.7 | 3.294146453 | -0.000002 | 0.019400% | 0.010304 | 3.328502748 | 1.0429% | 17757.8822 | 17756 | 0100010101011100  | 3.329148399 | 59.9497808 | 2% | 1.249781 |  |
|  | 58.8 | 3.302339864 | -0.000002 | 0.019400% | 0.010334 | 3.336882609 | 1.0460% | 17802.5896 | 17802 | 01000101000101010 | 3.337773136 | 60.0532776 | 2% | 1.253278 |  |
|  | 58.9 | 3.311127485 | -0.000001 | 0.019     |          |             |         |            |       |                   |             |            |    |          |  |

|      |              |          |           |          |             |         |            |       |                   |             |            |    |          |
|------|--------------|----------|-----------|----------|-------------|---------|------------|-------|-------------------|-------------|------------|----|----------|
| 60.9 | 3.488851994  | 0.000010 | 0.020500% | 0.010975 | 3.527614931 | 1.11114 | 18820.1648 | 18825 | 0100100101000101  | 3.529579782 | 32.3549574 | 2% | 1.454957 |
| 61   | 3.497490869  | 0.000011 | 0.020500% | 0.011005 | 3.536785993 | 1.11114 | 18868.7599 | 18873 | 0100100101011001  | 3.538579508 | 32.4629541 | 2% | 1.462954 |
| 61.1 | 3.50648962   | 0.000011 | 0.020600% | 0.011036 | 3.545669909 | 1.11739 | 18916.4887 | 18917 | 0100100101110010  | 3.546829256 | 32.5169151 | 2% | 1.461951 |
| 61.2 | 3.515605328  | 0.000012 | 0.020600% | 0.011066 | 3.554585924 | 1.12039 | 18966.2146 | 18971 | 0100100100000110  | 3.556953947 | 32.6834474 | 2% | 1.483447 |
| 61.3 | 3.524182798  | 0.000012 | 0.020700% | 0.011097 | 3.563775359 | 1.12359 | 19013.0842 | 19013 | 0100100100100001  | 3.564828707 | 32.7779445 | 2% | 1.477944 |
| 61.4 | 3.5327774952 | 0.000013 | 0.020700% | 0.011127 | 3.572573179 | 1.12659 | 19060.0114 | 19062 | 0100100100110110  | 3.574015926 | 32.8881911 | 2% | 1.488191 |
| 61.5 | 3.5413930552 | 0.000014 | 0.020800% | 0.011158 | 3.581492149 | 1.12969 | 19110.0007 | 19112 | 0100100101010000  | 3.58339064  | 32.0006877 | 2% | 1.500688 |
| 61.6 | 3.550550714  | 0.000014 | 0.020800% | 0.011188 | 3.590767572 | 1.13279 | 19157.0902 | 19162 | 0100100101010110  | 3.592765354 | 33.1131842 | 2% | 1.513184 |
| 61.7 | 3.559731972  | 0.000015 | 0.020900% | 0.011219 | 3.600164567 | 1.13589 | 19207.2241 | 19206 | 0100100100000010  | 3.601051102 | 32.2121812 | 2% | 1.512181 |
| 61.8 | 3.567833961  | 0.000015 | 0.020900% | 0.011249 | 3.609461586 | 1.13899 | 19251.5235 | 19252 | 0100100101010100  | 3.609639839 | 33.3156781 | 2% | 1.515678 |
| 61.9 | 3.57649681   | 0.000016 | 0.021000% | 0.011280 | 3.617347171 | 1.14209 | 19298.8661 | 19297 | 0100100101100001  | 3.618077082 | 33.416925  | 2% | 1.516925 |
| 62   | 3.58517365   | 0.000017 | 0.021000% | 0.011310 | 3.626226677 | 1.14519 | 19346.2677 | 19346 | 0100100110010010  | 3.627264301 | 33.5271716 | 2% | 1.527172 |
| 62.1 | 3.594400933  | 0.000017 | 0.021100% | 0.011341 | 3.635679499 | 1.14829 | 19396.6997 | 19398 | 0100100111000010  | 3.637014004 | 33.644168  | 2% | 1.544168 |
| 62.2 | 3.603110333  | 0.000018 | 0.021100% | 0.011371 | 3.644582933 | 1.15139 | 19444.2505 | 19446 | 0100100111110110  | 3.646013729 | 33.7521647 | 2% | 1.552165 |
| 62.3 | 3.611286935  | 0.000018 | 0.021200% | 0.011402 | 3.652977924 | 1.15459 | 19488.9963 | 19490 | 0100100100000100  | 3.654263477 | 33.8511661 | 2% | 1.551162 |
| 62.4 | 3.620101859  | 0.000019 | 0.021200% | 0.011432 | 3.661991595 | 1.15759 | 19536.6951 | 19538 | 0100100100010010  | 3.662362303 | 33.9591584 | 2% | 1.559158 |
| 62.5 | 3.628763583  | 0.000019 | 0.021300% | 0.011463 | 3.670881232 | 1.16079 | 19584.5043 | 19588 | 0100100100000010  | 3.672637197 | 34.071655  | 2% | 1.571655 |
| 62.6 | 3.638059215  | 0.000020 | 0.021300% | 0.011493 | 3.680399333 | 1.16379 | 19635.263  | 19640 | 0100100100111000  | 3.682387619 | 34.1886514 | 2% | 1.588651 |
| 62.7 | 3.646829647  | 0.000021 | 0.021400% | 0.011524 | 3.689832422 | 1.16689 | 19683.2099 | 19686 | 0100100101010010  | 3.691012356 | 34.2921483 | 2% | 1.592148 |
| 62.8 | 3.655613128  | 0.000021 | 0.021400% | 0.011554 | 3.698390547 | 1.16999 | 19731.2158 | 19732 | 0100100101000100  | 3.699637093 | 34.3956451 | 2% | 1.595645 |
| 62.9 | 3.663875341  | 0.000022 | 0.021500% | 0.011585 | 3.706854579 | 1.17319 | 19776.2586 | 19777 | 0100100100100001  | 3.708074335 | 34.496892  | 2% | 1.596892 |
| 63   | 3.672688099  | 0.000022 | 0.021500% | 0.011615 | 3.71588     | 1.17619 | 19824.5818 | 19827 | 0100100101000001  | 3.717449049 | 34.609386  | 2% | 1.609389 |
| 63.1 | 3.681508574  | 0.000023 | 0.021600% | 0.011646 | 3.724923136 | 1.17939 | 19872.8242 | 19873 | 0100100101010000  | 3.726073786 | 34.712854  | 2% | 1.612854 |
| 63.2 | 3.689281952  | 0.000023 | 0.021600% | 0.011676 | 3.732996654 | 1.18239 | 19915.3891 | 19914 | 0100100101010010  | 3.733760151 | 34.8051326 | 2% | 1.605133 |
| 63.3 | 3.698132034  | 0.000024 | 0.021700% | 0.011707 | 3.741972889 | 1.18559 | 19963.7838 | 19966 | 0100100101111110  | 3.743510754 | 34.922129  | 2% | 1.622129 |
| 63.4 | 3.707523476  | 0.000025 | 0.021700% | 0.011737 | 3.75158186  | 1.18859 | 20015.0837 | 20017 | 0100100100010000  | 3.753072923 | 35.0368755 | 2% | 1.636876 |
| 63.5 | 3.71869121   | 0.000025 | 0.021800% | 0.011768 | 3.760150895 | 1.19179 | 20060.7666 | 20060 | 0100100100100101  | 3.76132271  | 35.1358725 | 2% | 1.635873 |
| 63.6 | 3.724228949  | 0.000026 | 0.021800% | 0.011798 | 3.768723667 | 1.19479 | 20106.5031 | 20105 | 0100100101000101  | 3.769572458 | 35.2348695 | 2% | 1.634869 |
| 63.7 | 3.731281252  | 0.000026 | 0.021900% | 0.011829 | 3.776748232 | 1.19799 | 20155.1836 | 20159 | 0100100101011111  | 3.779697149 | 35.356365  | 2% | 1.653666 |
| 63.8 | 3.741515555  | 0.000027 | 0.021900% | 0.011859 | 3.786448367 | 1.20109 | 20201.714  | 20202 | 01001001010101010 | 3.787759403 | 35.4531128 | 2% | 1.653113 |
| 63.9 | 3.750440035  | 0.000027 | 0.022000% | 0.011890 | 3.795602099 | 1.20419 | 20249.8911 | 20251 | 01001001000100010 | 3.798946623 | 35.563395  | 2% | 1.663395 |
| 64   | 3.758584333  | 0.000028 | 0.022000% | 0.011920 | 3.804229063 | 1.20729 | 20295.9278 | 20298 | 01001001010100010 | 3.805758654 | 35.6691062 | 2% | 1.669106 |
| 64.1 | 3.767280243  | 0.000029 | 0.022100% | 0.011951 | 3.812876234 | 1.21039 | 20342.0613 | 20340 | 0100100101010110  | 3.814008602 | 35.7681032 | 2% | 1.668103 |
| 64.2 | 3.776241381  | 0.000029 | 0.022100% | 0.011981 | 3.821601806 | 1.21349 | 20391.0672 | 20392 | 01001001010101010 | 3.823008327 | 35.8760999 | 2% | 1.676100 |
| 64.3 | 3.784694082  | 0.000030 | 0.022200% | 0.012012 | 3.830736185 | 1.21659 | 20437.3459 | 20438 | 01001001010101010 | 3.832008053 | 35.9820496 | 2% | 1.684097 |
| 64.4 | 3.793678736  | 0.000030 | 0.022200% | 0.012042 | 3.839796818 | 1.21969 | 20486.4847 | 20486 | 01000000000000010 | 3.841007776 | 36.0909033 | 2% | 1.684093 |
| 64.5 | 3.802156773  | 0.000031 | 0.022300% | 0.012073 | 3.848647798 | 1.22289 | 20532.906  | 20536 | 0100000000110000  | 3.850382492 | 36.2045899 | 2% | 1.704590 |
| 64.6 | 3.810647816  | 0.000031 | 0.022300% | 0.012103 | 3.857359739 | 1.22589 | 20579.3851 | 20583 | 0100000000110011  | 3.859194723 | 36.3103367 | 2% | 1.710337 |
| 64.7 | 3.818636104  | 0.000032 | 0.022400% | 0.012134 | 3.86556861  | 1.22909 | 20623.1672 | 20624 | 01000000000000010 | 3.866881988 | 36.4025389 | 2% | 1.702584 |
| 64.8 | 3.827153826  | 0.000033 | 0.022400% | 0.012164 | 3.87305168  | 1.23209 | 20669.7906 | 20671 | 0100000001011111  | 3.87569422  | 36.5083306 | 2% | 1.708331 |
| 64.9 | 3.835682002  | 0.000033 | 0.022500% | 0.012195 | 3.881062641 | 1.23529 | 20716.5115 | 20718 | 0100000001010110  | 3.884506451 | 36.6140774 | 2% | 1.714077 |
| 65   | 3.844227115  | 0.000034 | 0.022500% | 0.012225 | 3.889368093 | 1.23829 | 20763.2766 | 20764 | 01000000000000010 | 3.893131177 | 36.7175742 | 2% | 1.717574 |
| 65.1 | 3.852782448  | 0.000034 | 0.022600% | 0.012256 | 3.896011633 | 1.24149 | 20810.1381 | 20812 | 01000000010001010 | 3.902318407 | 36.8278209 | 2% | 1.727821 |
| 65.2 | 3.861350085  | 0.000035 | 0.022600% | 0.012286 | 3.903043031 | 1.24459 | 20857.0425 | 20862 | 01000000010111110 | 3.911505627 | 36.9380675 | 2% | 1.738068 |
| 65.3 | 3.86992991   | 0.000035 | 0.022700% | 0.012317 | 3.918212823 | 1.24769 | 20904.4021 | 20905 | 0100000001010100  | 3.919567681 | 37.0348146 | 2% | 1.734815 |
| 65.4 | 3.878013293  | 0.000036 | 0.022700% | 0.012347 | 3.92651526  | 1.25079 | 20948.3364 | 20947 | 01000000010101010 | 3.92744264  | 37.1293117 | 2% | 1.729312 |
| 65.5 | 3.886618161  | 0.000037 | 0.022800% | 0.012378 | 3.935350206 | 1.25389 | 20995.4717 | 20995 | 01000000000000010 | 3.936442366 | 37.2373084 | 2% | 1.737308 |
| 65.6 | 3.895234995  | 0.000037 | 0.022800% | 0.012408 | 3.944194717 | 1.25699 | 21042.658  | 21046 | 01000000010101010 | 3.94600454  | 37.3520459 | 2% | 1.752055 |
| 65.7 | 3.903863303  | 0.000038 | 0.022900% | 0.012439 | 3.953054628 | 1.26019 | 21089.9965 | 21091 | 01000000010001000 | 3.954441816 | 37.4533018 | 2% | 1.753302 |
| 65.8 | 3.912503348  | 0.000038 | 0.022900% | 0.012469 | 3.961923853 | 1.26319 | 21137.2447 | 21141 | 01000000010001001 | 3.96381655  | 37.5567984 | 2% | 1.765798 |
| 65.9 | 3.920651431  | 0.000039 | 0.023000% | 0.012500 | 3.970298436 | 1.26639 | 21181.9239 | 21189 | 01000000010101110 | 3.971503796 | 37.6580455 | 2% | 1.758046 |
| 66   | 3.92931536   | 0.000040 | 0.023000% | 0.012530 | 3.979191994 | 1.26939 | 21229.3714 | 21229 | 01000000010101010 | 3.980316027 | 37.7637923 | 2% | 1.763792 |
| 66.1 | 3.937489113  | 0.000040 | 0.023100% | 0.012561 | 3.987598494 | 1.27259 | 21274.2004 | 21276 | 01000000010000010 | 3.989182258 | 37.8695391 | 2% | 1.769539 |
| 66.2 | 3.945175597  | 0.000041 | 0.023100% | 0.012591 | 3.995491917 | 1.27569 | 21316.3724 | 21317 | 01000000010000001 | 3.996815523 | 37.9617863 | 2% | 1.761786 |
| 66.3 | 3.953875995  | 0.000041 | 0.023200% | 0.012622 | 4.003436257 | 1.27889 | 21364.0524 | 21367 | 01000000010101010 | 4.006190237 | 38.0742828 | 2% | 1.774283 |
| 66.4 | 3.962587315  | 0.000042 | 0.023200% | 0.012652 | 4.011379528 | 1.28189 | 21411.7674 | 21412 | 01000000010101000 | 4.01642768  | 38.1755598 | 2% | 1.775530 |
| 66.5 | 3.970812019  | 0.000042 | 0.023300% | 0.012683 | 4.019386322 | 1.28509 | 21456.883  | 21463 | 01000000010101010 | 4.023814699 | 38.2857764 | 2% | 1.785776 |
| 66.6 | 3.979049419  | 0.000043 | 0.023300% | 0.012713 | 4.027308987 | 1.28809 | 21502.2408 | 21501 | 01000000010111111 | 4.031689459 | 38.3802735 | 2% | 1.780274 |
| 66.7 | 3.987794823  | 0.000044 | 0.023400% | 0.012744 | 4.039284726 | 1.29129 | 21549.9724 | 21552 | 01000000000000000 | 4.040876678 | 38.4905201 | 2% | 1.790520 |
| 66.8 | 3.996057427  | 0.000044 | 0.023400% | 0.012774 | 4.047775817 | 1.29439 | 21595.2732 | 21596 | 01000000010101010 | 4.049126427 | 38.5985171 | 2% | 1.798517 |
| 66.9 | 4.004330067  | 0.000045 | 0.023500% | 0.012805 | 4.056283026 | 1.29749 | 21640.66   | 21643 | 01000000010000010 | 4.057938658 | 38.6952639 | 2% | 1.795264 |
| 67   | 4.013108526  | 0.000045 | 0.023500% | 0.012835 | 4.065298893 | 1.30059 | 21686.7605 | 21691 | 01000000010101010 | 4.066938383 | 38.8032606 | 2% | 1.803261 |
| 67.1 | 4.02140568   | 0.000046 | 0.023600% | 0.012866 | 4.073830841 | 1.30379 | 21734.2792 | 21735 | 01000000010101011 | 4.075188131 | 38.9022576 | 2% | 1.802258 |
| 67.2 | 4.029714839  | 0.000046 | 0.023600% | 0.012896 | 4.082372365 | 1.30679 | 21779.8491 | 21782 | 01000000010100010 | 4.084000362 | 39.0008043 | 2% | 1.808004 |
| 67.3 | 4.038035887  | 0.000047 | 0.023700% | 0.012927 | 4.090920554 | 1.30999 | 21825.5207 | 2182  |                   |             |            |    |          |

|  |      |             |          |           |          |             |         |            |       |                   |             |            |    |          |
|--|------|-------------|----------|-----------|----------|-------------|---------|------------|-------|-------------------|-------------|------------|----|----------|
|  | 69.1 | 4.185943996 | 0.000057 | 0.024600% | 0.013476 | 4.243124561 | 1.3660% | 22637.4775 | 22636 | 0101100001101100  | 4.244120476 | 70.9294457 | 3% | 1.829446 |
|  | 69.2 | 4.194006373 | 0.000058 | 0.024600% | 0.013506 | 4.25142534  | 1.3691% | 22681.763  | 22685 | 0101100010001101  | 4.253307695 | 71.0396923 | 3% | 1.839692 |
|  | 69.3 | 4.201610495 | 0.000058 | 0.024700% | 0.013537 | 4.259267413 | 1.3723% | 22723.6012 | 22724 | 01011000011000100 | 4.260619972 | 71.1274397 | 3% | 1.827440 |
|  | 69.4 | 4.209696863 | 0.000059 | 0.024700% | 0.013567 | 4.267593517 | 1.3753% | 22768.0217 | 22768 | 0101100011110000  | 4.26886972  | 71.2264366 | 3% | 1.826437 |
|  | 69.5 | 4.21779457  | 0.000060 | 0.024800% | 0.013598 | 4.275935959 | 1.3785% | 22812.5295 | 22817 | 01011000100100001 | 4.27805694  | 71.3366833 | 3% | 1.836683 |
|  | 69.6 | 4.225903504 | 0.000060 | 0.024800% | 0.013628 | 4.284268976 | 1.3816% | 22857.0829 | 22856 | 01011000101001000 | 4.285369217 | 71.4244306 | 3% | 1.824431 |
|  | 69.7 | 4.234023552 | 0.000061 | 0.024900% | 0.013659 | 4.292653108 | 1.3847% | 22901.7171 | 22903 | 0101100010110111  | 4.294818448 | 71.5301774 | 3% | 1.830177 |
|  | 69.8 | 4.2421546   | 0.000061 | 0.024900% | 0.013689 | 4.301027573 | 1.3878% | 22946.3956 | 22949 | 0101100010100101  | 4.302806185 | 71.6336742 | 3% | 1.833674 |
|  | 69.9 | 4.249832509 | 0.000062 | 0.025000% | 0.013720 | 4.308946458 | 1.3910% | 22988.6436 | 22987 | 01011000111000101 | 4.309930967 | 71.7191716 | 3% | 1.819172 |
|  | 70   | 4.258449253 | 0.000063 | 0.025000% | 0.013750 | 4.317813389 | 1.3940% | 23033.9496 | 23035 | 01011000111111011 | 4.318930693 | 71.8271683 | 3% | 1.827168 |
|  | 70.1 | 4.266612633 | 0.000063 | 0.025100% | 0.013781 | 4.326226562 | 1.3972% | 23080.8347 | 23085 | 0101101000010101  | 4.328350407 | 71.9396649 | 3% | 1.839665 |
|  | 70.2 | 4.274325413 | 0.000064 | 0.025100% | 0.013811 | 4.334177944 | 1.4003% | 23123.2561 | 23125 | 0101101001010101  | 4.335805178 | 72.0296621 | 3% | 1.829662 |
|  | 70.3 | 4.282510746 | 0.000064 | 0.025200% | 0.013842 | 4.342614412 | 1.4035% | 23168.2654 | 23169 | 01011010100000001 | 4.344054926 | 72.1286591 | 3% | 1.828659 |
|  | 70.4 | 4.290247169 | 0.000065 | 0.025200% | 0.013872 | 4.350590759 | 1.4065% | 23210.82   | 23215 | 0101101010101011  | 4.352679663 | 72.232156  | 3% | 1.832156 |
|  | 70.5 | 4.298454004 | 0.000065 | 0.025300% | 0.013903 | 4.359050071 | 1.4097% | 23255.9512 | 23256 | 01011010110110100 | 4.360366928 | 72.3244031 | 3% | 1.824403 |
|  | 70.6 | 4.306670948 | 0.000066 | 0.025300% | 0.013933 | 4.367514722 | 1.4128% | 23301.111  | 23301 | 01011011000000101 | 4.368804171 | 72.42565   | 3% | 1.825650 |
|  | 70.7 | 4.314441509 | 0.000067 | 0.025400% | 0.013964 | 4.375531622 | 1.4159% | 23343.8819 | 23347 | 01011011001100101 | 4.377428907 | 72.5291469 | 3% | 1.829147 |
|  | 70.8 | 4.321768431 | 0.000067 | 0.025400% | 0.013994 | 4.38309566  | 1.4190% | 23384.2368 | 23388 | 01011011010101100 | 4.385116173 | 72.6213941 | 3% | 1.821394 |
|  | 70.9 | 4.330017884 | 0.000068 | 0.025500% | 0.014025 | 4.39159925  | 1.4222% | 23429.6042 | 23431 | 01011011100000111 | 4.393178427 | 72.7181411 | 3% | 1.818141 |
|  | 71   | 4.338277002 | 0.000068 | 0.025500% | 0.014055 | 4.400109729 | 1.4253% | 23475.0085 | 23478 | 0101101101101010  | 4.401990565 | 72.8238879 | 3% | 1.823888 |
|  | 71.1 | 4.346093091 | 0.000069 | 0.025600% | 0.014086 | 4.408174824 | 1.4284% | 23510.0365 | 23520 | 0101101111100000  | 4.409865417 | 72.918385  | 3% | 1.818385 |
|  | 71.2 | 4.353920518 | 0.000069 | 0.025600% | 0.014116 | 4.416248462 | 1.4315% | 23561.1102 | 23562 | 01011100000001010 | 4.417740177 | 73.0128821 | 3% | 1.812882 |
|  | 71.3 | 4.36175917  | 0.000070 | 0.025700% | 0.014147 | 4.424337451 | 1.4347% | 23604.2657 | 23609 | 01011100001110001 | 4.426552408 | 73.1186289 | 3% | 1.818629 |
|  | 71.4 | 4.369608935 | 0.000071 | 0.025700% | 0.014177 | 4.432433726 | 1.4378% | 23647.4601 | 23648 | 0101110001100000  | 4.433864865 | 73.2063762 | 3% | 1.806376 |
|  | 71.5 | 4.377918511 | 0.000071 | 0.025800% | 0.014208 | 4.44100245  | 1.4410% | 23693.1751 | 23698 | 0101110010010010  | 4.443239399 | 73.3188728 | 3% | 1.818873 |
|  | 71.6 | 4.385789227 | 0.000072 | 0.025800% | 0.014238 | 4.449120986 | 1.4440% | 23736.4882 | 23738 | 01011100101101010 | 4.45073917  | 73.40887   | 3% | 1.808870 |
|  | 71.7 | 4.393223792 | 0.000072 | 0.025900% | 0.014269 | 4.456803087 | 1.4472% | 23777.473  | 23779 | 01011100111000011 | 4.458426435 | 73.5011172 | 3% | 1.801172 |
|  | 71.8 | 4.401116893 | 0.000073 | 0.025900% | 0.014299 | 4.464945317 | 1.4503% | 23820.9126 | 23822 | 0101110100000110  | 4.466486869 | 73.5978643 | 3% | 1.797864 |
|  | 71.9 | 4.409020552 | 0.000073 | 0.026000% | 0.014330 | 4.473104305 | 1.4535% | 23864.4415 | 23869 | 01011101001110101 | 4.47530092  | 73.703611  | 3% | 1.803611 |
|  | 72   | 4.417378778 | 0.000074 | 0.026000% | 0.014360 | 4.481719436 | 1.4565% | 23910.4041 | 23910 | 0101110101100100  | 4.482988186 | 73.7958582 | 2% | 1.795858 |
|  | 72.1 | 4.424859096 | 0.000075 | 0.026100% | 0.014391 | 4.489448921 | 1.4597% | 23951.6416 | 23950 | 0101110110000110  | 4.490487957 | 73.885855  | 2% | 1.785855 |
|  | 72.2 | 4.432793763 | 0.000075 | 0.026100% | 0.014421 | 4.49763634  | 1.4628% | 23995.3223 | 23996 | 01011101101101100 | 4.499112694 | 73.9893523 | 2% | 1.789352 |
|  | 72.3 | 4.440297222 | 0.000076 | 0.026200% | 0.014452 | 4.505390284 | 1.4660% | 24036.6903 | 24038 | 0101110111100100  | 4.506987453 | 74.0838494 | 2% | 1.783849 |
|  | 72.4 | 4.448252942 | 0.000076 | 0.026200% | 0.014482 | 4.513600058 | 1.4691% | 24080.4903 | 24082 | 0101110000010010  | 4.515237202 | 74.1828464 | 2% | 1.782846 |
|  | 72.5 | 4.455779093 | 0.000077 | 0.026300% | 0.014513 | 4.52137801  | 1.4722% | 24121.9864 | 24121 | 01011100001110001 | 4.522549479 | 74.2705937 | 2% | 1.770594 |
|  | 72.6 | 4.463755425 | 0.000077 | 0.026300% | 0.014543 | 4.529609696 | 1.4753% | 24165.9032 | 24169 | 01011110011010001 | 4.531549204 | 74.3785904 | 2% | 1.778590 |
|  | 72.7 | 4.471303823 | 0.000078 | 0.026400% | 0.014574 | 4.537411209 | 1.4785% | 24207.5251 | 24212 | 01011110100010100 | 4.539611458 | 74.4753375 | 2% | 1.775337 |
|  | 72.8 | 4.478863641 | 0.000079 | 0.026400% | 0.014604 | 4.545220189 | 1.4815% | 24249.1867 | 24254 | 0101111010111100  | 4.547486217 | 74.5698346 | 2% | 1.769835 |
|  | 72.9 | 4.48667053  | 0.000079 | 0.026500% | 0.014635 | 4.553488992 | 1.4847% | 24293.3016 | 24292 | 0101111011100100  | 4.554611    | 74.655332  | 2% | 1.755332 |
|  | 73   | 4.494886771 | 0.000080 | 0.026500% | 0.014665 | 4.561762162 | 1.4878% | 24337.4397 | 24340 | 01011111000010100 | 4.563610725 | 74.7633287 | 2% | 1.763329 |
|  | 73.1 | 4.502478335 | 0.000080 | 0.026600% | 0.014696 | 4.569610478 | 1.4910% | 24379.3113 | 24383 | 01011111001111111 | 4.571672979 | 74.8600758 | 2% | 1.760076 |
|  | 73.2 | 4.510080875 | 0.000081 | 0.026600% | 0.014726 | 4.577464767 | 1.4941% | 24421.2146 | 24425 | 01011111011010001 | 4.579547739 | 74.9545729 | 2% | 1.754573 |
|  | 73.3 | 4.518126366 | 0.000081 | 0.026700% | 0.014757 | 4.585774789 | 1.4973% | 24465.5494 | 24467 | 01011111100100011 | 4.587422499 | 75.04907   | 2% | 1.749070 |
|  | 73.4 | 4.52574961  | 0.000082 | 0.026700% | 0.014787 | 4.593651071 | 1.5003% | 24507.5701 | 24509 | 01011111101101101 | 4.595297258 | 75.1435671 | 2% | 1.743567 |
|  | 73.5 | 4.53383501  | 0.000083 | 0.026800% | 0.014818 | 4.60154331  | 1.5035% | 24549.676  | 24553 | 01011111110100001 | 4.603547007 | 75.2425641 | 2% | 1.742564 |
|  | 73.6 | 4.541027928 | 0.000083 | 0.026800% | 0.014848 | 4.609443077 | 1.5066% | 24591.822  | 24594 | 0110000000010010  | 4.611234272 | 75.3348113 | 2% | 1.734811 |
|  | 73.7 | 4.548254354 | 0.000084 | 0.026900% | 0.014879 | 4.616922684 | 1.5098% | 24631.7264 | 24635 | 0110000000110101  | 4.618921537 | 75.4270584 | 2% | 1.727058 |
|  | 73.8 | 4.556347954 | 0.000084 | 0.026900% | 0.014909 | 4.62527938  | 1.5129% | 24676.3102 | 24679 | 0110000001100111  | 4.627171286 | 75.5260554 | 2% | 1.726055 |
|  | 73.9 | 4.564023335 | 0.000085 | 0.027000% | 0.014940 | 4.633215711 | 1.5160% | 24718.6513 | 24719 | 0110000010001111  | 4.634671057 | 75.6160527 | 2% | 1.716053 |
|  | 74   | 4.570857421 | 0.000086 | 0.027000% | 0.014970 | 4.64029375  | 1.5191% | 24756.4133 | 24760 | 0110000010110000  | 4.642358322 | 75.7082999 | 2% | 1.708300 |
|  | 74.1 | 4.578554707 | 0.000086 | 0.027100% | 0.015001 | 4.6482543   | 1.5223% | 24798.8836 | 24801 | 0110000011000001  | 4.650045588 | 75.8005471 | 2% | 1.700547 |
|  | 74.2 | 4.586685761 | 0.000087 | 0.027100% | 0.015031 | 4.656649996 | 1.5254% | 24843.6755 | 24847 | 0110000100000111  | 4.658670324 | 75.9040439 | 2% | 1.704044 |
|  | 74.3 | 4.593978818 | 0.000087 | 0.027200% | 0.015062 | 4.664201142 | 1.5286% | 24883.9615 | 24886 | 0110000100101010  | 4.665982601 | 75.9917912 | 2% | 1.691791 |
|  | 74.4 | 4.601705426 | 0.000088 | 0.027200% | 0.015092 | 4.6721872   | 1.5316% | 24926.5679 | 24925 | 0110000101010101  | 4.673294878 | 76.0795385 | 2% | 1.679539 |
|  | 74.5 | 4.609020417 | 0.000088 | 0.027300% | 0.015123 | 4.679761574 | 1.5348% | 24966.9779 | 24969 | 01100001100001001 | 4.681544626 | 76.1785355 | 2% | 1.678536 |
|  | 74.6 | 4.616766937 | 0.000089 | 0.027300% | 0.015153 | 4.687768819 | 1.5379% | 25009.6974 | 25008 | 0110000110110000  | 4.688856903 | 76.2662828 | 2% | 1.666283 |
|  | 74.7 | 4.624522802 | 0.000090 | 0.027400% | 0.015184 | 4.695790809 | 1.5411% | 25052.4955 | 25051 | 0110000110101011  | 4.696919157 | 76.3630299 | 2% | 1.663030 |
|  | 74.8 | 4.631869424 | 0.000090 | 0.027400% | 0.015214 | 4.703393978 | 1.5442% | 25093.0591 | 25095 | 01100010000000111 | 4.705168905 | 76.4620269 | 2% | 1.662027 |
|  | 74.9 | 4.639226971 | 0.000091 | 0.027500% | 0.015245 | 4.711012472 | 1.5474% | 25133.7045 | 25138 | 0110001000110010  | 4.713231159 | 76.5587739 | 2% | 1.658774 |
|  | 75   | 4.647012023 | 0.000091 | 0.027500% | 0.015275 | 4.719061806 | 1.5505% | 25176.6485 | 25178 | 01100010010101010 | 4.72073093  | 76.6487712 | 2% | 1.648771 |
|  | 75.1 | 4.654390197 | 0.000092 | 0.027600% | 0.015306 | 4.726702214 | 1.5536% | 25217.4108 | 25219 | 01100010100000011 | 4.728481896 | 76.7410183 | 2% | 1.641018 |
|  | 75.2 | 4.66177897  | 0.000092 | 0.027600% | 0.015336 | 4.734350073 | 1.5567% | 25258.2128 | 25262 | 0110001010101010  | 4.73648045  | 76.8377654 | 2% | 1.637765 |
|  | 75.3 | 4.669178231 | 0.000093 | 0.027700% | 0.015367 | 4.742012851 | 1.5599% | 25299.0945 | 25298 | 0110001011010010  | 4.743230244 | 76.9187629 | 2% | 1.618763 |
|  | 75.4 | 4.677000998 | 0.000094 | 0.027700% | 0.015397 | 4.750101417 | 1.5630% | 253        |       |                   |             |            |    |          |

|      |             |          |           |          |             |         |            |       |                   |             |            |    |          |
|------|-------------|----------|-----------|----------|-------------|---------|------------|-------|-------------------|-------------|------------|----|----------|
| 77.3 | 4.816426028 | 0.001004 | 0.028700% | 0.015977 | 4.894578803 | 1.6227% | 26113.0539 | 26112 | 0110011000000000  | 4.898580568 | 78.750207  | 2% | 1.450207 |
| 77.4 | 4.82323576  | 0.001005 | 0.028700% | 0.016007 | 4.905169519 | 1.6257% | 26115.7661 | 26112 | 0110011000100100  | 4.903350357 | 78.842043  | 2% | 1.440204 |
| 77.5 | 4.830846648 | 0.001006 | 0.028800% | 0.016038 | 4.905366787 | 1.6289% | 26129.8512 | 26195 | 0110011001010011  | 4.911421611 | 78.9369513 | 2% | 1.436951 |
| 77.6 | 4.838071822 | 0.001006 | 0.028800% | 0.016068 | 4.917029581 | 1.6320% | 26232.8256 | 26234 | 01100110011111010 | 4.918724888 | 79.0246987 | 2% | 1.424698 |
| 77.7 | 4.844913332 | 0.001007 | 0.028900% | 0.016099 | 4.924137651 | 1.6352% | 26270.7448 | 26269 | 0110011001101111  | 4.925287188 | 79.1034663 | 2% | 1.403466 |
| 77.8 | 4.852158466 | 0.001007 | 0.028900% | 0.016129 | 4.931611134 | 1.6383% | 26310.8333 | 26312 | 0110011001100100  | 4.933349442 | 79.2001933 | 2% | 1.400193 |
| 77.9 | 4.85941307  | 0.001008 | 0.029000% | 0.016160 | 4.939179277 | 1.6415% | 26350.9963 | 26350 | 0110011001101110  | 4.940474224 | 79.2856907 | 2% | 1.385691 |
| 78   | 4.866677032 | 0.001009 | 0.029000% | 0.016190 | 4.946712366 | 1.6446% | 26391.1861 | 26393 | 0110011001100011  | 4.948536478 | 79.3824377 | 2% | 1.382438 |
| 78.1 | 4.873560604 | 0.001009 | 0.029100% | 0.016221 | 4.953865313 | 1.6478% | 26429.3477 | 26428 | 01100110011001110 | 4.955098778 | 79.4611853 | 2% | 1.361185 |
| 78.2 | 4.881232623 | 0.001010 | 0.029100% | 0.016251 | 4.961814122 | 1.6508% | 26471.7554 | 26473 | 01100110011010101 | 4.96353602  | 79.5624322 | 2% | 1.362432 |
| 78.3 | 4.888136096 | 0.001010 | 0.029200% | 0.016282 | 4.968898218 | 1.6540% | 26501.0299 | 26515 | 01100110011001011 | 4.97141078  | 79.6569294 | 2% | 1.356929 |
| 78.4 | 4.89505021  | 0.001011 | 0.029200% | 0.016312 | 4.976167508 | 1.6571% | 26548.3321 | 26552 | 01100110011011000 | 4.978348068 | 79.7401768 | 2% | 1.340177 |
| 78.5 | 4.902361114 | 0.001011 | 0.029300% | 0.016343 | 4.983756701 | 1.6603% | 26588.8212 | 26592 | 01100110011011000 | 4.985847839 | 79.8301741 | 2% | 1.330174 |
| 78.6 | 4.909680757 | 0.001012 | 0.029300% | 0.016373 | 4.991349167 | 1.6634% | 26629.3277 | 26632 | 01101000000001000 | 4.99334761  | 79.9201713 | 2% | 1.320174 |
| 78.7 | 4.916624467 | 0.001013 | 0.029400% | 0.016404 | 4.998564975 | 1.6666% | 26677.8247 | 26671 | 01101000000101111 | 5.000659887 | 80.0079186 | 2% | 1.307919 |
| 78.8 | 4.923962124 | 0.001013 | 0.029400% | 0.016434 | 5.006177695 | 1.6697% | 26708.4393 | 26707 | 01101000000100011 | 5.007409681 | 80.0889162 | 2% | 1.288916 |
| 78.9 | 4.930925325 | 0.001014 | 0.029500% | 0.016465 | 5.013414236 | 1.6729% | 26747.4047 | 26749 | 01101000000111001 | 5.015284441 | 80.1834433 | 2% | 1.283413 |
| 79   | 4.93789835  | 0.001014 | 0.029500% | 0.016495 | 5.02065321  | 1.6760% | 26785.6895 | 26789 | 01101000000110011 | 5.022784212 | 80.2734105 | 2% | 1.273411 |
| 79.1 | 4.944504036 | 0.001015 | 0.029600% | 0.016526 | 5.027527395 | 1.6792% | 26822.342  | 26822 | 01101000000110001 | 5.028971523 | 80.3476583 | 2% | 1.247658 |
| 79.2 | 4.95148184  | 0.001015 | 0.029600% | 0.016556 | 5.034792245 | 1.6823% | 26861.1007 | 26865 | 01101000011000110 | 5.037037777 | 80.4444053 | 2% | 1.234405 |
| 79.3 | 4.958497626 | 0.001016 | 0.029700% | 0.016587 | 5.042071504 | 1.6855% | 26899.9363 | 26900 | 01101000010001010 | 5.043596077 | 80.5231529 | 2% | 1.223153 |
| 79.4 | 4.965889368 | 0.001017 | 0.029700% | 0.016617 | 5.049740948 | 1.6886% | 26940.8535 | 26940 | 01101000010111010 | 5.051095848 | 80.6131502 | 2% | 1.213150 |
| 79.5 | 4.972531575 | 0.001017 | 0.029800% | 0.016648 | 5.056656426 | 1.6918% | 26977.7482 | 26977 | 01101000010100001 | 5.058033136 | 80.6963976 | 2% | 1.196398 |
| 79.6 | 4.979565076 | 0.001018 | 0.029800% | 0.016678 | 5.063986867 | 1.6948% | 27016.718  | 27016 | 01101000010001000 | 5.065345413 | 80.784145  | 2% | 1.184145 |
| 79.7 | 4.986606256 | 0.001018 | 0.029900% | 0.016709 | 5.071281327 | 1.6981% | 27055.7735 | 27059 | 01101000010101010 | 5.073407667 | 80.880892  | 1% | 1.168892 |
| 79.8 | 4.993656614 | 0.001019 | 0.029900% | 0.016739 | 5.078605419 | 1.7011% | 27094.8482 | 27097 | 01101000010110010 | 5.08053245  | 80.9663984 | 1% | 1.160639 |
| 79.9 | 5.000341479 | 0.001019 | 0.030000% | 0.016770 | 5.085564422 | 1.7043% | 27131.9527 | 27132 | 01101010000000000 | 5.080322221 | 81.0563867 | 1% | 1.156387 |
| 80   | 5.002267689 | 0.001020 | 0.030000% | 0.016800 | 5.087676427 | 1.7074% | 27173.2497 | 27177 | 01101010000000010 | 5.088969692 | 81.0676363 | 1% | 1.067636 |